



The Actualization State

Artist's Proof 01

Foundations

The spine — viability geometry, agency, coupled corridors

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Artist's Note

This is Artist's Proof 01 of The 420 Code — the foundational paper of the work. It is the second chapter of Notebook I, The Premise. The paper installs the four formal Papers from which the rest of the work hangs.

Five sections carry the work. Paper 0 is the structural narrative that motivated the work; it is not load-bearing. Paper A defines the Actualization State — the operational measure of record-structured irreversibility. Paper B characterises selection as a costly, rate-limited, irreversible exclusion process. Paper C develops agency as a geometric control quantity. Paper D extends coupling to multi-agent systems and derives structural filtering, hierarchy, cooperation, and deterrence as geometric consequences.

The dependency is one-way. Each paper is independently falsifiable. Failure of D does not invalidate C; failure of C does not invalidate B; failure of B does not invalidate A. The kill switches are published. The experiments are specified.

Twenty-four kill switches are engaged in this paper. Every load-bearing claim attaches to a structural condition under which the claim would fail. KS-V.1 (the operational invariance of AS itself) is the global kill switch — if it fires, the entire structure fails. The conditions are stated, the falsifiers are specified, and the registry is maintained at the420code.org.

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Orientation

Where this paper fits in the work

The 420 Code is organised across eight Notebooks. This paper is the second chapter of Notebook I — The Premise.

Notebook I carries the foundation: what the axiom is, what the operational quantities the axiom names actually are, and the proof that the axioms hold in physical reality. Three papers carry this Notebook — AP40 (The Irrational), AP01 (The Actualization State), AP20 (The Proof). AP40 examines the axiom as a statement. AP01 installs the four formal Papers of the spine. AP20 closes the central conditional of the work.

Read in order, the Notebook reads: what the axiom is, what it produces operationally, and that it holds.

AP01 sits second because the apparatus the axiom produces must be installed before it can be proven. AP40 names the axiom's structural character. AP01 installs the four operational quantities — actualisation state, selection cost, agency, joint viability under coupling — through which every later paper does its work. AP20 then closes the central conditional: the apparatus holds in physical reality, unconditionally within the work's derivation chain.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Paper 0 is the structural narrative that motivated the work and is explicitly non-load-bearing — read it as the story of how nothing becomes structure, and how structure, once written, cannot be unwritten. Papers A, B, C, D carry the formal spine in dependency order — A defines the state; B characterises selection; C develops agency; D extends to coupling. Section 6 is the falsification register (twenty-four kill switches). Section 7 is the

synthesis — what this paper closes, what it leaves open, and where the philosophical-register companions live.

Two voices alternate as the material requires. Narrative passages — Paper 0, openings, transitions, interpretive paragraphs immediately after a proof — speak directly. Formal sections — definitions, theorems, proofs, postulates, kill switches — speak in the precision the mathematics requires. The gear-changes are intentional.

A note on what this paper does and does not do. AP01 installs the four operational quantities and proves the foundational theorems about them. It does not derive the fundamental constants. It does not close every empirical question. What it does is install the four formal quantities, prove the theorems they obey, and tie each load-bearing claim to at least one published falsifier with a specified observable.

0 — Dependency and Scope

0.1 — What this paper does

AP01 installs the four formal Papers through which The 420 Code does its operational work. Each Paper defines an operational quantity, proves the theorems that quantity obeys, and ties each load-bearing claim to at least one published falsifier with a specified observable.

Four claims, each derived.

First. There is an operational measure of record-structured irreversibility — the Actualization State (AS) — definable on any system that admits a physically realizable coarse-graining, monotone under decohering dynamics, and bounded between zero and one. Paper A produces the definition, the monotonicity theorem (T1), and the no-return-surface result (T2).

Second. Selection between alternatives — the transition from multiplicity to definiteness — is a costly, rate-limited, irreversible exclusion process. The selection cost is non-zero, the selection rate is bounded above by $\Delta E_G/\hbar$ for gravitationally distinguishable records (under the gravity-limited hypothesis), and selection cannot precede operational irreversibility. Paper B produces the definitions, the structural requirements on the selection channel, and the rate inequality with its kill switches.

Third. Agency is the operator-relative measure of the viability kernel reachable from a current state under admissible control. Agency decays under irreversible drift unless control compensates; the viability kernel has no-return surfaces that act as absorbing boundaries. Paper C produces the geometric definition, the control-budget and fatigue results, the slack and redundancy structure, and the exit-as-control-outcome treatment.

Fourth. Joint agency under coupling is non-additive. Stable configurations under drift satisfy specific structural conditions — impedance matching, anti-resonance avoidance, non-cascade propagation. Emergent order — hierarchy, cooperation, deterrence — falls out of the geometry, not from imported preferences. Paper D produces the joint-agency formalism, the structural-filtering result, and the falsifiers for each derived ordering.

The paper makes one further methodological claim. The four quantities are defined operationally — they can be measured — and tied to twenty-four falsifiers that specify the observable conditions under which each load-bearing claim would fail. KS-V.1 (operational invariance of AS itself) is the global kill switch. If it fires, the entire structure fails.

0.2 — Dependencies

The axiom $1:1 + 1 \times \varepsilon @ AS$ is taken as primitive. AP01 inherits the axiom and installs the operational apparatus the axiom produces under $\{S, B, R, C\}$.

AP40 (The Irrational). The first chapter of Notebook I. AP40 examines the structural character of the governing axiom — its irrationality, its pre-duality exemption, its necessity as the only kind of axiom that can found the framework it generates. AP01 inherits AP40's establishment of the axiom and proceeds to install the operational apparatus. Not load-bearing for AP01's own derivations — AP01's theorems hold whether the axiom is read structurally or arithmetically — but named for completeness as the immediately prior chapter in Notebook I.

The axioms $\{S, B, R, C\}$ (formal statement carried at AP20 section 3.3; completeness and minimality obligations fenced at KS-P.1/KS-P.2). Load-bearing across all four formal Papers. Axiom R is principal — records do not unwrite; operational irreversibility is the structural fact R produces in the presence of finite C. Axiom B carries Paper B (selection as the break between alternatives). Axiom C carries the bounded propagation that makes operator horizons and finite viability kernels possible. Axiom S provides the substrate within which records are written and distinguishable sectors exist.

The work's later derivations inherit this paper's apparatus. AP07 (The Record Measure) inherits the record-algebra coarse-graining framework.

AP08 (The Identity — Einstein Field Equations) inherits the irreversibility theorems. AP25 (The Measure) inherits Paper A’s operational coarse-graining apparatus and the dephased-entropy structure. AP43 (The Gravity of Possibilities) inherits the Actualization State as the operator’s records-side anchor and develops the structural dual to it on the pre-state side. The Ø Models catalogue’s philosophical-register treatments of measurement (Dissolutions Chapter 12), choice (Resolutions Chapter 8.5), other minds (Dissolutions Chapter 6), free will and determinism (Dissolutions Chapter 7), and other intelligences (Resolutions Chapter 10) all stand on the apparatus this paper installs.

0.3 — Axiom mapping

Axiom R is principal load-bearing across Papers A, B, C, and D. Records do not unwrite. Operational irreversibility is the structural fact R produces in the presence of finite C. Without R, the Actualization State has nothing to measure, no-return surfaces collapse, and selection has no irreversibility to be costly relative to.

Axiom B is load-bearing for Paper B. The break — selection between alternatives — is what the costly exclusion process accomplishes. Without B, multiplicity does not resolve to definiteness, and the operational distinction between the pre-measurement ensemble and the post-measurement individual realization does not exist.

Axiom C is load-bearing for Papers A, C, and D. The bounded propagation through which records reach is what produces the operator horizon (T2), the no-return surfaces in viability theory, and the finite viability kernels on which agency is defined. Without C, no-return surfaces don’t exist and agency collapses to a uniform unbounded predicate.

Axiom S provides the structure of the substrate within which records are written and distinguishable sectors exist. Without S, the physically realizable coarse-graining **Θ** of Paper A has nothing to range over, and the record algebra has no distinct sectors to be selected between.

0.4 — Epistemic status per section

Paper 0 (Foundations). NARRATIVE. The structural story that motivated the work. Explicitly non-load-bearing. The papers that follow stand or fall on their own logic, independent of the narrative.

Paper A (Actualization State: An Operational Measure of Record-Structured Irreversibility). DEFINITION + DERIVATION. Five core definitions (D1–D5), monotonicity theorem T1 with rate corollary T1a and converse T1b, operator-horizon theorem T2, no-return surface results (D7–D10, P3.3), quantum-mechanical instantiation (D11–D16, postulate P, postulate G, proposition P4.1), and seven falsifiers (F0 global, F1–F3 selection, G1–G3 gravitational). Worked examples in appendices.

Paper B (Selection as Irreversible Exclusion). STRUCTURAL DERIVATION. Definiteness reframed as operational individual-realization status. Nonlinearity requirement and selection cost made operational (B2.3, B2.4). Six structural requirements on selection dynamics (B3.1–B3.6). Gravitational rate-limiter hypothesis (B4) with its own kill switches. Five experimental regimes (BT1–BT5).

Paper C (Agency as Constrained Control). DEFINITION + DERIVATION. Agency as the operator-relative ratio of reachable-set-within-viability to viability (C1). Drift under irreversibility (C2). Necessary conditions for agency preservation (C3). No-return geometry (C4). Control budgets and fatigue (C5 + theorem C5.1). Noise and silence (C6). Coupling and rescue (C7). Slack and robustness (C8). Exit as control outcome (C9). Falsifiers and closure (C10).

Paper D (Coupled Viability). DEFINITION + DERIVATION. Shared constraint environments (D1) with the Geometric Exclusion Principle. Composition of agency (D2) with non-additivity, impedance matching, anti-resonance. Stable configurations under drift (D3) with necessary and sufficient conditions. Emergent order without design (D4) — structural filtering, hierarchy, cooperation, deterrence. Experimental instantiations and falsifiers (D5). Structural closure (D6).

Section 6 (Kill Switches). FALSIFICATION REGISTER. Twenty-four kill switches engaged. KS-V.1 is the global kill switch (operational invariance of

AS itself). KS-V.2-V.4 target the selection postulate; KS-V.5-V.7 target the gravitational rate limiter; KS-V.8 targets selection-irreversibility ordering. KS-V.9-V.13 target the agency apparatus. KS-V.14-V.24 target the coupled-viability apparatus.

Section 7 (What This Establishes and What Remains Open). SYNTHESIS.

0.5 — Kill switch summary

Twenty-four kill switches are engaged in this paper. The full register is in section 6. Numbering follows the master registry: KS-V.1 through KS-V.24. Legacy labels (F0, F1-F3, G1-G3, B2, FC1-FC5, FD0-FD4.4b) are preserved at first mention in the consolidated section and remain attached to their formal blocks in the body of the Papers.

KS-V.1: Operational invariance of AS (F0). GLOBAL KILL SWITCH. LIVE — EMPIRICAL.

KS-V.2: Pointer-basis targeting (F1). LIVE — EMPIRICAL.

KS-V.3: Born violation (F2). LIVE — EMPIRICAL.

KS-V.4: Context dependence (F3). LIVE — EMPIRICAL.

KS-V.5: Selection rate exceeds gravitational bound (G1). LIVE — EMPIRICAL.

KS-V.6: Selection in gravitationally degenerate regime (G2). LIVE — EMPIRICAL.

KS-V.7: Non-gravitational rate scaling (G3). LIVE — EMPIRICAL.

KS-V.8: Pre-irreversibility selection (B2). LIVE — EMPIRICAL.

KS-V.9: Agency increase without control (FC1). LIVE — EMPIRICAL.

KS-V.10: Irreversible loss reversed (FC2). LIVE — EMPIRICAL.

KS-V.11: Stable control past no-return surface (FC3). LIVE — EMPIRICAL.

KS-V.12: Free lunch (FC4). LIVE — EMPIRICAL.

KS-V.13: Resurrection (FC5). LIVE — EMPIRICAL.

KS-V.14: Multi-agent persistence violation (FD0). LIVE — EMPIRICAL.

KS-V.15: No free survival (FD1). LIVE — EMPIRICAL.

KS-V.16: Additivity under coupling (FD2.1). LIVE — EMPIRICAL.

KS-V.17: Impedance-independent efficiency (FD2.2). LIVE — EMPIRICAL.

KS-V.18: Anti-resonant optimality (FD2.3). LIVE — EMPIRICAL.

KS-V.19: Cascade non-propagation (FD3.3). LIVE — EMPIRICAL.

KS-V.20: Order indistinguishable from noise (FD4). LIVE — EMPIRICAL.

KS-V.21: Persistent violators (FD4.2). LIVE — EMPIRICAL.

KS-V.22: Hierarchy inversion (FD4.3). LIVE — EMPIRICAL.

KS-V.23: Cooperation nonexistence (FD4.4a). LIVE — EMPIRICAL.

KS-V.24: Deterrence exit (FD4.4b). LIVE — EMPIRICAL.

0.6 — Structural debts owed

None outstanding from this paper. AP01 installs operational apparatus and its falsifiers; its load-bearing claims are derived within the paper or are framed as postulates with explicit kill switches.

The gravity-limited rate hypothesis (postulate G in Paper A, B4 in Paper B, with falsifiers G1-G3 / F_G1-F_G3) is a falsifiable postulate carrying its own kill switches, not a structural debt. If it fails, Papers A and B retreat to weaker claims about selection rate; Paper C's agency apparatus and Paper D's coupling apparatus are unaffected.

The objective-actualisation-channel postulate (postulate P in A4.2) is a falsifiable postulate, not a structural debt. Its kill switches are KS-V.2 through KS-V.4 (pointer-basis targeting, Born statistics, context independence). If it fires, the work retreats to decoherence-without-actualisation, and Papers A and B's irreversibility results survive at the ensemble level.

0.7 — Items held open without claiming debt status

Three items are bracketed inside this paper. They are real and live, but they are not structural debts — they are operational specifications, parameter identifications, and cross-AP integration work.

The relativistic extension of Paper A's quantum-mechanical instantiation. Section A4 is non-relativistic; the gravitational rate bound is treated in the weak-field limit. The relativistic form is open work. This is operational specification, not structural derivation.

The first-principles derivation of the coupling parameters in Papers C and D — the control budget B_0 , the cost function $c(u)$, the impedance Z_i , the resonance phase. These are parameterised in the structural account so the theorems can be stated and the falsifiers specified; their physical instantiations in specific embodied systems are open work for domain experts. This is parameter identification, not structural derivation.

The relationship between AP01's records-side apparatus and AP43's pre-state-side apparatus. AP01 installs the operational measure of record-structured irreversibility — the body has already written records, and AS measures how irreversibly. AP43 installs the operator's interface with the pre-state through coupling capacity, pattern, pattern coherence, tracking coherence — the body reads the slope of what could be written next. The two are duals across the now: AP08 (gravity from records on the manifold) and AP43 (gravity of possibilities on the pre-state) inherit this dual structure. The formal integration of AS-on-records with AP43's amplitude-landscape apparatus is open work for the relativistic field-equation paper (closing AP43's D1).

0.8 — Structural relationships

This paper opens the formal spine of the work. Every later derivational paper that uses the words actualisation, irreversibility, selection, agency, viability, coupling, joint viability, no-return surface, viability kernel, or operator horizon inherits its definitions from one of the four Papers here.

This paper is the second chapter of Notebook I — The Premise. AP40 (Chapter 1) examines what the axiom is; AP01 (Chapter 2) installs the

operational quantities the axiom produces; AP20 (Chapter 3) closes the central conditional by proving the apparatus holds in physical reality.

This paper provides the formal-register apparatus that the Ø Models catalogue's philosophical-register treatments build on. Dissolutions Chapter 12 (The Measurement Problem) is the philosophical-register companion to Papers A and B — same observation-as-forcing-function structure, two vocabularies. Dissolutions Chapter 6 (Other Minds) and Chapter 7 (Free Will and Determinism), with Resolutions Chapter 8.5 (Choice) and Chapter 10 (Other Intelligences), are the philosophical-register companions to Papers C and D — the operator, agency, joint coupling, substrate-independence. The direction-of-acknowledgment is named explicitly in section 7.

This paper provides the records-side anchor for AP43's gravity-of-possibilities. AP01 names the operational measure of record-structured irreversibility on the manifold; AP43 names the operator's interface with the amplitude landscape on the pre-state. The two together open the dual structure AP08 and the future relativistic field-equation paper formalise.

Paper 0 — Foundations

P0.1 — Before the Beginning

Close your eyes. Try to imagine nothing.

Not darkness — darkness is something. Not silence — silence is something.
Not empty space — space is something.

Nothing.

The absence of everything, including the absence itself.

You cannot do it. Your mind keeps producing something to fill the void. That inability is not a failure of imagination. It is the first clue.

Begin with nothing. Not empty space. Not vacuum fluctuation. Not a quantum field in its ground state. Nothing. No topology, no dimension, no time, no observer. The empty set: $\emptyset$.

$\emptyset$ is not a place. It has no properties to describe. But it is not incoherent. Mathematics begins with the empty set and builds everything from it. Set theory constructs the integers, the reals, topology, and eventually the structures physicists use to describe the universe. The empty set is the seed.

The question is not whether $\emptyset$ is real. You cannot answer that.

The question is whether the transition from $\emptyset$ to structure has a shape — and whether that shape leaves traces in what we observe.

P0.2 — The Fracture

You have seen symmetry break. A glass falls from a table. Before it falls, all directions are equally possible. After it falls, one direction is actual. The glass cannot un-fall.

The first distinction is binary. From $\emptyset$, two values in perfect balance: 1:1. Not numbers yet — just the minimal possible differentiation. A fracture in undifferentiated potential.

One side is absence (0). The other is presence (1). But perfect balance is not structure.

Structure requires the smallest possible perturbation — a deviation from symmetry so slight that it could not be smaller and still exist. Call it ϵ .

The fracture is not 0 and 1 alone. It is **1:1 + 1 $\times\epsilon$ @ AS**.

This is the axiom from which the argument proceeds. It is not a physical event. It is the structural minimum from which everything in the four papers follows.

The @ AS in the axiom names the actualizing structural prior — the now at which the substrate is held and the break is processed. Without AS named in the axiom, the formula describes a balance and a crack with no agent and no time at which any of it could occur. With AS, the axiom names the foundation: the substrate is held, the break persists, the cycle continues.

A note on terminology, to prevent confusion downstream. AS in the axiom — the prior, the now — is structurally distinct from the Actualization State (also abbreviated AS) defined operationally in Paper A as a measurable quantity ranging from 0 to 1. The relationship is precise. AS-the-prior is the dimension along which actualisation happens. AS-the-quantity, defined in Paper A, measures how far along that dimension a system has progressed. The axiom names the prior. Paper A quantifies progress along it. Both abbreviations are retained because the work uses them in their respective contexts; where ambiguity could arise, the text disambiguates explicitly.

The simplest thing that can happen to nothing is that it becomes two things. The simplest thing that can happen to two things in balance is that the balance breaks.

Call the 0-side **orientation**. It is the residue of what was not chosen — the background against which presence is defined.

Call the 1-side **actualisation**. It is the fact of record. The commitment that something, rather than nothing, has occurred.

The perturbation ε is what makes the difference between potential and record. Without it, the two sides are indistinguishable and no structure exists.

A crucial point. The break does not distinguish two pre-existing sides. There are no sides before ε . The break creates the sides by breaking the symmetry that made them indistinguishable. Orientation and actualisation are consequences of the break, not preconditions of it.

A second crucial point. Actualization is not merely a label applied to one side of the fracture. It is a dimension — a degree of freedom as real as any spatial direction that will later emerge.

If the manifold that forms has three spatial dimensions and one temporal dimension, actualisation is the fifth: the dimension of possibility from which records are written into the four. The canvas is not less real than the painting. It is what makes the painting possible.

The 0-side and the 1-side do not sit inside the manifold. The manifold sits inside them. Every record is written from the actualisation dimension into the manifold.

This observation is structural, not formal. It is developed operationally in Paper A, where AS quantifies movement along this dimension, and formally in the dimensional analysis of AP10.

The fracture is silent. No energy is released, because energy has not yet been defined. No observer records it, because recording requires structure that does not yet exist.

The symmetry breaks and there is no sound. This is the silent pop.

What follows — the expansion of structure, the differentiation of forces, the emergence of spacetime — is the Big Bang. The silent pop precedes it: the break that makes the bang possible.

Forward reference. The fracture named here — the silent pop where $1:1$ becomes $1:1 + 1 \times \varepsilon$ — is examined as a statement in AP40 (The Irrational), the first chapter of Notebook I. AP40 shows the equals sign in the governing axiom is identity, not equality: the substrate before the break and the

substrate after the break are the same substrate, held at one prior. The break is exempt from the duality it generates (AP40 section 2 — pre-duality exemption), which is a second independent proof path to the one-I. The narrative here in Paper 0 walks the structure; AP40 examines what kind of statement that structure rests on.

P0.3 — The First Force

The fracture is not passive. It does something. You know this from experience — every time equilibrium breaks, motion follows.

If structure can emerge from potential, the first question is what mediates between them. What is the interaction between actualized structure and the undifferentiated background from which it emerged?

Gravity has a unique property among the known forces.

It is universal. It couples to all energy, not just to specific charges. It is unscreenable. There is no gravitational insulator. And it is always attractive. It draws structure together rather than separating it into types.

These properties make gravity the only known interaction that could plausibly serve as the first mediator between differentiated structure and undifferentiated background.

This is not a derivation. It is a structural observation: if you need one force to emerge first, and that force must couple to everything that exists simply by virtue of its existence, gravity is the only candidate in the known inventory.

Whether the observation is deep or coincidental is precisely the kind of question that cannot be settled by argument. It is settled by what the rest of the work can produce.

P0.4 — Accumulation

You have never undone a moment. Not one.

Once the fracture has occurred and structure begins to actualize, the process has a direction. Records form. Alternatives are excluded. Irreversibility accumulates.

This is the upper half of the hourglass: potential converting into record, the 0-side draining into the 1-side.

The formal version of this process is Actualization State increasing under decohering dynamics — Theorem T1 of Paper A. But the intuition precedes the formalism. The universe, once it has begun to differentiate, does not spontaneously undifferentiate. Records, once formed, do not unform.

The arrow is structural, not thermodynamic. Thermodynamics inherits it.

During accumulation, the available space for new records is vast. Branching is cheap. Alternatives proliferate. The viability kernel — defined in Paper A — is large relative to the occupied state.

Agency, in the control-theoretic sense of Paper C, is near its maximum. There is room to move.

P0.5 — Saturation

Everything fills up. Your hard drive. Your patience. The universe.

Accumulation cannot continue without limit. Every record consumes capacity. Every actualisation forecloses alternatives. The viability kernel shrinks. The no-return surface — defined in Paper A — advances inward.

Saturation is the state in which the capacity for new record-structured branching approaches zero. The system has committed nearly all of its available degrees of freedom. New differentiation requires old structure to be recycled — and recycling requires energy that is itself subject to the same capacity constraints.

Black holes are the extreme expression of saturation. They represent states of maximal gravitational commitment — configurations from which no further internal differentiation is accessible to any external agent.

In the language of Paper A, they are deep within the capture basin: states from which exit is impossible under all admissible controls. They are not reset buttons. They are endpoints of the accumulation process.

P0.6 — The Turn

Here the narrative enters territory you cannot test. Hold it lightly.

The Turn is the most speculative element of this narrative. It is included because the question it addresses — what happens when accumulation is complete? — is unavoidable once you take the argument seriously. It is not included because there is evidence for it.

A companion document, Artist's Proof 04: The Loop Hypothesis, develops this speculation into a formal conjecture with explicit falsification conditions. What follows here is the intuition that preceded that conjecture.

At saturation, two things are true.

First, all capacity has been consumed. No further branching is possible.

Second, the structure that has been built is real. It consists of irreversible records that cannot be undone.

The question is whether there exists any admissible transformation that restores capacity without violating the irreversibility of existing records.

Paper A addresses this in Section A6 as an optional module. The formal conditions are specific: no reversal of realized selections, no bypass of the selection mechanism, and restoration of the effective record-algebra dimensionality. Conformal rescaling — a transformation insensitive to absolute scale — is one candidate satisfying these conditions at extreme dilution.

In general relativity, there exists a structural correspondence between the interior geometry of a collapsing configuration at maximum compression and the geometry of an expanding configuration at its origin. This correspondence is not a temporal sequence but a geometric identity: the two descriptions may refer to the same structure read from different sides.

Whether this identity is physically realized is an empirical question. AP04 addresses it.

The intuitive image is the bottom of the hourglass. Sand has accumulated. The bulb is full. But the bottom of the hourglass is also the top of the next one — not because the glass has been turned, but because the geometry at maximum compression is structurally identical to the geometry at the origin of expansion.

The old records remain as boundary conditions. The capacity renews. The structure continues, with the prior record intact.

Whether this actually happens is not a question this argument can answer. It is flagged here because the structure of the argument makes the question well-posed, and because intellectual honesty requires acknowledging the places where the intuition reaches beyond what the formalism can support.

P0.7 — The Loop

If the structural identity holds, the process is not cyclic in time but identical in geometry: compression $\equiv$ origin. Each side inherits the record structure of the other as a boundary condition. Nothing is erased.

The loop is not a repetition. It is a structure with memory, read differently from each side of the identity.

The most provocative reading of this structure is that a universe is operationally defined by its record structure. The records produced by actualisation constitute the only evidence that anything happened at all. A universe with no records is indistinguishable from $\emptyset$. A universe with records is, precisely and only, those records.

A note on terminology. Terms such as "witness" or "observe," wherever they appear in this narrative, mean record formation only. They do not mean consciousness, inner experience, or subjective awareness. The spine invokes none of those concepts.

This is where the artist's intuition ends and the physicist's discipline begins. The preceding paragraphs are a story — a structural story, constrained by the mathematics that follows, but a story nonetheless.

Stories do not have truth values. They have coherence, and they have consequences.

The consequences of this story are the four papers that follow.

P0.8 — A Conjecture on Energy and Actualization

The following conjecture is retained for historical completeness. It is not a current claim of the argument.

Subsequent work — AP04: The Loop Hypothesis — indicates the conjecture is likely incorrectly formulated: systems at maximum compression represent states of maximal coarse-grained entropy, not minimal energy contribution.

The conjecture is included because it was the original compact expression of the argument's intuition, and because intellectual honesty requires preserving the record of what was thought before it was corrected.

The simplest such relationship would be: $E = mc^2 \times AS$, where $AS \in [0, 1]$ is the Actualization State defined in Paper A.

At $AS = 0$, no record structure exists and the system contributes nothing to the energy budget of actualized reality. At $AS = 1$, the system is maximally actualized and its full mass-energy is committed.

The original intuition was that reality is not given but earned, one irreversible record at a time. That intuition survives. This particular formulation does not.

The conjecture does not appear in, and is not referenced by, any of the four formal papers. The spine is unaffected by its status.

P0.9 — Bridge to the Spine

The preceding sections describe an intuition. The four papers that follow formalize a set of consequences that are consistent with that intuition, but do not depend on it.

No definition, theorem, proposition, or falsifier in Papers A through D requires anything from Paper 0. The spine is self-supporting.

Paper A defines Actualization State as an operational measure of record-structured irreversibility. It proves that AS increases under decohering dynamics, establishes no-return surfaces from bounded capacity, and specifies falsifiable experimental tests. It depends on nothing outside standard quantum mechanics and viability theory.

Paper B characterizes selection — the transition from multiplicity to definiteness — as a costly, rate-limited exclusion process. It derives structural requirements and a falsifiable gravitational rate bound. It depends on Paper A and nothing else.

Paper C develops agency as a control-theoretic quantity: the fraction of the viability kernel reachable from where you currently stand under admissible control. It formalizes drift, fatigue, coupling, and exit as consequences of irreversibility. It depends on Papers A and B and nothing else.

Paper D extends coupling to multi-agent systems operating in shared constraint environments. It derives structural filtering, hierarchy, cooperation, and deterrence as geometric consequences. Every power structure you have ever encountered — every hierarchy, every alliance, every threat — has this geometry of irreversible drift underneath it. It depends on Papers A, B, and C and nothing else.

Each paper is independently falsifiable. You can kill any one of them. Each contains explicit conditions under which it fails. The dependency chain is one-way: failure of D does not invalidate C; failure of C does not invalidate B; failure of B does not invalidate A.

The papers stand or fall on their own logic, independent of the narrative that motivated them.

In the symbolic notation that motivates the formal development:

A record is the actualisation state of an irreversible symmetry-breaking event applied to the vacuum — where $\emptyset_0$ is the undifferentiated potential of P0.1 and Crack is the symmetry-breaking fracture of P0.2. This notation is evocative, not formal. Paper A defines all quantities operationally.

End of Paper 0. Non-Falsifiable · Structural Narrative · Complete.

Paper A

Actualization State: An Operational Measure of Record-Structured Irreversibility

Reference Document · Canonical

A0 — Front Matter

A0.1 — Title Block and Abstract

Title. Actualization State (AS): An Operational Measure of Record-Structured Irreversibility.

You are reading this sentence. That is a record. Photons hit your retina, neurons fired, a pattern was recognised. The event cannot unhappen.

This paper builds a tool to measure how far that process has gone — and proves that, under the right conditions, it can only go in one direction.

Abstract.

Actualization State (AS) is an operational measure of irreversible record formation in quantum systems. It is defined relative to physically realisable coarse-grainings induced by system–environment interactions, and it quantifies the degree to which mutually exclusive classical alternatives have become durably encoded.

Irreversibility here is not about entropy. It is about reachability — the boundary beyond which you cannot get back, no matter what you do.

The paper establishes criteria under which AS is well defined, operationally invariant, and falsifiable. The proof shows that AS is monotonic under decohering, record-forming dynamics within a precisely delimited scope. A domain-neutral no-return theorem then shows that bounded maintenance capacity generically induces irreversible loss of reachability, independent of quantum mechanics or gravity.

Together these results provide a falsifiable, interpretation-agnostic framework that isolates irreversible record formation as a measurable physical process — independent of collapse, gravity, or consciousness. You need no interpretation of quantum mechanics to use this tool. You need only the measurements.

No collapse mechanism, gravitational hypothesis, or cosmological assumption is invoked. The argument isolates the definitional and theorem layers required for any subsequent theory of selection or definiteness.

A0.2 — What This Paper Does and Does Not Do

It does:

- Define AS as a physically meaningful measure of irreversible record formation.
- Prove AS increases under decohering dynamics (Theorem T1).
- Establish no-return surfaces from bounded capacity (Theorem T2).
- Require operational invariance — and die if that requirement fails (Kill Switch F0).

It does not:

- Propose a collapse mechanism.
- Derive the Born rule.
- Appeal to gravity or cosmology.
- Solve the measurement problem.
- Explain consciousness.

Sections A0–A3 are self-contained. Sections A4–A5 add independently falsifiable postulates. If A4–A5 fail, A0–A3 are untouched.

A1 — Problem Statement

A1.1 — The Actualization Problem

You have never experienced a superposition. Every moment of your life has been definite — this room, this chair, this sentence.

Yet quantum mechanics says that before measurement, systems exist in superpositions of all possible outcomes. Something bridges the gap between *all possible* and *one actual*. That bridge is the subject of this paper.

Quantum theory describes closed systems by unitary evolution in Hilbert space. Experiments report records: mutually exclusive, persistent, classical facts. Between these descriptions lies a structural gap.

Standard measurement language tries to bridge the gap with observers, projections, or epistemic updates. None of these specifies a physical transition. They describe when an agent updates a description, not when a system becomes unable to support alternatives.

Decoherence explains the suppression of interference. By itself it does not quantify how much irreversible structure has formed, nor does it specify when alternative histories cease to be operationally recoverable.

What is missing is a quantity that refers only to physically accessible degrees of freedom, distinguishes loss of coherence from mere ignorance, and measures the accumulation of durable record structure prior to any claim about outcome definiteness.

That quantity is Actualization State (AS).

The argument is intentionally minimal. It does not ask why the universe permits records. It asks when records become irreversible. It does not explain the *feltness* of outcomes. It explains the structural conditions under which multiple outcomes cease to be simultaneously accessible.

By isolating the transition from quantum coherence to classical record, the paper provides a shared phenomenological target for any deeper theory of definite outcomes.

A1.2 — What Is New: Positioning Relative to Existing Notions

Actualization State is not a redefinition of decoherence, entropy, or thermodynamic irreversibility. The distinctions are structural, and they matter.

AS vs. Decoherence. Decoherence is a dynamical process that suppresses interference between alternatives. AS is an operational quantity that measures the extent of record-structured commitment resulting from such decoherence.

The two are distinct. Decoherence can occur without significant growth of AS, and AS can increase even when total entropy change is negligible. A concrete demonstration of their independence is given in the worked comparison below.

AS vs. Entropy. Entropy quantifies total uncertainty or mixedness, including contributions from unobserved degrees of freedom. AS deliberately discards such contributions and tracks only inter-sector branching relative to the physically realizable record algebra.

A system may have high entropy and low AS, or low entropy and high AS. The von Neumann entropy comparison below makes this independence explicit.

AS vs. Quantum Darwinism. Quantum Darwinism (Zurek) quantifies the redundancy with which information is imprinted in environmental fragments. AS measures the informational richness of committed classical branching, not the number of copies of that information.

The two quantities are operationally independent. Each can be maximized or minimized independently of the other. The worked comparison below demonstrates exactly this.

AS vs. Consistent Histories. The history-based representation AS_h (Section A2.4) is restricted to single-time record histories under complete decoherence. This is a deliberate narrowing relative to the full consistent-histories framework.

The full framework permits multi-time, multi-branching history sets. AS_h does not. AS_h serves as an equivalence bridge to the primary definition, not as a replacement for consistent histories.

Worked Comparison: Where AS and Quantum Darwinism Redundancy Diverge

The preceding distinctions are structural. Their force is best seen in a concrete system where AS and Quantum Darwinism redundancy move independently. The two cases below share identical quantum dynamics; they differ only in the number of environmental fragments and the number of pointer sectors.

Case 1: High redundancy, zero AS.

A qubit S with pointer basis $\mathcal{O} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$ is prepared in the pure pointer state $|0\rangle$. The environment consists of $N = 1000$ fragments, each independently recording that the system is in sector $|0\rangle$.

The Quantum Darwinism redundancy is $R\delta \approx 1000$. The classical information *the system is in $|0\rangle$* is broadcast across a thousand environmental fragments. Any small fraction of the environment suffices to determine the system state.

The sector weights, however, are $p_0 = 1, p_1 = 0$. The Shannon entropy $H(\{p_i\}) = 0$ and therefore $AS = 0$. No branching exists.

The environment has recorded a single, definite outcome with extreme redundancy. There is no record-structured irreversibility to measure.

Case 2: Zero redundancy, maximal AS.

A four-level system S with pointer basis $\mathcal{O} = \{\Pi_1, \Pi_2, \Pi_3, \Pi_4\}$ is prepared in an equal superposition and then fully dephased by coupling to a single environmental fragment E .

The sector weights are $p_i = 1/4$ for all i . Quantum Darwinism redundancy is $R_\square = 1$: only one fragment carries the classical information, and loss of that fragment destroys access.

But $AS = H(\{1/4, 1/4, 1/4, 1/4\}) / \log 4 = \log 4 / \log 4 = 1$. Maximal record-structured branching exists across four mutually exclusive alternatives.

In Case 1, the system is classically definite and robustly broadcast — but it has no actualisation structure. In Case 2, the system is maximally branched but fragile in the Darwinist sense.

AS and redundancy are not merely different in definition. They are operationally independent quantities that can be maximized or minimized independently of each other.

AS vs. von Neumann entropy. A similar divergence arises with respect to the von Neumann entropy $S(\rho)$.

Consider a single pointer sector Π_i of rank $d_i = 100$, with the system state confined entirely to that sector in a maximally mixed intra-sector state. The von Neumann entropy is $S(\rho) = \log 100$ — large. But $AS = H(\{1\}) / \log 1 = 0$, since all weight resides in one sector.

Conversely, a two-sector system with rank-1 projectors and equal weights $p_0 = p_1 = 1/2$ has $S(\rho) = \log 2$ and $AS = 1$.

Von Neumann entropy tracks total uncertainty, including intra-sector degeneracy. AS tracks only inter-sector branching. They answer different questions.

Operator Horizon vs. the Second Law.

The Second Law expresses the typical growth of entropy under macroscopic dynamics. The Operator Horizon introduced in Section A3 instead defines irreversibility as a boundary of operational accessibility — a geometric limit in state space beyond which recovery is impossible given bounded control capacity.

Irreversibility here is not a statement about likelihood or typicality. It is a statement about reachability under admissible operations.

The notion applies equally to quantum, classical, and abstract control systems, and it is independent of thermodynamic assumptions.

A1.3 — Scope Clarification

The paper does not propose a mechanism of collapse, derive the Born rule, or assume any cosmological or gravitational hypothesis.

It isolates the minimal definitional and theorem-level structure required to make irreversible record formation a well-defined, operationally testable concept.

Any subsequent theory of outcome selection or definiteness must be built on this foundation — or fail against it.

A2 — Definitions

What follows are the tools. Each definition names a specific thing and says exactly what it does.

If you lose track, come back here. The definitions do not move.

A2.1 — D1: Physically Realizable Coarse-Graining $\mathcal{O}$

Let $\mathcal{H}_s$ be the system Hilbert space and $\mathcal{H}_e$ its environment.

Definition D1. A physically realizable coarse-graining $\mathcal{O}$ is a finite set of mutually orthogonal projectors $\mathcal{O} = \{\Pi_i\}$ satisfying all of the following:

- The projectors are orthogonal and complete on the support of the reduced state.
- The set is selected by the physics of system-environment coupling, not by observer choice.
- The associated dephasing map is approximately stable under accessible Hamiltonian evolution within tolerance ε .

The critical point: $\mathcal{O}$ is not your choice. It is nature's choice. The physics of the interaction determines what gets measured. You do not pick the basis. The coupling picks it for you.

Computability remark. In practice, the physically realizable coarse-graining is identified as the stable algebra generated by the interaction Hamiltonian's pointer observables — for instance, via the predictability sieve (Zurek, 1993) or stability analysis under the system-environment coupling.

Definition D5 (Operational Invariance) then tests robustness across any co-admissible candidates that survive this selection.

The identification of the pointer algebra for a given Hamiltonian is a research problem, not a closed algorithm. D5 converts this openness into a falsifiable condition rather than leaving it as an ambiguity.

A2.2 — D2: Dephasing Map $\Delta\mathcal{O}$

Definition D2. Given a density matrix ρ on $\mathcal{H}_s$, the dephasing map relative to $\mathcal{O}$ is

$$\Delta\mathcal{O}(\rho) \equiv \sum_i \Pi_i \rho \Pi_i.$$

$\Delta\mathcal{O}$ removes quantum interference between record sectors while preserving classical probabilities.

Critical clarification. Pay attention to this. It is where most confusion enters.

$\Delta\mathcal{O}$ does not measure ignorance. It enforces projection onto the record algebra, isolating entropy attributable to irreversible branching rather than lack of knowledge.

This step prevents conflating classical uncertainty with physical actualisation.

A2.3 — D3: Actualization State — Primary Definition

Statement of the Definition.

Let ρ be the reduced density matrix of a system after tracing over inaccessible degrees of freedom. Let $\mathcal{O} = \{\Pi_i\}$ be a physically realizable coarse-graining selected by system–environment interaction (Definition D1).

Define the dephasing map relative to this record algebra: $\Delta\mathcal{O}(\rho) \equiv \sum_i \Pi_i \rho \Pi_i$.

The Actualization State is defined as

$$\mathbf{AS}(\rho; \mathcal{O}) \equiv \mathbf{S}_{\text{eff}}(\rho; \mathcal{O}) / \log d\mathcal{O}$$

where $\mathbf{S}_{\text{eff}}$ is the effective record entropy defined below and $d\mathcal{O}$ is the total record-algebra dimension.

Notation note (work-wide reading). The abbreviation AS is used in two related but distinct senses across The 420 Code, and you are asked to keep them apart.

In the foundational axiom 1:1 + 1 $\times$ ε @ AS (P0.2), AS names the actualizing structural prior — the now at which actualisation happens, the dimension along which records are written.

In Paper A, AS denotes the Actualization State as defined here: the operational measure of how far along that dimension a system has progressed, normalized to the interval [0, 1].

The two readings are not in conflict. They refer to different aspects of the same structure. AS-the-prior is the dimension. AS-the-quantity, defined here, measures progress along it. Where context makes the distinction load-bearing, the text disambiguates explicitly.

Effective Entropy.

When record sectors Π_i have rank greater than one, the dephased entropy decomposes as

$$\mathbf{S}(\Delta\mathcal{O}(\rho)) = H(\{p_i\}) + \sum_i p_i \mathbf{S}(\sigma_i),$$

where $p_i = \text{Tr}(\Pi_i \rho)$ and σ_i is the normalised intra-sector state.

AS tracks inter-sector branching only. What happens inside each sector is invisible to AS — deliberately so.

The effective entropy entering AS is defined as

$$\mathbf{S_eff}(\rho; \mathcal{O}) \equiv H(\{p_i\}),$$

with intra-sector entropy discarded by construction. For rank-1 sectors, $S(\Delta\mathcal{O}(\rho)) = H(\{p_i\})$ and no distinction arises. A formal derivation is given in Appendix A.

Rank-1 Simplification.

For rank-1 sectors (pure pointer states), the definition simplifies to

$$AS(\rho; \mathcal{O}) = H(\{p_i\}) / \log N,$$

where H is the Shannon entropy and $N = |\mathcal{O}|$ is the number of record sectors.

Normalization and Physical Bounds.

Let $\mathcal{O} = \{\Pi_i\}$ be the physically realizable record algebra, with total record-algebra dimension $d\mathcal{O} \equiv \sum_i \text{rank}(\Pi_i)$.

The maximum effective entropy achievable for a state confined to $\mathcal{O}$ is $S_eff^{\max} = \log d\mathcal{O}$ (uniform distribution over the record algebra).

Therefore $AS(\rho; \mathcal{O}) \in [0, 1]$ for all admissible ρ .

Interpretation: The AS Inversion.

You should expect, from background, that "more actualized" means "more definite." It does not. The convention is the opposite, and the reason is structural.

$AS = 0$ when one record sector carries all the weight. The system is completely definite with respect to $\mathcal{O}$.

$AS = 1$ when the weight is distributed maximally across all record sectors. The system is maximally branched.

The reason for this convention: AS measures *capacity for branching that has been committed*. Definiteness is the absorption of that capacity into a single sector. Branching is the active state in which all sectors remain operationally distinguishable.

You do not lose information when $AS = 1$. You hold maximum record-structured multiplicity. Selection — Paper B's territory — is what reduces AS by collapsing branching into definiteness.

Why Dephasing-Relative Entropy (and Not Purity Alone).

Purity, $\text{Tr}(\rho^2)$, measures total mixedness. It cannot distinguish a system mixed across record sectors (genuinely branched) from a system mixed within a single sector (intra-sector ignorance).

The dephasing map removes the intra-sector contribution by construction. The Shannon entropy of the resulting sector weights is what counts.

If you used purity, you would be measuring something the framework explicitly excludes.

Normalization Discipline.

The denominator $\log d\mathcal{O}$ is fixed by the record algebra, not by ρ . The normalization is independent of the state. AS values for different states are directly comparable when computed against the same $\mathcal{O}$.

When $\mathcal{O}$ changes, you change the denominator. The kill switch F0 (D5) is what enforces consistency across choices of $\mathcal{O}$.

Operational Meaning.

AS quantifies the informational richness of the committed record structure. High AS means many distinguishable alternatives have been written. Low AS means few have. The branching is what the environment has actually committed to recording.

Read this against your experience. Before a coin lands, the world is in superposition over outcomes — but the record algebra has not yet resolved. After it lands, AS is 0 with respect to the head/tail algebra: definiteness is achieved. During the moment of falling, when both possibilities are still operationally live in the environment, AS is high.

Scope and Kill-Switch.

AS is well defined whenever $\mathcal{O}$ is identified by D1. Where $\mathcal{O}$ is ambiguous — where two co-admissible coarse-grainings exist and yield different AS values beyond tolerance — Kill Switch F0 (D5) fires. The framework dies cleanly.

This is intentional. AS is operationally meaningful only when the physics has selected the record algebra. Where the physics is silent, the framework refuses to invent an answer.

Worked Example: Two-Qubit Dephasing.

Two qubits S_1, S_2 . Pointer basis $\mathcal{O} = \{|00\rangle\langle 00|, |01\rangle\langle 01|, |10\rangle\langle 10|, |11\rangle\langle 11|\}$. Initial state $|\psi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ (Bell state). Reduced state on the joint system: $\rho = |\psi\rangle\langle\psi|$.

After dephasing in $\mathcal{O}$: $\Delta\mathcal{O}(\rho) = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$. Sector weights: $p_{00} = \frac{1}{2}, p_{01} = 0, p_{10} = 0, p_{11} = \frac{1}{2}$.

$H(\{p_i\}) = \log 2$. Normalization: $\log d\mathcal{O} = \log 4 = 2 \log 2$.

Therefore $AS = \log 2 / (2 \log 2) = \frac{1}{2}$.

Two of the four record sectors carry weight; the AS reading reflects half-occupied branching.

A2.4 — D4: History-Based Representation of AS (AS_h)

Definition D4. Let D be a set of consistent histories indexed by α , with weights $w(\alpha)$ and decoherence functional $D(\alpha, \beta)$. Define the history-based AS as

$$AS_h(D) \equiv H(\{w(\alpha)\}) / \log |D|$$

where H is the Shannon entropy and $|D|$ is the cardinality of the history set.

Conventions on Approximation and Operational Criteria.

The dephasing condition $D(\alpha, \beta) \approx 0$ for $\alpha \neq \beta$ is taken to hold within operational tolerance $\varepsilon > 0$ — the same ε used throughout the paper for finite-resolution measurements. Effective orthogonality, inaccessibility, and recoverability are all defined relative to ε .

Where the histories correspond one-to-one with the record sectors of D1 at a single time slice and decoherence is complete within ε , $AS(\rho; \mathcal{O})$ and $AS_h(D)$ coincide. The equivalence is proved in Appendix A.

AS_h is provided as a bridge for readers fluent in the consistent-histories framework. The primary definition is $AS(\rho; \mathcal{O})$.

A2.5 — D5: Operational Invariance Test (Kill Switch F0)

Setup.

Let $\mathcal{O}_1$ and $\mathcal{O}_2$ be two physically realizable coarse-grainings selected by the same system-environment coupling. Both must satisfy D1.

Definition D5 (Operational Invariance Requirement).

For any system ρ admitting more than one physically realizable coarse-graining $\{\mathcal{O}_k\}$,

$$|AS(\rho; \mathcal{O}_k) - AS(\rho; \mathcal{O}_{k'})| \leq \delta_{\text{exp}}$$

for all admissible pairs $(\mathcal{O}_k, \mathcal{O}_{k'})$, where δ_{exp} is the experimental tolerance set by the precision of the apparatus.

Falsifier F0 (Global Kill Switch).

If, in any laboratory implementation, two co-admissible coarse-grainings $\mathcal{O}_k, \mathcal{O}_{k'}$ yield AS values differing persistently beyond δ_{exp} , then AS as defined here is not an operationally invariant quantity. The entire framework of Paper A fails.

This is the global kill switch. F0 is non-negotiable. AS is meaningful only if the value is the same regardless of which valid coarse-graining you use to compute it.

Notes on Scope.

F0 does not require that *every* mathematically possible coarse-graining yield the same AS. It requires that all *co-admissible* coarse-grainings — those

satisfying D1 for the same physical coupling — agree within experimental tolerance.

If two coarse-grainings disagree because one fails D1 (it is not physically selected by the coupling), F0 has not fired. F0 has fired only when two valid candidates persistently disagree beyond δ_{exp} .

Worked Example: Operational Invariance in Circuit QED.

A flux-tunable transmon coupled to a microwave cavity. The coupling Hamiltonian selects two co-admissible pointer bases at different operating points: charge-basis $\{|0\rangle, |1\rangle\}$ at one tuning, and flux-basis $\{|+\rangle, |-\rangle\}$ at another.

Compute AS in both bases for a calibration state. If the values agree within the experimental noise floor, F0 has not fired and the framework survives this test. If they persistently disagree by more than the calibration uncertainty, F0 fires.

R0 in Section A5.2 is the explicit test programme. It is the first test to run, because if F0 fires, no further test is meaningful.

A3 — Theorems: Irreversibility and No-Return

The definitions are set. Now the proofs.

What follows cannot be argued with. It can only be tested.

A3.1 — Theorem T1: Monotonicity of AS Under Decohering Dynamics

Setup.

Let $\rho(t)$ be the reduced state of a system evolving under a completely positive trace-preserving (CPTP) dynamical semigroup $\{\mathcal{E}_t\}_{t \geq 0}$ with generator $\mathcal{L}$. Let $\mathcal{O} = \{\Pi_i\}$ be a physically realizable record algebra (Definition D1), and let $\Delta\mathcal{O}(\rho)$ denote the associated dephasing map.

Theorem T1 (Statement).

Here is the central result of Paper A. Everything before it was preparation. Everything after it is consequence.

The result says: under three precisely stated conditions, the branching can only grow. The coin cannot un-land. The ink cannot un-dry. The record cannot un-write.

Possibility becomes fact, and the transition is one-way. Not because physics prohibits reversal — but because the conditions that create records are the conditions that make the inequality hold.

Actualization State is monotonic nondecreasing along the evolution $\rho(t)$ provided the following minimal sufficient conditions hold:

(1) Decoherence relative to the record algebra. Interference between record sectors is not regenerated:

$$(d/dt) C\mathcal{O}(\rho(t)) \leq 0,$$

where $C\mathcal{O}$ is any coherence monotone that vanishes on $\Delta\mathcal{O}(\rho)$. Equivalently, off-diagonal terms in the $\mathcal{O}$ -basis decay monotonically.

(2) Closure of the record algebra.

$$\Delta\mathcal{O} \circ \mathcal{E}_t = \mathcal{E}_t \circ \Delta\mathcal{O} \text{ for all } t \geq 0.$$

This ensures that populations in the record sectors evolve autonomously once decohered.

(3) Unital (mixing) dynamics on the record algebra. The sector weights $p_i(t) = \text{Tr}(\Pi_i \rho(t))$ evolve under a doubly stochastic map:

$$p(t) = M(t) p(0),$$

where M preserves the uniform distribution.

In plain language: the mixing is fair. No sector is favoured. The dynamics spread probability out, not funnel it in.

Conclusion.

Under conditions (1)–(3),

$$(d/dt) AS(\rho(t); \mathcal{O}) \geq 0.$$

That is, Actualization State is monotonic along decohering, record-forming dynamics.

Scope of the theorem.

T1 is a conditional statement. Its domain of applicability is exactly the set of dynamics satisfying conditions (1)–(3).

Dynamics outside this domain — including dissipative, non-unital, or feedback-controlled evolution — may cause AS to decrease. This does not contradict the theorem. It indicates the dynamics are not purely record-forming in the sense defined above.

The explicit scope boundary and converse are given below.

Note on novelty and scope.

Conditions (1)–(3) are sufficient but not necessary. The monotonicity of Shannon entropy under doubly stochastic mixing is a standard result (Schur convexity), and this paper does not claim otherwise.

What is new is not the mathematical inequality. It is the physical application:

- The identification of conditions under which doubly stochastic mixing is the correct effective description of decoherence in a physically realized record algebra.
- The isolation of inter-sector branching entropy (via the dephasing map $\Delta\mathcal{O}$) from thermodynamic entropy.
- The explicit rate bound (T1a) and converse (T1b) that turn the monotonicity into a diagnostic tool for identifying record-forming dynamics.

The theorem is a known inequality applied in a new physical context. The contribution is the context, not the inequality.

You are seeing a known mathematical tool applied to a new physical question — and the answer it gives is decisive.

Lemma T1.1 (Doubly stochastic mixing).

Under conditions (2) and (3), the sector weights $p(t)$ evolve under a doubly stochastic matrix $M(t)$ for all $t \geq 0$.

Proof. By condition (2), $\Delta\mathcal{G} \circ \mathcal{E}_t = \mathcal{E}_t \circ \Delta\mathcal{G}$, so the evolution commutes with dephasing. Therefore the diagonal elements evolve autonomously: there exists a linear map $M(t)$ such that $p(t) = M(t) p(0)$.

The map is stochastic because $\mathcal{E}_t$ is trace-preserving: $\sum_i p_i(t) = 1$ for all t .

Unitality (condition 3) means $\mathcal{E}_t(I/d) = I/d$. Applying $\Delta\mathcal{G}$ to both sides and using condition (2):

$$\Delta\mathcal{G}(\mathcal{E}_t(I/d)) = \mathcal{E}_t(\Delta\mathcal{G}(I/d)) = \mathcal{E}_t(I/d) = I/d.$$

In terms of sector weights, the uniform distribution is fixed: $M(t)(1/N, \dots, 1/N)^T = (1/N, \dots, 1/N)^T$.

A stochastic matrix that preserves the uniform distribution is doubly stochastic.

The lemma is short. Its consequence is not.

Once you know the mixing is doubly stochastic, Shannon's inequality does the rest. With Lemma T1.1 established, the monotonicity of Shannon entropy under doubly stochastic mixing is a standard result (Schur-convexity of $-H$). The theorem follows.

The proof is standard mathematics applied to a new physical context. What is new is the recognition that record-forming dynamics satisfy exactly the conditions that make the inequality hold.

Interpretation.

When interference between record-distinguishable alternatives is suppressed, the record algebra is dynamically closed, and record-sector probabilities mix without coherent backflow, the informational richness of committed classical branching cannot decrease.

This monotonic increase defines the arrow of actualisation.

You have lived inside this arrow your entire life. Every moment has been forward. Every record has been permanent. The theorem says why: under the conditions that create records, the branching can only grow.

Explicit Scope Boundary.

Outside these three conditions, the guarantee is void. AS may decrease under cooling, decay, relaxation, feedback control, or any dynamics that funnel probability into fewer sectors.

The theorem does not claim universality. It claims precision. These cases do not contradict T1; they lie outside its scope by construction.

Why the Scope Is Exact.

Pay attention to this. It is the difference between a real theorem and a hand-wave.

AS measures branching, not definiteness. Decoherence creates branching — AS increases. Selection resolves branching — AS decreases. T1 applies only to the creation phase.

The theorem is not claiming that AS always increases everywhere forever. It is claiming something much more precise: under exactly these three conditions, AS cannot decrease. Violate the conditions and the guarantee is void.

The conditions are not technicalities. They are the physics. And the physics is testable.

Quantitative Sharpening: Rate Bounds and Converse.

T1 establishes monotonicity but does not quantify the rate at which AS approaches its equilibrium value, nor does it characterize conditions under which AS must decrease. The two results below sharpen T1 in both directions.

Corollary T1a (Rate of convergence).

Under conditions (1)–(3), let the induced classical dynamics on the sector weights be governed by a continuous-time doubly stochastic rate matrix W , so that $dp/dt = Wp$. Let $\lambda_2 < 0$ denote the second-largest eigenvalue of W (the spectral gap). Then the deviation of AS from its equilibrium value $AS_{eq} = 1$ satisfies

$$|AS(t) - 1| \leq C \cdot \exp(\lambda_2 t),$$

where C depends on the initial condition.

Derivation. For any initial distribution $p(0)$, the deviation from uniformity satisfies $\|p(t) - \pi\|_1 \leq \sqrt{N} \cdot \exp(\lambda_2 t)$ (by standard contraction bounds for reversible Markov chains).

The Shannon entropy $H(p)$ is Lipschitz in the L_1 norm on the probability simplex: $|H(p) - H(q)| \leq \|p - q\|_1 \cdot \log N$ (continuity bound for entropy; Cover and Thomas 2006).

Therefore $|AS(t) - 1| = |H(p(t)) - \log N| / \log N \leq \|p(t) - \pi\|_1 \leq \sqrt{N} \cdot \exp(\lambda_2 t)$, where the constant $C = \sqrt{N}$ absorbs the dimension dependence.

The rate of convergence to maximal branching is controlled by the spectral gap of the mixing dynamics, not by any property intrinsic to the AS definition.

For the symmetric d -sector model with uniform inter-sector mixing rate w , the spectral gap is $\lambda_2 = -dw$, and the convergence time to $AS \approx 1$ scales as $\tau \sim 1/(dw)$.

This connects AS growth directly to the physical decoherence rate of the record algebra.

The faster the environment records the system, the faster AS climbs. You can measure this. The spectral gap is a physical quantity. The convergence rate is a prediction.

Proposition T1b (Converse: conditions for AS decrease).

If condition (3) is violated — specifically, if the induced classical dynamics on sector weights are governed by a stochastic matrix M that is not doubly stochastic, with stationary distribution $\pi \neq \text{uniform}$ — then there exist initial distributions $p(0)$ for which AS is strictly decreasing.

In particular, if $p(0)$ majorizes π , the Shannon entropy $H(p(t))$ may increase, but if π majorizes $p(0)$, then $H(p(t))$ can decrease toward $H(\pi) < \log d$.

Physically, T1b corresponds to dissipative dynamics that preferentially funnel population into a subset of record sectors — for example, amplitude damping or spontaneous decay into a ground-state sector. Such dynamics violate the unital condition (3) and drive AS downward.

The converse confirms that condition (3) is not merely a technical convenience. It is a physical requirement: monotonicity of AS is a signature of symmetric, environment-driven mixing, not of dissipative relaxation.

When AS decreases, something is funnelling probability into fewer branches — cooling, decay, relaxation. When AS increases, the environment is writing records. The direction tells you which process is dominant.

Summary.

T1, T1a, and T1b together establish that AS monotonicity is the exact fingerprint of doubly stochastic record-forming dynamics. T1 gives the direction. T1a gives the rate. T1b gives the converse. No further characterization of the monotonicity is needed or claimed.

A3.2 — The Operator Horizon: No-Return as an Inequality Setup.

Let $x(t) \geq 0$ be a scalar variable representing the degree of maintained structure of a system — displacement from its unmaintained equilibrium baseline.

Assume deterministic dynamics:

$$dx/dt = -a x + u, u(t) \in [0, u_{\max}],$$

where $a > 0$ is an intrinsic decay/drift rate toward equilibrium, u is a control/maintenance input, and $u_{\max} \geq 0$ is a hard upper bound on control capacity.

Theorem T2 (Operator Horizon).

You have felt this theorem in your body. Every system with limited resources — every body, every business, every civilisation — has a point beyond which no strategy can save it.

The theorem names that point.

Define the operator horizon

$$x_h \equiv u_{\max} / a.$$

If at some time t_0 the system satisfies $x(t_0) > x_h$, then for all admissible controls $u(t)$,

$$dx/dt < 0 \text{ whenever } x > x_h,$$

and $x(t)$ decreases monotonically toward x_h .

Once you cross the horizon, $x(t)$ decreases no matter what you do. Maximum effort slows the decline but cannot reverse it. Recovery is impossible by control alone.

Proof.

From the dynamics, $dx/dt = -a x + u \leq -a x + u_{\max}$.

If $x > u_{\max}/a$, then $-a x + u_{\max} < 0$, hence $dx/dt < 0$. At $x = x_h$, maximal control yields $dx/dt = 0$; continuity implies monotone approach to x_h from above.

Interpretation.

x_h is a capacity boundary, not a physical wall. It is the maximum sustainable structure given maximum maintenance effort.

Beyond x_h , the system decays toward the horizon regardless of strategy. Irreversibility arises from insufficiency of admissible control, not from prohibition of reverse dynamics.

Generalizations.

Nonlinear decay. If $dx/dt = -g(x) + u$ with $g(0) = 0$ and $g(x)$ increasing, the horizon is implicitly defined by $g(x_h) = u_{\max}$.

Your body operates under nonlinear decay — the maintenance cost of health increases with age, and the horizon shifts. The qualitative no-return result is unchanged.

Time-dependent capacity. If $a(t)$ or $u_{\max}(t)$ varies, $x_h(t) = u_{\max}(t)/a(t)$ defines a time-dependent horizon. Exceeding the instantaneous horizon for sufficient time yields practical no-return behavior.

You know this. You have watched a garden become overgrown past the point where you could maintain it. You have watched debt compound past the point where income could service it. You have watched a body deteriorate past the point where medicine could restore it.

The mathematics is confirming what your experience already knows. There is a line, and once you cross it, effort is not enough.

Classical analogy. Consider a leaking bucket with a bounded refill rate. The horizon is the maximum fill level sustainable at maximum pumping. Once above it, the bucket drains regardless of effort.

A3.3 — No-Return Surfaces and Operational Irreversibility

Setup.

The scalar horizon is the simple case. Real systems have many dimensions. The generalisation uses viability theory (Aubin, 1991) — the mathematics of survival under constraint.

Definition D6: State Space and Admissible Dynamics. Let $X \subseteq \mathbb{R}^n$ be the state space. Let admissible controls satisfy $u(t) \in U$, where U is compact. Dynamics: $dx/dt = f(x, u)$, with f locally Lipschitz in x . Let $R \subset X$ be the recoverable (safe) set.

Definition D7: Viability Kernel.

$$\text{Viab}(R) \equiv \{ x_0 \in R \mid \exists u(\cdot) \in U \text{ such that } x(t; x_0, u) \in R \text{ for all } t \geq 0 \}.$$

States from which the system can be kept inside R indefinitely using admissible control.

Definition D8: Capture Basin.

$$\text{Cap}(R) \equiv \{ x_0 \in X \mid \forall u(\cdot) \in U, \exists t \geq 0 \text{ such that } x(t; x_0, u) \notin R \}.$$

States from which exit from R is inevitable under all admissible controls.

Definition D9: No-Return Surface.

$$\Sigma_h \equiv \partial \text{Viab}(R).$$

This is the geometric generalization of the scalar horizon x_h .

Proposition P3.3.

Assume the standard viability conditions (local Lipschitz continuity of f , compact U). Then:

(1) $\text{Viab}(R)$ consists of states from which at least one admissible control avoids loss indefinitely. $\text{Cap}(R)$ consists of states from which all admissible controls lead to loss in finite time.

(2) Crossing Σ_h transfers the system from a region where recovery is reachable to a region where it is not.

Caveat. $\text{Viab}(R)$ and $\text{Cap}(R)$ partition X up to the boundary Σ_h . Boundary states may be marginal.

Definition D10: Operational Irreversibility (Quantified).

A state x_0 is operationally irreversible with respect to R iff $x_0 \notin \text{Viab}(R)$. The reverse transition to R does not exist under admissible controls.

Operational irreversibility depends on reachability under constraints, not on microscopic time-reversal symmetry. The distinction matters.

A shattered vase is not irreversible because physics forbids reassembly. It is irreversible because you do not have the resources, precision, or time to reassemble it. Irreversibility is about what you can do. Not about what nature prohibits.

Consistency with A3.2.

A3.2 is recovered as the special case $n = 1$, $f(x, u) = -ax + u$, $R = [0, x_h]$. Then $\text{Viab}(R) = [0, x_h]$ and $\Sigma_h = \{x_h\}$. The scalar horizon is exactly the one-dimensional no-return surface.

You now have the complete no-return geometry. The scalar horizon (T2) is the simple case. The viability kernel (D7) is the general case. The no-return surface (D9) is the boundary between where you can still recover and where you cannot.

Every system you have ever cared about — your body, your relationships, your work — has this geometry. The mathematics names what your experience already knows.

Robustness Notes.

With stochastic forcing, replace viability by almost-sure or probabilistic viability; Σ_h becomes a probabilistic boundary. The concept is unchanged; only the quantifier changes. $\text{Viab}(R)$ and Σ_h are computable via Hamilton-Jacobi reachability methods.

A4 — Quantum Mechanical Instantiation

Sections A0–A3 are complete. They stand alone. What follows adds independently falsifiable postulates — each one an invitation to destroy a specific claim.

If any postulate in this section falls, everything above it survives. You lose the extension, not the foundation.

Sections A0–A3 establish definitions and theorems that are self-contained: they depend only on operational definitions, standard quantum mechanics, and viability theory. Nothing in A0–A3 requires the content of A4 or A5.

This section introduces postulates that extend the argument to address selection and gravitational rate constraints. These postulates are independently falsifiable. Each has explicit conditions under which it fails (F1–F3, G1–G3). Failure of any postulate here does not invalidate the definitions, theorems, or irreversibility results of A0–A3.

The transition from theorem to postulate is marked explicitly at each point of introduction.

A4.1 — Operational Irreversibility in Open Quantum Systems

The viability geometry of Section A3 now meets quantum mechanics. The abstract becomes concrete. You will recognise the structure.

Setup: Accessible Dynamics.

You have access to the system but not the environment. That restriction is the source of irreversibility.

Let the total Hilbert space factor as $\mathcal{H} = \mathcal{H}_s \otimes \mathcal{H}_e$, with joint state evolving unitarily under H_{se} .

The accessible system state — the one you can actually measure — is

$$\rho_s(t) = \text{Tr}_e[U(t) \rho_{se}(0) U^\dagger(t)].$$

Admissible operations are restricted to system-local CPTP maps — the operations you can actually perform on the system without accessing the environment:

$$\Lambda: \mathcal{B}(\mathcal{H}_s) \rightarrow \mathcal{B}(\mathcal{H}_s),$$

optionally employing fresh ancilla, but with no access to the original environment E .

Definition D11: Coherence-Recoverable States.

A system state ρ_s is coherence-recoverable relative to $\mathcal{O}$ iff there exists an admissible CPTP map Λ such that

$$\|\Lambda(\rho_s) - \rho_{\text{coh}}\|_1 \leq \varepsilon,$$

for some state ρ_{coh} satisfying $\Delta\mathcal{O}(\rho_{\text{coh}}) \neq \rho_{\text{coh}}$, and for fixed operational tolerance $\varepsilon > 0$.

Definition D12: Recoverable Set and Viability Kernel.

$$\text{K}\varepsilon(\mathcal{O}) \equiv \{ \rho_s \mid \rho_s \text{ is coherence-recoverable relative to } \mathcal{O} \}.$$

$\text{K}\varepsilon(\mathcal{O})$ is the viability kernel of coherence under admissible control. States outside $\text{K}\varepsilon(\mathcal{O})$ are operationally irreversible with respect to $\mathcal{O}$.

Mapping to viability theory (D6-D9).

The abstract viability framework of Section A3.3 instantiates in the quantum setting as follows.

- *State space* X : the convex set of density operators on $\mathcal{H}_s$, equipped with the trace-norm topology.
- *Admissible controls*: the set of system-local CPTP maps (optionally with fresh ancilla), representing all operations an agent can perform on S alone.
- *Dynamics* $f(x, u)$: the discrete-time or continuous-time evolution generated by applying admissible operations.
- *Recoverable set* R : the set of states from which coherence between record sectors can be restored (Definition D11).

- *Viability kernel* $Viab(R)$: exactly $K\varepsilon(\mathcal{O})$ — the set of states for which there exists an admissible control strategy that maintains coherence-recoverability indefinitely.
- *Capture basin* $Cap(R)$: the set of states from which, under any admissible control strategy, the system eventually becomes operationally irreversible.
- *No-return surface* Σ_h : the boundary $\partial K\varepsilon(\mathcal{O})$, separating recoverable from irreversibly decohered states.

The scalar horizon $x_h = u_{\max}/a$ of T2 is the one-dimensional special case of this general geometric construction.

Proposition P4.1: Tracing Induces Loss of Recoverability.

If system-environment interaction produces correlations such that distinct record sectors become correlated with orthogonal (or trace-distance-separated) environment states, then for sufficiently small ε ,

$$\rho_s(t) \notin K\varepsilon(\mathcal{O}).$$

Once which-record information is encoded in inaccessible degrees of freedom, no admissible operation on S alone can restore coherence between record sectors.

You have felt this. Once the words leave your mouth, you cannot un-say them. The environment has recorded them — in the other person’s memory, in the vibrations of the air, in the electromagnetic radiation that left the room at the speed of light. No operation on your mouth alone can undo what the environment now holds.

Proof sketch. Orthogonality of environment states implies non-injectivity of the reduced dynamics on $\mathcal{H}_s$. Multiple globally distinct states map to the same ρ_s , and no CPTP map on S can reconstruct the lost phase information.

Definition D13: Operational No-Return Surface (Quantum).

The operational no-return surface relative to $\mathcal{O}$ is the boundary $\partial K\varepsilon(\mathcal{O})$.

Crossing this boundary transfers the system from a region where coherence is recoverable to a region where it is not.

Irreversibility is identified with loss of reachability. Not with energy dissipation. Not with entropy increase. Not with violation of microscopic reversibility. With the fact that you cannot get back.

Interpretation.

Operational irreversibility in quantum mechanics arises not from non-unitarity, but from restricted access. Tracing over inaccessible degrees of freedom removes states from the recoverable set K_ε , producing a no-return surface in state space exactly analogous to the operator horizons of Section A3.

Irreversibility is a geometric property of admissible control. It is not a statement about time reversal at the microscopic level.

A4.2 — Objective Actualization Channels and Selection Dynamics

Impossibility of Deterministic Selection by Linear CPTP Dynamics.

Proposition. No deterministic, linear CPTP map acting on the system state can transform a diagonal mixture over record sectors into a single realized sector in individual runs.

Linear CPTP evolution preserves convex mixtures: $\mathcal{E}(\sum_i p_i \rho_i) = \sum_i p_i \mathcal{E}(\rho_i)$. Any linear CPTP map that acts identically on each component preserves the mixture structure and cannot yield single-outcome definiteness in individual realizations.

Any mechanism that resolves a decohered mixture — any mechanism that produces the definiteness you experience — into a single realized branch must involve either a stochastic unraveling or an explicitly nonlinear effective evolution at the level of single trajectories.

Read that again. Standard quantum mechanics — linear, deterministic, trace-preserving — cannot produce definiteness in individual runs. Something else is required. The proof is not complicated. Linear maps preserve mixtures. If you feed a mixture in, you get a mixture out. Something nonlinear or stochastic must intervene for one outcome to become THE outcome.

Definition D14: Decoherence vs. Selection.

Decoherence suppresses interference between record-distinguishable alternatives and yields a stable diagonal mixture in the record algebra $\mathcal{O}$.

Selection is the further transition from a diagonal mixture to a single realized branch.

Decoherence is sufficient for irreversibility. Selection is required for definiteness.

You live in a definite world. Something selects.

Decoherence and selection are distinct processes. You experience them as distinct. The formalism confirms your experience.

Operational Meaning of "Realized Branch."

A branch i is realized iff, for all subsequent times and for any sequence of admissible operations confined to $\mathcal{H}_s$, the accessible state behaves as if it had been prepared in the conditional state

$$\rho_i \equiv (\Pi_i \rho_s \Pi_i) / p_i$$

at the onset of selection — in the sense that for all admissible POVMs M on S ,

$$|\text{Tr}(M \rho_s(t)) - \text{Tr}(M \rho_i)| \leq \varepsilon$$

for operational tolerance ε .

Decoherence separates the branches. Selection picks one. You have felt both — the moment when the options became clear (decoherence) and the moment when one became actual (selection). The physics mirrors the experience.

Postulate P: Objective Actualization Channel.

The paper posits an objective actualisation channel acting on the reduced state after decoherence has established operational irreversibility.

The ensemble evolution of the reduced state is written schematically as

$$d\rho_s/dt = \mathcal{L}_{\text{unitary}}(\rho_s) + D\mathcal{G}(\rho_s) + A\mathcal{G}(\rho_s),$$

where $D\mathcal{G}$ is the standard pointer-dephasing channel and $A\mathcal{G}$ is a selection channel responsible for definiteness.

The master equation governs ensemble dynamics. Single-run definiteness requires a stochastic or nonlinear unraveling of $A\mathcal{G}$.

Structural Requirements on the Selection Channel.

(S0) Activation Condition. $A\mathcal{G} \approx 0$ until the decoherence condition (D13) holds within tolerance ε . Selection activates only after branches are operationally distinct.

(S1) Record-algebra locality. $A\mathcal{G}(\rho_s) = A\mathcal{G}(\Delta\mathcal{G}(\rho_s))$. Selection never creates interference.

(S2) Sector fixed points. $A\mathcal{G}(\Pi_i \rho_s \Pi_i) = 0$ for all i . Once a branch is realized, dynamics cease.

(S3) Contractivity (single-run resolution). Under the stochastic unraveling, the Shannon entropy $H(\{p_i\})$ is a supermartingale: it decreases along individual trajectories almost surely.

(S4) Born boundary condition. For an ensemble of identical preparations at the onset of selection, the distribution of realized branches converges to $\{p_i\}$.

S4 is a boundary condition, not a derivation.

Notice what S4 does and does not say. It does not explain *why* the Born rule holds. It says: whatever selection is, it must produce Born statistics at the ensemble level. The constraint is structural. The explanation is someone else's problem.

Five structural requirements. Not a mechanism — an interface. Whatever selection IS, it must satisfy these five constraints. The constraints are testable. The mechanism is nature's business.

Observational consequence. Any selection channel satisfying S0-S4 produces, at the single-trajectory level, dynamics that are not reproducible

by any linear CPTP map acting on $\mathcal{H}_s$. This is the empirical signature of selection: post-decoherence statistics that violate linearity.

Concretely, monitoring a single system through the selection process should reveal either diffusive wandering or discrete jumps in sector weights — neither of which is consistent with a Lindblad master equation applied after dephasing is complete. This prediction is the target of test R5 (A5.2).

Toy Model: Stochastic Selection on a Qubit.

Abstract requirements are convincing only when you can see them work in a concrete case. Here is the simplest possible example — a single qubit. Watch how all five requirements are satisfied simultaneously.

The structural requirements S0–S4 constrain the selection channel but do not uniquely determine it. To demonstrate that the requirements are jointly satisfiable and to anchor the postulate in a concrete mathematical object, exhibit a minimal toy model: diffusive selection on a single qubit.

Setup. Let the system be a qubit with pointer basis $\mathcal{O} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$. After decoherence is complete, the reduced state is $\rho_s = \text{diag}(p, 1-p)$, with $p \in [0, 1]$. The selection channel acts on the single free parameter p via the Itô stochastic differential equation

$$dp = \sqrt{\gamma} \cdot p(1 - p) \cdot dW(t),$$

where $W(t)$ is a standard Wiener process and $\gamma > 0$ is the selection rate parameter.

The diffusion coefficient $\sigma(p) = \sqrt{\gamma} \cdot p(1 - p)$ vanishes at the boundaries $p = 0$ and $p = 1$, making both endpoints absorbing states.

Verification of S0–S4.

(S0) Activation condition. γ is set to zero until the decoherence threshold (D13) is satisfied. Prior to activation, p evolves only under standard quantum dynamics.

(S1) *Record-algebra locality*. The SDE acts entirely on the diagonal sector weight p . No off-diagonal (coherence) terms are generated. The map is record-algebra local by construction.

(S2) *Sector fixed points*. At $p = 0$ and $p = 1$, the diffusion coefficient vanishes identically: $\sigma(0) = \sigma(1) = 0$. Both sector-pure states are absorbing. Once a branch is realized, dynamics cease.

(S3) *Contractivity (full Itô calculation)*. Define $H(p) = -p \log p - (1-p) \log(1-p)$. The selection SDE is $dp = \sqrt{\gamma} \cdot p(1-p) dW$. Apply Itô's formula: $dH = H'(p) dp + \frac{1}{2} H''(p) (dp)^2$.

Compute: $H'(p) = -\log p + \log(1-p) = \log[(1-p)/p]$. $H''(p) = -1/p - 1/(1-p) = -1/[p(1-p)]$.

The quadratic variation is $(dp)^2 = \gamma p^2(1-p)^2 dt$. Substituting:

$$dH = \log[(1-p)/p] \cdot \sqrt{\gamma} \cdot p(1-p) dW + \frac{1}{2} \cdot [-1/(p(1-p))] \cdot \gamma p^2(1-p)^2 dt.$$

The drift term simplifies: $\frac{1}{2} \cdot [-1/(p(1-p))] \cdot \gamma p^2(1-p)^2 = -(\gamma/2) p(1-p)$. Therefore

$$dH = -(\gamma/2) p(1-p) dt + \sqrt{\gamma} \cdot p(1-p) \cdot \log[(1-p)/p] dW.$$

The drift term $-(\gamma/2) p(1-p)$ is strictly negative for all $p \in (0, 1)$, with equality only at the absorbing boundaries. The dW term is a martingale (zero expectation). Therefore $E[dH] = -(\gamma/2) p(1-p) dt \leq 0$, establishing that H is a supermartingale.

Shannon entropy decreases along individual trajectories almost surely, with rate proportional to γ .

The mathematics just proved that the branching resolves. Not on average. Not in expectation. Along every single trajectory. The coin lands.

(S4) *Born boundary condition*. The process $dp = \sqrt{\gamma} \cdot p(1-p) dW$ is a martingale, since the drift term is zero: $E[p(t)] = p(0)$ for all t .

Because p converges almost surely to $\{0, 1\}$, the probability of absorption at $p = 1$ is exactly $p(0)$, and absorption at $p = 0$ occurs with probability $1 - p(0)$.

The ensemble statistics of realised branches reproduce the Born weights without any additional assumption. No extra postulate. No interpretation. The martingale property of the process — the same mathematics that governs fair games in probability theory — forces the Born rule.

The universe plays fair.

Connection to the gravitational limiter. Postulate G (A4.3) constrains the selection rate: $\gamma \leq \Delta E_G / \hbar$.

In the toy model, the mean time to absorption scales as $\tau \sim 1/\gamma$, so the gravitational bound imposes $\tau \geq \hbar/\Delta E_G$. For gravitationally indistinguishable records ($\Delta E_G = 0$), $\gamma = 0$ and no selection occurs.

What the toy model does not determine.

The diffusive form of the SDE is a choice, not a derivation. A jump (Poisson) unraveling would also satisfy S0–S4, as would other diffusion coefficients of the form $\sigma(p) = f(p)$ with $f(0) = f(1) = 0$.

The toy model demonstrates joint satisfiability of the structural requirements and provides a concrete anchor for falsification predictions — for example, the distinction between diffusive and jump statistics in R5. It does not claim to be the unique or correct selection dynamics.

The postulate specifies an interface. The toy model is one implementation.

You do not need to know which implementation nature uses. You need to know that any implementation satisfying S0–S4 reproduces what you observe.

This underdetermination is a feature of the postulate’s generality, but it has experimental consequences. Diffusive unravelings predict continuous, Brownian-like wandering of sector weights before absorption, yielding a characteristic $1/f$ noise spectrum in single-trajectory monitoring. Jump unravelings predict sudden, discrete transitions between sector weights, yielding telegraph noise.

These are operationally distinguishable in systems where single trajectories can be monitored — continuously measured superconducting qubits, fluorescence-monitored trapped ions. Test R5 is designed, among other

things, to make this distinction. Until such data exist, the postulate correctly remains agnostic about which unraveling nature selects.

Explicit Distinction from Spontaneous Collapse Models.

Postulate P shares surface features with spontaneous collapse models such as GRW (Ghirardi, Rimini, Weber) and CSL (Continuous Spontaneous Localization), but it differs from them in structure, scope, and commitment. The distinctions are precise.

Basis of collapse. GRW and CSL collapse the wavefunction in the position basis by construction. The localization operators are spatial: they multiply the wavefunction by a Gaussian centered at a random point in physical space.

The selection postulate instead acts on whatever record algebra $\mathcal{O}$ is selected by the system-environment coupling.

In systems where the pointer basis is not position — superconducting qubits, photon polarization, spin ensembles — GRW/CSL and Postulate P make different predictions. That is the content of falsifier F1 and test R1.

Order of operations. In GRW/CSL, collapse is a universal, always-on process that competes with unitary evolution at all times.

The selection postulate requires an activation condition (S0): selection is negligible until decoherence has rendered record sectors operationally distinct. This ordering is falsifiable via test R5.

If collapse signatures appear before decoherence is complete, the selection postulate fails. If they appear only after, GRW/CSL's always-on character is unnecessarily strong.

Rate structure. GRW introduces a universal collapse rate λ (one collapse per particle per $\sim 10^{16}$ seconds) and a localization width $a \sim 10^{-7}$ m as free parameters. CSL replaces the discrete hits with continuous diffusion but retains the same two parameters.

The selection postulate introduces no universal rate. Instead, the rate is bounded from above by the gravitational self-energy distinguishability of the record sectors (Postulate G). The selection rate is system-dependent and

vanishes for gravitationally indistinguishable records — a prediction GRW/CSL do not make.

Ontological commitment. GRW/CSL modify the Schrödinger equation at the fundamental level. They add a stochastic, nonlinear term to the dynamical law governing all matter.

The selection postulate does not modify quantum mechanics. It posits an additional channel that acts on the reduced state after decoherence, without specifying whether this channel is fundamental, emergent, or an effective description of deeper physics.

The limitation is deliberate, not an oversight: the postulate specifies an interface, not an ontology.

Summary of experimental discriminants.

Three tests distinguish the selection postulate from GRW/CSL:

- (R1) whether selection targets the pointer basis or always position;
- (R2) whether selection occurs between gravitationally indistinguishable records;
- (R5) whether selection requires prior decoherence or operates independently of it.

Agreement on all three would be surprising and informative. Disagreement on any one is decisive.

Relation to Actualization State.

Selection reduces AS. Decoherence increases AS by creating record-structured branching (A3.1). Selection decreases AS by collapsing that branching into a single realized history.

There is no contradiction. AS measures branching richness, not outcome definiteness.

Decoherence opens the fan. Selection closes it. AS tracks the fan.

Falsifiability.

F1 (Pointer failure). Selection does not respect the record algebra $\mathcal{O}$.

F2 (Born violation). Ensemble statistics of realized branches deviate from $\{p_i\}$.

F3 (Context dependence). Selection depends on observer intervention rather than objective dynamics.

Failure of the postulate does not invalidate AS, T1–T2, or A4.1.

A4.3 — Physical Constraints on Selection Rates (Gravity as Limiter)

Definition D15: Gravitational Self-Energy Distinguishability.

Let two decohered record sectors i and j correspond to mass-energy densities $\rho_i(x)$ and $\rho_j(x)$. Define the gravitational self-energy difference

$$\Delta E_G[i, j] \equiv (G/2) \iint [\rho_i(x) - \rho_j(x)] [\rho_i(y) - \rho_j(y)] / |x - y| d^3x d^3y.$$

$\Delta E_G = 0$: the two records are gravitationally indistinguishable. Larger ΔE_G : stronger gravitational distinguishability.

Definition D16: Inter-Sector Selection Rate.

Let τ_{ij} be the characteristic time for an individual realization to become operationally indistinguishable from the conditional state within tolerance ε . Define the inter-sector selection rate

$$\lambda_{ij} \equiv 1/\tau_{ij}.$$

Postulate G: Gravity-Limited Selection Rate.

The objective selection rate between two record sectors is bounded above by their gravitational distinguishability:

$$\lambda_{ij} \leq \Delta E_G[i, j] / \hbar.$$

The bound is a limiting inequality, not an equality. Selection may be slower. Selection cannot be faster without invoking a coupling stronger than gravity.

Physical motivation.

This bound is not derived from first principles. It is a postulate. But it is not arbitrary.

Two record sectors with distinct mass-energy distributions source distinct gravitational fields. If these sectors are in superposition, the gravitational field itself is in a superposition of distinguishable configurations.

The gravitational self-energy difference ΔE_G quantifies the degree to which the two field configurations are distinguishable. It is the interaction energy of the difference mass distribution with itself.

The energy-time relation $\Delta E \cdot \Delta t \geq \hbar$ then implies that the minimum time required for any physical process to resolve this distinguishability is $\Delta t \sim \hbar/\Delta E_G$.

The bound $\lambda_{ij} \leq \Delta E_G/\hbar$ therefore says: selection cannot resolve two record sectors faster than the gravitational field configurations sourced by those sectors can be distinguished.

The bound is not a derivation of gravitational collapse. It is a consistency constraint. Whatever mechanism performs selection, it cannot outrun gravitational distinguishability unless it couples to the system more strongly than gravity does.

The bound is testable (R3). Its failure would indicate either that selection couples to a non-gravitational degree of freedom, or that the energy-time reasoning does not apply to the selection process.

Think about what this means for everyday objects.

A cat in a box has enormous gravitational self-energy difference between *alive* and *dead* configurations — different mass distributions, different gravitational fields. The bound says selection happens almost instantly. You never see a cat in superposition because gravity resolves it before you could notice.

An electron spin has essentially zero gravitational self-energy difference between *up* and *down* — same mass, same distribution. The bound says selection is negligible. Electrons remain in superposition indefinitely.

One inequality. Both the classical and the quantum world explained.

Distinction from Diósi-Penrose.

The Diósi-Penrose proposal argues that gravity *causes* collapse due to spacetime ambiguity in superposed mass configurations. The present framework makes a weaker claim: gravity *limits the rate* of selection.

The mechanism of selection is not specified. Gravity provides only a ceiling on how fast it can proceed.

This is the difference between *gravity collapses the wavefunction* and *whatever performs selection cannot do so faster than gravitational distinguishability allows*.

Special Cases.

Gravitationally indistinguishable records. $\Delta E_G = 0 \Rightarrow \lambda_{ij} = 0$. Gravity forbids objective selection between them. A superposition of such records can persist indefinitely unless another interaction provides a limiter.

Macroscopic spatial superpositions. For spatially separated mass distributions, ΔE_G grows with mass and separation, yielding $\tau_{ij} \geq \hbar/\Delta E_G$. Macroscopic records resolve quickly but not instantaneously.

Falsifiability.

G1. Selection occurs faster than $\Delta E_G/\hbar$ for gravitationally distinguishable records.

G2. Selection occurs between records with $\Delta E_G = 0$.

G3. Selection rates scale universally with non-gravitational parameters across macroscopic records.

Failure here invalidates only the gravitational-limiter hypothesis, not the selection postulate or prior results.

A5 — Experimental Regimes and Falsification Paths

A5.1 — Orientation: Exclusions Before Fits

Up to A4.3, the argument specifies what must be true if the theory is correct. A5 specifies how it can fail, and how that failure would be observed.

The principles are simple. Qualitative exclusions before quantitative fits. Absence tests before rate tests. Pointer-basis dependence before gravitational scaling. Operational signatures before interpretation.

If any regime below fails, the corresponding theoretical component is dead — cleanly and locally.

A5.2 — Test Map (R0-R5)

R0 — Operational Invariance of AS (Global Kill Switch).

Prepare a system with two physically realizable coarse-grainings $\mathcal{O}_1$ and $\mathcal{O}_2$. Compute $AS(\rho; \mathcal{O}_1)$ and $AS(\rho; \mathcal{O}_2)$.

Prediction: $|AS(\rho; \mathcal{O}_1) - AS(\rho; \mathcal{O}_2)| \leq \delta_{\text{exp}}$.

Falsifier F0: Persistent disagreement beyond tolerance $\rightarrow$ entire structure fails.

Priority ordering.

The tests are listed in order of logical dependency but should be executed in order of discriminatory power.

R0 is the global kill switch and must be tested first. If R0 fails, the entire framework is dead and no further test is meaningful.

If R0 passes, R2 (gravitational null case) is the most discriminating test of Postulate G, because it probes the sharpest prediction — zero selection rate for gravitationally indistinguishable records. R5 (order of operations) tests

the selection postulate directly. R3 (rate bound) provides quantitative constraint. R1 and R4 test secondary predictions.

In short: R0 first; if R0 passes, R2 and R5 next; then R3; then R1 and R4.

R1 — Pointer Basis vs. Position Basis.

Setup. Systems where the environment-selected pointer algebra $\mathcal{O}$ is not position. Concrete examples: superconducting qubits (e.g., flux-tunable transmons), cavity QED (e.g., circuit QED, Schuster et al. 2007), collective spin ensembles.

Prediction. Selection targets $\mathcal{O}$, not position.

Falsifier F1. If definiteness consistently appears in position despite $\mathcal{O} \neq$ position, the selection postulate fails.

R2 — Gravitational Null Case ($\Delta E_G = 0$).

Setup. Decohered records that differ only by internal degrees of freedom with identical mass distributions. Candidates: nuclear spin states, photon polarization states, hyperfine states with identical spatial profiles.

Prediction. $\Delta E_G = 0 \Rightarrow \lambda_{ij} = 0$. No selection dynamics beyond standard decoherence.

Falsifier G2. Observation of objective selection between gravitationally indistinguishable records.

This does not forbid decoherence between such states via other interactions (e.g., EM). It forbids only objective selection on gravitational timescales.

R3 — Upper-Bound Rate Test (Speed Limit).

Setup. Mesoscopic/macrosopic superpositions with controlled mass distributions (levitated nanospheres, optomechanical resonators). Compute: $\Delta E_G \Rightarrow \tau_{\min} = \hbar/\Delta E_G$.

Prediction. $\tau_{ij} \geq \tau_{\min}$.

Falsifier G1. Selection on timescales shorter than $\hbar/\Delta E_G$ — rates exceeding $\Delta E_G/\hbar$.

Concrete estimates. For a spherical nanoparticle of radius $R = 100$ nm composed of a high-density material (tungsten, $\rho \approx 19$ g/cm³), with two record sectors separated by $\Delta x \sim R$, the gravitational self-energy scales as $\Delta E_G \propto \rho^2 R^5$.

This yields $\tau_{\min} \sim 1$ – 10 seconds, within reach of current cryogenic optical/magnetic levitation platforms and proposed microgravity free-fall experiments.

For comparison, silica nanoparticles ($\rho \approx 2$ g/cm³) at the same radius give $\tau_{\min} \sim 10^2$ – 10^3 seconds, pushing toward the boundary of current coherence times. High-density materials are therefore strongly preferred for near-term tests.

R4 — Born Boundary Condition.

Repeated preparations of identical decohered mixtures $\{p_i\}$.

Prediction. The final ensemble of realized branches converges to $\{p_i\}$. Intermediate-time biases allowed; only the asymptotic ensemble is constrained.

Falsifier F2. Systematic deviation from Born weights.

R5 — Order-of-Operations Test.

Continuously tune environmental coupling to control degree of decoherence.

Prediction. Selection is negligible until decoherence has rendered pointer sectors operationally distinct (D13).

Falsifier F3. Detection of selection signatures prior to operational irreversibility.

This test kills models where collapse is invoked to *cause* decoherence.

A5.3 — Operational Signature of Selection

Selection corresponds to nonlinear or stochastic dynamics at the single-trajectory level after decoherence is complete, producing effects not reproducible by any linear CPTP map acting on $\mathcal{H}_s$.

Detectable signatures include single-trajectory anomalies (jump or diffusion statistics inconsistent with any linear Lindbladian after dephasing), irreversible loss of interference-revival capacity even under idealized system-only control, and telegraph-like stabilization (once a branch is realized, subsequent measurements behave as if prepared in the conditional state within operational tolerance).

A5.4 — What Counts as Confirmation vs. Survival

Passing a test does not confirm the argument. It only allows it to survive. Confirmation would require joint success across multiple regimes — and even then, what is established is structure, not interpretation.

Failure, by contrast, is immediate and final.

A5.5 — Timeline to Falsification

The estimates below reflect the state of experimental capability as of 2025. They are orientation, not prediction.

R0 (Operational Invariance). Near-term (0–2 years). Circuit QED platforms already produce systems with multiple co-admissible pointer bases. Verification of AS invariance across such bases requires only measurement and classical post-processing of existing data.

R1 (Pointer Basis vs. Position). Near-term (0–3 years). Superconducting qubit and cavity QED experiments routinely prepare states where the pointer basis is energy or charge, not position. Checking whether definiteness tracks the pointer basis requires monitoring which observable resolves first under controlled decoherence.

R5 (Order of Operations). Near-term to medium-term (1–5 years). Requires continuously tunable environmental coupling with single-trajectory readout. Superconducting qubits with adjustable coupling to engineered reservoirs are the most promising platform. The key observable is whether selection-like signatures (telegraph stabilization, non-Lindbladian statistics) appear only after decoherence is complete.

R4 (Born Boundary Condition). Medium-term (2–5 years). Requires large ensembles of identically prepared, fully decohered systems with high-fidelity single-shot readout. Trapped-ion and superconducting qubit arrays are approaching the required scale and fidelity.

R2 (Gravitational Null Case). Medium-term to long-term (3–10 years). Requires decohered records that differ only by gravitationally indistinguishable internal degrees of freedom — for example, nuclear spin states with identical spatial profiles. The challenge is isolating such systems from all non-gravitational decoherence sources long enough to confirm the absence of selection.

R3 (Upper-Bound Rate Test). Long-term (5–15 years). Requires maintaining spatial superpositions of high-density nanoparticles (~ 100 nm, tungsten or osmium) for seconds in a decoherence-free environment, then measuring whether selection occurs faster than $\hbar/\Delta E_G$. Cryogenic levitation and proposed space-based platforms (e.g., MAQRO mission concept) are plausible but not yet operational at the required scale.

Ordering logic. Tests are listed in order of experimental accessibility, not theoretical importance. R0 and R1 can be performed with existing hardware. R5 and R4 require modest extensions. R2 and R3 require dedicated experimental programs.

A negative result on R0 terminates the entire framework before any other test is needed.

A5.6 — Stop Condition

Definitions are operational. You can measure every one of them. Theorems are scoped. Each tells you exactly where it applies and where it does not. Postulates are isolated. Kill one and the rest survive. Bounds are testable. You can check them with existing equipment. Falsifiers are explicit. You know exactly what would kill each claim.

Nothing further can be settled by argument. The mathematics has spoken. The experiments are specified. The kill switches are published.

What remains is nature's answer.

The argument has told you everything it can tell you. It has defined the measuring tool. It has proven the monotonicity. It has established the no-return surface. It has characterised selection. It has given the gravitational rate bound. It has named the operational signatures.

Every claim is stated. Every kill switch is published. Every experiment is specified.

What remains is nature's answer. The argument cannot give you that. Only measurement can. Only contact with reality can determine whether the definitions are operationally invariant, whether the gravitational bound holds, whether selection respects the pointer basis, whether the Born statistics emerge.

The argument waits. It has placed itself at the mercy of experiment.

That is the only place an honest argument belongs.

A6 — Optional Module: The Turn

Status. Optional module. Not load-bearing. Included for conceptual completeness; failure leaves the entire laboratory program intact.

A6.1 — Capacity Saturation

Let K be the viability kernel of maintainable structure (A3.3).

Capacity saturation obtains when the accessible state space for creating new durable record sectors has measure zero under admissible controls.

Operationally: further decoherence may occur, but no new independent records can be written.

Capacity saturation corresponds to thermodynamic heat death in the limit where all free-energy gradients are exhausted. It is not identical to heat death in general. Saturation may occur locally or structurally before global thermal equilibrium.

A6.2 — Restoration Without Reversal

Any admissible Turn must satisfy:

- (1) **No reversal.** Previously realized selections are not undone.
- (2) **No selection bypass.** A4.2 remains valid locally.
- (3) **Capacity restoration.** The effective record algebra regains room for new, independent branches.

This is an interface specification, not a dynamical law.

A6.3 — Conformal Rescaling

At extreme dilution, dynamics become insensitive to absolute scale.

A conformal identification can map a capacity-saturated configuration to an initial configuration with renewed branching capacity without reversing causal order.

Existence proof, not assertion of actuality.

The phrase *records are compressed, not erased* means: distinguishable record sectors at late times map to a lower-resolution effective record algebra under the conformal identification, preserving orthogonality relations and causal precedence while reducing accessible distinguishability.

A6.4 — Relation to Heat Death

Black holes are capacity sinks, not reset buttons. They localize saturation and demonstrate no-return boundaries (A3.3).

Any global Turn, if it exists, must respect the same non-reversal constraint.

A6.5 — Why This Module Is Optional

The core program answers how irreversibility arises, how definiteness arises, and how fast definiteness may arise.

Module T addresses only whether global capacity can ever be restored.

Failure of Module T leaves the entire laboratory program intact.

Appendices

Appendix E — Glossary of Terms and Notation

Actualization State (AS). An operational scalar $\in [0, 1]$ measuring the degree of inter-sector branching in the record algebra. $AS = 0$: all weight in one record sector. $AS = 1$: maximal branching across all record sectors. Defined in D3 (A2.3).

Admissible operation. A CPTP map acting on the accessible system, optionally with fresh ancilla, but with no access to the original environment. The set of admissible operations defines what an agent can do, and thereby what counts as operationally irreversible.

Capture basin, $Cap(R)$. The set of states from which exit from the recoverable set R is inevitable under all admissible controls. Defined in D8 (A3.3).

Coherence-recoverable state. A system state from which coherence between record sectors can be restored by admissible operations within tolerance ϵ . Defined in D11 (A4.1).

Coarse-graining, physically realizable ($\mathcal{O}$). A finite set of mutually orthogonal projectors selected by the physics of system–environment coupling, not by observer choice. Defined in D1 (A2.1).

Dephasing map, $\Delta\mathcal{O}$. The map $\Delta\mathcal{O}(\rho) \equiv \sum_i \Pi_i \rho \Pi_i$ that removes quantum interference between record sectors while preserving classical probabilities. Defined in D2 (A2.2).

Effective entropy, S_{eff} . The Shannon entropy $H(\{p_i\})$ of the sector weights, with intra-sector entropy discarded. The entropy entering the AS definition. Defined in A2.3.

Falsifier (F0, F1, F2, F3, G1, G2, G3). An experimentally observable condition whose occurrence would invalidate a specific component of the argument. F0 is global (kills AS itself); F1–F3 target the selection postulate; G1–G3 target the gravitational limiter. Listed in A5.2.

Gravitational self-energy distinguishability, ΔE_G . The Newtonian self-energy of the difference mass distribution between two record sectors. Defined in D15 (A4.3). Appendix C provides explicit forms.

No-return surface, Σ_h . The boundary of the viability kernel. States beyond this surface cannot return to the recoverable set under any admissible control. Defined in D9 (A3.3) and D13 (A4.1).

Operator horizon, x_h . The scalar specialization of the no-return surface: $x_h \equiv u_{\max}/a$, the maximum sustainable structure given maximum maintenance effort. Defined in T2 (A3.2).

Operational invariance. The requirement that AS values computed from different physically realizable coarse-grainings of the same system agree within experimental tolerance. Defined in D5 (A2.5). Violation triggers the global kill switch F0.

Operational irreversibility. A state is operationally irreversible with respect to a recoverable set R iff it lies outside the viability kernel of R. Irreversibility is defined by loss of reachability under admissible control, not by entropy increase or violation of time-reversal symmetry. Defined in D10 (A3.3).

Record algebra. The algebra generated by the physically realizable coarse-graining $\mathcal{O} = \{\Pi_i\}$. States in this algebra are diagonal in the record basis. The record algebra defines what the environment can physically distinguish.

Selection. The transition from a diagonal mixture over record sectors (post-decoherence) to a single realized branch (definiteness). Distinguished from decoherence in D14 (A4.2). The mechanism is specified by Postulate P; the rate is bounded by Postulate G.

Viability kernel, $\text{Viab}(R)$. The set of states from which the system can be kept inside the recoverable set R indefinitely using admissible control. Defined in D7 (A3.3).

ε (operational tolerance). A fixed positive parameter representing experimental resolution. All operational definitions (effective orthogonality, inaccessibility, recoverability) are quantified relative to ε . Defined in A2.4.

Appendix A — Equivalence of AS Representations

A.1 Purpose.

This appendix establishes the precise conditions under which the primary $AS(\rho; \mathcal{O})$ coincides with the history-based representation $AS_h(D)$. No equivalence is assumed in the main text.

A.2 Objects and Restrictions.

The histories $\{\alpha\}$ are taken to correspond one-to-one with the record sectors $\{\Pi_i\}$ at a single time slice. No multi-time or branching-tree histories are included. Under this restriction: $N = d\mathcal{O}$.

This restriction is explicit and intentional.

A.3-A.4 Decoherence Condition and Block Entropy.

Assume complete decoherence: $D(\alpha, \beta) \approx 0$ for $\alpha \neq \beta$. Under this condition, the dephased state has block form $\Delta\mathcal{O}(\rho) = \sum_i p_i \sigma_i$.

The von Neumann entropy decomposes exactly as

$$S(\Delta\mathcal{O}(\rho)) = H(\{p_i\}) + \sum_i p_i S(\sigma_i).$$

No assumption of maximal mixing within sectors is made.

A.5-A.6 Equivalence Statement.

Under the restriction of A.2 and the decoherence condition of A.3-A.4, $AS_h(D) = H(\{w(\alpha)\}) / \log |D|$ with $w(\alpha) = p_i$ for the corresponding sector. AS uses $S_{\text{eff}}(\rho; \mathcal{O}) \equiv H(\{p_i\})$ with normalization $AS(\rho; \mathcal{O}) = H(\{p_i\}) / \log d\mathcal{O}$.

Where the histories are exactly the record sectors and decoherence is complete, the two normalizations coincide (since $N = d\mathcal{O}$), and $AS_h(D) = AS(\rho; \mathcal{O})$ by construction.

A.7 Non-Equivalence Regimes.

The equivalence fails when decoherence is incomplete, when histories span multiple time slices, or when one attempts to include intra-sector entropy as *actualisation*.

In these regimes, $AS(\rho; \mathcal{O})$ remains well-defined. $AS_h(D)$ ceases to be a faithful representation.

$AS(\rho; \mathcal{O})$ is preferred in all ambiguous cases because it requires only the reduced state and record algebra, not a full history space.

Appendix B — Viability Theory Background

Dynamics: $dx/dt = f(x, u)$, $u \in U$.

Viability kernel: $Viab(K) = \{x_0 \in K \mid \exists u(t): x(t) \in K \forall t \geq 0\}$.

Capture basin: $Cap(K^c) = \{x_0 \mid \forall u(t), \exists t: x(t) \notin K\}$.

No-return surface: $\Sigma_{NR} = \partial Viab(K)$.

The partition holds up to boundary sets of measure zero. For further background, see Aubin (1991), *Viability Theory*.

Appendix C — Gravitational Self-Energy

C.1 Definition.

For two record sectors i, j with mass densities $\mu_i(x)$, $\mu_j(x)$:

$$\Delta E_G[i, j] \equiv (G/2) \iint [\mu_i(x) - \mu_j(x)] [\mu_i(y) - \mu_j(y)] / |x - y| d^3x d^3y.$$

C.2 Weighted Multi-Sector Form.

For a multi-sector decomposition with weights p_i :

$$\Delta E_G^{\text{total}} \equiv \sum_{i < j} p_i p_j \Delta E_G[i, j].$$

C.3 Special Cases.

Identical mass distributions. $\Delta E_G = 0$.

Rigid sphere (radius R , displacement $\Delta x \ll R$). $\Delta E_G \sim Gm^2/R \cdot (\Delta x/R)^2$ (order estimate).

C.4 Positivity.

By construction, $\Delta E_G \geq 0$.

Appendix D — Experimental Feasibility Estimates

D.1 High-Density Nanoparticle Regime.

Consider a spherical nanoparticle of radius $R = 100$ nm composed of a high-density material (tungsten or osmium, $\rho \approx 19\text{--}22$ g/cm³). For comparison, silica has $\rho \approx 2$ g/cm³.

With two record sectors separated by $\Delta x \sim R$, and using $\Delta E_G \sim Gm^2/\Delta x$ with $m \propto \rho R^3$, the self-energy scales as $\Delta E_G \propto \rho^2 R^5$.

Tenfold density increase yields $\sim 100\times$ increase in ΔE_G . Resulting timescale: $\tau_{\text{min}} \sim \hbar/\Delta E_G$ yields $\tau_{\text{min}} \sim 1\text{--}10$ s for $R \sim 100$ nm high-density particles.

D.2 Experimental Platforms.

Timescales in the second-to-tens-of-seconds range are within reach of:

- Cryogenic optical or magnetic levitation of heavy nanoparticles (e.g., Delić et al., 2020; Tebbenjohanns et al., 2021).
- Hybrid optomechanical traps with active feedback cooling (e.g., Aspelmeyer group, Vienna).

- Space-based or microgravity proposals (long-coherence free-fall platforms, e.g., MAQRO mission concept).

D.3 Interpretation.

The purpose of this estimate is to demonstrate that the relevant regime is experimentally accessible.

Observation of selection faster than the bound falsifies the gravity-limiter hypothesis. Absence of selection constrains gravity's relevance without undermining the core AS framework.

Appendix F — Worked Example: AS Calculation for a Dephasing Qubit

This appendix provides a complete, explicit AS calculation for the simplest nontrivial case, intended as a pedagogical anchor.

F.1 Setup.

System: a single qubit S with Hilbert space $\mathcal{H}_s = \mathbb{C}^2$, coupled to an environment E .

Pointer basis (selected by coupling): $\mathcal{O} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$.

Record algebra dimension: $d\mathcal{O} = 2$.

F.2 Initial State.

$$|\psi(0)\rangle = (|0\rangle + |1\rangle)/\sqrt{2} \otimes |E_0\rangle.$$

The system is in a coherent superposition. The reduced state is $\rho_s(0) = |+\rangle\langle +|$.

After dephasing: $\Delta\mathcal{O}(\rho_s(0)) = \text{diag}(1/2, 1/2)$. Sector weights: $p_0 = p_1 = 1/2$.

F.3 Before Decoherence.

Off-diagonal elements are present in $\rho_s(0)$. However, for the purpose of computing AS, you apply $\Delta\mathcal{O}$ first.

$S_{\text{eff}} = H(\{p_i\}) = H(1/2, 1/2) = \log 2$. Normalization: $S_{\text{max}} = \log 2$.

Therefore $AS = \log 2 / \log 2 = 1$.

Even before decoherence is complete, the sector weights already distribute maximally. AS measures the branching structure of the dephased state, not whether decoherence has physically occurred.

F.4 After Decoherence.

Environment records which-path: $|\psi\rangle \rightarrow (|0\rangle|E_0\rangle + |1\rangle|E_1\rangle)/\sqrt{2}$ with $\langle E_0|E_1\rangle \approx 0$.

Reduced state: $\rho_s = \text{diag}(1/2, 1/2)$. Off-diagonals have been physically suppressed.

$AS = 1$. Same numerical value, but now the system has crossed the no-return surface: coherence is not recoverable. Operational irreversibility has been established.

F.5 After Selection.

Selection resolves the mixture into sector $|0\rangle$ (say). Now $p_0 = 1$, $p_1 = 0$. $H = 0$.

$AS = 0$.

One story remains. The system has transitioned from maximal branching to definiteness.

F.6 Summary.

AS tracks branching richness, not definiteness. It rises during decoherence (branching phase) and falls during selection (definiteness phase).

The worked example illustrates that $AS \approx 0$ and $AS \approx 1$ are both physically meaningful endpoints of different processes — not a hierarchy of *more actualized* vs. *less actualized*.

Paper A — Canonical Reference. Locked · Execution-Complete.

Paper B

Selection as Irreversible Exclusion

Rates, Costs, and Constraints on Definiteness

Depends on Paper A

Paper A measured the branching. It proved the branching can only grow under the right conditions. It established the point of no return.

But it left one question unanswered — the question that haunts every interpretation of quantum mechanics.

If definiteness occurs in individual runs — and it does, every experiment ever conducted says so — what must selection be?

Not what *might* it be. What *must* it be.

What structural requirements does any selection mechanism have to satisfy? What must it cost? How fast can it act? And is there a universal bound on that speed?

No collapse mechanism is proposed here. No interpretation is invoked. No result of Paper A is rederived. All hypotheses introduced are independently falsifiable. Failure of any does not invalidate Paper A.

B0.1 — Dependency Statement

This paper is a strict continuation of Paper A. Everything in Paper A is assumed established and is not rederived.

In particular, the following are taken as given:

- The operational definition and validity of Actualization State (AS).
- Operational irreversibility as loss of reachability under admissible control.
- The existence of no-return surfaces induced by bounded capacity.
- The separation between branching (AS increase) and definiteness.

For reference: the record-sector algebra $\mathcal{R}$ is the algebra generated by the projectors $\{\Pi_i\}$ of a physically realizable coarse-graining $\mathcal{O}$ (Paper A, Definition D1). It represents the set of operationally accessible record observables.

All norms and maps in this paper that reference $\mathcal{R}$ act on this algebra.

B0.2 — Purpose

Paper A establishes irreversibility without definiteness. After decoherence and loss of recoverability, multiple mutually exclusive record sectors can persist simultaneously in the reduced description.

This paper addresses what is left:

If definiteness occurs, what must selection be, given the constraints already established?

A second question follows necessarily:

What physical resources must be expended to enforce such selection?

The argument does not claim that selection must exist. It characterizes the structure and constraints of selection if it exists at all.

B0.3 — Hard Non-Claims

This paper does not:

- Redefine Actualization State.
- Propose a collapse mechanism.
- Derive or assume the Born rule.
- Invoke observers, consciousness, or epistemic update.
- Introduce agency, decision-making, or control.
- Claim gravity *causes* selection.

Failure of this paper does not invalidate Paper A.

B1 — The Definiteness Problem (Reframed)

B1.1 — What Remains After Paper A

After Paper A, the following are established:

- (1) Interference is suppressed between record-distinguishable alternatives (Paper A, T1).
- (2) Recoverability is lost once record information is encoded in inaccessible degrees of freedom (Paper A, D13).
- (3) Actualization State increases during the branching phase, quantifying record-structured multiplicity (Paper A, T1).

None of these imply that, in an individual experimental trial, only one record persists.

A diagonal reduced state of the form

$$\rho = \sum_i p_i \Pi_i \rho \Pi_i,$$

where $\{\Pi_i\}$ are the record-sector projectors of D1 and $\{p_i\}$ are the diagonal coefficients inherited from decoherence (no probabilistic interpretation assumed here), is fully consistent with all results of Paper A.

You experience definiteness anyway. Every single time. The reduced description says branches persist. Your experience says only one obtained. The gap between these two is what this paper takes seriously.

B1.2 — Why Decoherence Is Not Definiteness

Decoherence explains why interference terms become inaccessible. It does not explain why alternatives are excluded.

Operationally:

- *Decoherence* answers: why alternatives cannot interfere.
- *Definiteness* asks: why alternatives are no longer reachable.

These are distinct constraints. Paper A resolves the first and intentionally stops there. Paper B picks up where Paper A stopped.

B1.3 — Individual Realizations (Operational Definition)

An **individual realization** (or individual run) is defined as a single experimental trial producing a definite, time-ordered record stream in the environment, which subsequently constrains all future accessible system behavior.

This definition is purely operational. It refers only to record structure. No mention of consciousness. No mention of measurement. No mention of observers.

A run is a record stream. Records constrain what comes next. That is all.

B1.4 — Selection as Irreversible Exclusion

If definiteness exists, it must correspond to a physical exclusion process acting after irreversibility is established. The reason is direct: all subsequent experimental outcomes in the record stream depend causally on which sector obtains (B1.3).

Definition (Selection).

Selection is the transition of the system state into a restricted reachable region of state space — under admissible control — in which only one record sector remains reachable.

Equivalently: selection is the irreversible removal of alternative record sectors from operational accessibility in an individual realization.

Once selection has occurred, no admissible system-local operation can restore reachability of excluded sectors.

The branches do not vanish from the universe. They vanish from your access. That is the entire physical content of selection.

B1.5 — Consequence for Actualization State

Selection has a precise consequence for AS.

Decoherence increases AS by creating record-structured branching (Paper A, T1).

Selection reduces the **accessible AS** of an individual realization by restricting reachability to a single record sector.

This does not imply erasure of environmental records. It reflects the collapse of future operational accessibility, not the destruction of past structure.

The fan opens (decoherence). One blade is picked (selection). The other blades are still in the room — you just cannot reach them anymore.

B1.6 — Selection Cost (Foreshadowing)

Exclusion is not free.

Any process that removes reachability of alternatives must expend physical resources to enforce that restriction. Call this the **cost of selection**: the minimal physical resource expenditure required to enforce irreversible exclusion.

The cost need not be thermal energy. It may appear as time, interaction strength, or consumption of distinguishability capacity. Its precise form depends on the limiting interaction. Sections B2 and B4 quantify it.

B2 — The Nonlinearity Requirement and Selection Cost

B2.1 — Linearity Constraint

Deterministic linear completely positive trace-preserving (CPTP) dynamics acting on the reduced system state preserve convex structure. Linear ensemble evolution cannot, by itself, enforce single-sector definiteness in individual realizations.

Formally, for any deterministic linear CPTP map $\mathcal{E}$:

$$\mathcal{E}(\sum_i p_i \rho_i) = \sum_i p_i \mathcal{E}(\rho_i).$$

Linearity preserves convex mixtures. No such map can select a single component from a diagonal mixture in individual runs.

This is a structural consequence of linearity and does not depend on interpretation.

You feed a mixture in. You get a mixture out. The mixture cannot resolve itself by linear means alone.

B2.2 — Ensemble Linearity vs. Trajectory Resolution

The implication is precise.

Ensemble-level evolution may remain linear and CPTP. Selection, if it occurs, must act at the trajectory level — resolving individual realizations via stochastic or effectively nonlinear dynamics.

There is no contradiction with quantum linearity at the ensemble level.

Average over many runs and you see linear evolution. Watch a single run and you see resolution. Both descriptions hold simultaneously, at different levels of the same physics.

B2.3 — Quantifying Selection Deviation

Selection is a trajectory-level phenomenon. Its signature is not a deviation of the ensemble state. Ensemble consistency is required (B3.5), and the ensemble state is preserved by construction.

The signature of selection is that individual trajectories resolve to outcomes that no deterministic CPTP map could produce from the same initial state.

Let $\mathcal{E}_{\text{ens}}$ denote the ensemble-level CPTP evolution acting on the reduced state.

Let Φ denote a stochastic selection process that, for an initial state ρ , produces a random trajectory with realization-dependent final state $\rho^{\omega}(\rho)$

indexed by trajectory ω . Ensemble consistency (B3.5) requires $\mathbb{E}[\rho^\omega] = \mathcal{E}_{\text{ens}}(\rho)$.

The quantity that distinguishes selection from deterministic CPTP evolution is not the ensemble mean. It is the trajectory spread.

Define the **selection deviation**:

$$\delta_{\text{sel}}(\rho) \equiv \mathbb{E}[\|\rho^\omega - \mathbb{E}[\rho^\omega]\|_{\mathcal{R}}^2],$$

where $\|\cdot\|_{\mathcal{R}}$ is an operational norm restricted to the record-sector algebra $\mathcal{R}$.

This is the expected squared deviation of individual trajectory outcomes from the ensemble mean, measured in the record algebra.

For any deterministic CPTP map, all trajectories produce the same output, so $\delta_{\text{sel}} = 0$. For selection, trajectories resolve to different record sectors, so $\delta_{\text{sel}} > 0$.

All subsequent results hold for any choice of norm satisfying:

- (i) contractivity under admissible system-local CPTP maps, and
- (ii) sensitivity to distinguishability between record sectors.

Compatibility with ensemble consistency.

δ_{sel} measures the spread of trajectory outcomes, not the deviation of their mean. Ensemble consistency (B3.5) constrains the first moment: $\mathbb{E}[\rho^\omega] = \mathcal{E}_{\text{ens}}(\rho)$. The selection deviation constrains the second moment.

These are independent. A process can have zero mean deviation and nonzero trajectory spread. Selection is precisely such a process.

Verification (qubit toy model).

For the selection SDE $dp = \sqrt{\gamma} \cdot p(1-p) dW$ (Paper A, Section A4.2), trajectories resolve to $p = 0$ or $p = 1$ with probabilities $1-p_0$ and p_0 respectively.

The ensemble mean is preserved: $\mathbb{E}[\rho^\omega] = \text{diag}(p_0, 1-p_0)$. The trajectory variance is $\delta_{\text{sel}} = p_0(1-p_0) > 0$ for any nontrivial initial mixture, confirming that selection produces nonzero δ_{sel} while satisfying ensemble consistency.

Selection requires $\delta_{\text{sel}} > 0$. If $\delta_{\text{sel}} = 0$ for all accessible states, every trajectory produces the same output as the ensemble map, and no resolution to individual sectors has occurred.

B2.4 — Definition: Selection Cost

The **cost of selection** is defined as the minimal physical resource expenditure required to produce nonzero trajectory variance ($\delta_{\text{sel}} > 0$) sufficient to resolve individual realizations into single record sectors.

The cost may be expressed as:

- A timescale (rate of exclusion).
- A coupling strength to enforcing interactions.
- A resource budget required to maintain exclusion.

Section B4 identifies universal constraints on this cost.

B2.5 — Falsifier B2: Pre-Irreversibility Selection

If exclusion signatures appear before the system has crossed the no-return surface (Paper A, D13), the entire selection model is dead.

Selection must wait for irreversibility. If it does not wait, the model is wrong.

Selection cannot precede irreversibility.

B3 — Structural Requirements on Selection Dynamics

Any admissible selection dynamics must satisfy a minimal set of structural requirements. These are implied jointly by Paper A and Sections B1-B2.

The requirements are necessary, not sufficient. Failure of any requirement falsifies the selection hypothesis without affecting the irreversibility results of Paper A.

B3.1 — Post-Irreversibility Activation

Selection may act only after operational irreversibility is established.

Let $K\varepsilon(\mathcal{O})$ denote the ε -recoverable set defined in Paper A (Definition D12). For all states $\rho \in K\varepsilon(\mathcal{O})$, admissible dynamics must satisfy

$$\delta_{\text{sel}}(\rho) = 0.$$

No selection deviation is permitted while recovery remains reachable under admissible control. Selection dynamics may become active only once $\rho \notin K\varepsilon(\mathcal{O})$.

B3.2 — Record-Algebra Locality

Selection must act only on degrees of freedom that distinguish record sectors, and only during active selection.

Let $\Delta\mathcal{O}$ denote the dephasing map onto the record algebra $\mathcal{O}$. During active selection (i.e., after $\rho \notin K\varepsilon(\mathcal{O})$):

$$\Phi(\rho) = \Phi(\Delta\mathcal{O}(\rho)).$$

This condition is consistent with B3.1: by the time selection activates, off-diagonal terms in the record basis are already operationally inaccessible.

Selection may not generate or reintroduce interference. It may not act on unmonitored intra-sector degrees of freedom.

B3.3 — Absorbing Record Sectors

Selection is an absorbing process. Once a record sector Π_i is realized, sector membership must remain fixed under subsequent selection dynamics.

Formally:

$$\Phi(\Pi_i \rho \Pi_i / p_i) \subseteq \Pi_i \mathcal{B}(\mathcal{H}_s) \Pi_i,$$

up to operational tolerance set by experimental resolution.

This condition enforces irreversible confinement to the realized sector while allowing arbitrary intra-sector evolution.

B3.4 — Contractivity of Multiplicity

Selection resolves multiplicity. It must not amplify it.

Let $\{p_i(t)\}$ denote the diagonal coefficients of the reduced state in the record-sector basis, treated here as record weights. The Shannon entropy

$$H(\{p_i\}) = - \sum_i p_i \log p_i$$

is used strictly as a multiplicity measure, not as a thermodynamic or epistemic entropy.

Any admissible selection dynamics must be such that, along individual trajectories generated by those dynamics, $H(\{p_i(t)\})$ is a supermartingale:

$$\mathbb{E}[H(\{p_i(t+dt)\}) \mid \mathcal{F}_t] \leq H(\{p_i(t)\}),$$

with strict decrease during active selection. The expectation is taken with respect to the trajectory measure induced by the selection dynamics.

This is a requirement on the class of admissible dynamics, not a property derived from a specific generator. Any candidate selection process whose trajectories amplify multiplicity is excluded.

No assumption about the Born rule is made here. H is used purely as a multiplicity measure.

Contractivity of H along trajectories is a consequence of the trajectory-level character established in B2.2. The stochastic process Φ must resolve the mixture, and resolution requires H to decrease along individual realizations.

B3.5 — Ensemble Consistency

While individual trajectories resolve to single sectors, the ensemble description must remain consistent with linear evolution. Averaging over all trajectory realizations must reproduce the ensemble map:

$$\mathbb{E}[\rho^\omega] = \mathcal{E}_{\text{ens}}(\rho).$$

This requirement constrains the first moment of the trajectory distribution. It does not constrain the second moment. Trajectory spread ($\delta_{\text{sel}} > 0$, B2.3) is fully compatible with ensemble consistency.

Selection is characterized by the combination of preserved ensemble mean and nonzero trajectory variance.

B3.6 — Boundary Condition on Outcomes (BC1)

This paper does not derive outcome statistics. Admissible selection dynamics must produce a well-defined distribution over realized record sectors.

The analysis restricts to the class of selection dynamics that are **Born-consistent**: under the trajectory measure induced by the dynamics, the marginal distribution over realized sectors converges to the diagonal weights $\{p_i\}$ inherited from decoherence.

This is a defining constraint of the model class studied here, not a derived result. The existence and properties of Born-inconsistent selection dynamics are a separate question outside the present scope.

Born-consistency is experimentally testable. Repeated preparations of identically decohered systems must yield realized-sector frequencies converging to $\{p_i\}$. Persistent deviation falsifies the Born-consistent class, not selection itself.

B3.7 — Summary of Structural Requirements

Selection dynamics, if they exist, must be:

Post-irreversibility — inactive while recovery remains reachable.

Record-local — acting only on the record algebra during active selection.

Absorbing — once a sector is realized, sector membership remains fixed.

Contractive — monotonically reducing multiplicity along trajectories.

Ensemble-consistent — preserving linear ensemble evolution.

Any candidate process violating these conditions is not a physically admissible form of selection under the argument established by Paper A.

Five requirements. Not a mechanism. An interface.

Worked Example: Structural Requirements Applied to Qubit Selection

Paper A (Section A4.2) defines a toy model of stochastic selection on a qubit with pointer basis $\mathcal{O} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$ and selection dynamics

$$dp = \sqrt{\gamma} \cdot p(1-p) dW.$$

The five structural requirements all hold for this toy model. Verification, in order.

B3.1 — Post-irreversibility activation. The selection rate γ is set to zero while the system remains within the recoverable set $K\varepsilon(\mathcal{O})$. The SDE activates only after decoherence has rendered pointer sectors operationally distinct. Prior to activation, $\delta_{\text{sel}} = 0$ identically.

B3.2 — Record-algebra locality. The SDE acts entirely on the diagonal sector weight p . No off-diagonal coherence terms are generated or accessed. The process is record-algebra local by construction: $\Phi(\rho) = \Phi(\Delta\mathcal{O}(\rho))$.

B3.3 — Absorbing sectors. At $p = 0$ and $p = 1$, the diffusion coefficient $\sigma(p) = \sqrt{\gamma} \cdot p(1-p)$ vanishes identically. Both sector-pure states are absorbing fixed points. Once realized, sector membership is permanent.

B3.4 — Contractivity. By the full Itô calculation (Paper A, A4.2, requirement S3):

$$dH = -(\gamma/2) p(1-p) dt + \sqrt{\gamma} \cdot p(1-p) \cdot \log[(1-p)/p] dW.$$

The drift term $-(\gamma/2) p(1-p)$ is strictly negative for $p \in (0, 1)$. H is a supermartingale with strict decrease during active selection, as required.

B3.5 — Ensemble consistency. The SDE $dp = \sqrt{\gamma} \cdot p(1-p) dW$ is a martingale: $\mathbb{E}[p(t)] = p(0)$ for all t . Averaging over trajectories reproduces the ensemble state $\rho_{\text{ens}} = \text{diag}(p(0), 1-p(0))$ at all times. The ensemble map is linear and CPTP.

The structural requirements B3.1–B3.5 are jointly satisfiable. The toy model is not the unique solution. It is a proof of existence.

Any candidate selection dynamics must pass all five requirements to be admissible.

B4 — Universal Rate Constraints on Selection

Selection, if it exists, cannot occur arbitrarily fast.

This section establishes necessary upper bounds on the rate at which admissible selection dynamics may act, without introducing new physics and without exceeding the scope of Paper A.

B4.1 — Selection Rate as an Operational Quantity

For two record sectors i and j , define the **inter-sector selection time** τ_{ij} as the minimal duration required, in an individual realization, for the system’s accessible behavior to become operationally indistinguishable from confinement to sector i rather than j , within experimental tolerance.

The corresponding selection rate is

$$\lambda_{ij} \equiv 1/\tau_{ij}.$$

This rate is operationally measurable. It characterizes how rapidly exclusion is enforced between competing record sectors.

B4.2 — Requirements on a Universal Rate Limiter

Any candidate universal rate limiter on selection must satisfy three constraints.

Universality. The bound must apply across all macroscopic records, independent of composition or charge.

Context Independence. The bound must not depend on observer intervention, measurement choice, or apparatus-specific tuning.

Discriminatory Relevance. The bound must couple directly to the physical features that distinguish record sectors.

These three constraints do not uniquely determine a limiter, but they strongly restrict admissible candidates.

B4.3 — Gravity as a Candidate Universal Limiter (Hypothesis)

Among known interactions, gravity satisfies all three requirements above.

It is universal. It is unscreenable. It is directly sensitive to mass-energy configuration, which is exactly what distinguishes macroscopic records.

The hypothesis: gravity provides a universal upper bound on selection rates.

This is an empirical claim about known interactions. It is not a proof of uniqueness. It does not assert that gravity *causes* selection.

B4.4 — Gravitational Distinguishability of Record Sectors

The gravitational self-energy distinguishability ΔE_G between record sectors i and j is defined in Paper A (Definition D15, Appendix C). It measures the Newtonian self-energy of the difference mass distribution between two record sectors and is zero when the sectors are gravitationally indistinguishable.

The definition, explicit integral form, and positivity proof are given in Paper A and are not repeated here.

B4.5 — Rate Inequality

The gravity-limited selection bound (Paper A, Postulate G) states

$$\lambda_{ij} \leq \Delta E_G[i, j] / \hbar.$$

Paper A introduces this bound as a physical constraint. Here it is elevated to a candidate universal rate limiter by demonstrating that gravity satisfies the universality, context-independence, and discriminatory relevance requirements of B4.2 — requirements that were not articulated in Paper A.

The experimental consequences of this elevation are developed in B5.

The bound is limiting, not exact. Selection may be slower. Selection cannot be faster without invoking a coupling stronger than gravity to mass-energy distinguishability.

B4.6 — Null Case (Conditional)

Under the gravity-limited hypothesis, if two record sectors are gravitationally indistinguishable — $\Delta E_G[i, j] = 0$ — then the gravity-constrained contribution to the selection rate vanishes:

$$\lambda_{ij} = 0.$$

If no alternative limiter satisfying the requirements of B4.2 applies, such superpositions persist indefinitely. If a non-gravitational mechanism consistent with B4.2 exists, it would supply an independent rate bound not covered by the present hypothesis.

This is a testable conditional prediction of the gravity-limited framework.

B4.7 — Consistency with Prior Results

The gravity-limited bound is consistent with all earlier sections.

It applies only after operational irreversibility (B3.1). It constrains rates, not outcome statistics (B3.6). It preserves ensemble linearity (B3.5). It does not explain decoherence or branching (Paper A).

Gravity here functions solely as a rate limiter, not a causal mechanism.

B4.8 — Falsifiers (Rate-Level)

The gravity-limited hypothesis is falsified if any of the following are observed.

F_G1. Selection occurs with $\lambda_{ij} > \Delta E_G/\hbar$.

F_G2. Selection occurs between records with $\Delta E_G = 0$ in the absence of any alternative limiter satisfying B4.2.

F_G3. Selection rates scale universally with non-gravitational parameters across macroscopic records.

Failure here invalidates the limiter hypothesis only. It does not invalidate selection as defined in this paper, nor irreversibility as defined in Paper A.

B4.9 — Closing

If selection occurs, it is constrained by physical limits on how rapidly alternatives can be distinguished.

Definiteness cannot emerge arbitrarily fast. It can emerge no faster than a universal interaction can discriminate between competing records.

B5 — Experimental Regimes and Discriminating Tests

This section translates the structural and rate constraints of B1–B4 into experimentally discriminable regimes.

The aim is not parameter fitting. The aim is to specify what observations would count as confirmation, survival, or falsification of selection as defined in this paper.

B5.1 — Principle of Test Construction

Experiments testing selection must satisfy three criteria.

Post-Irreversibility Regime. Decoherence and loss of recoverability must already be established (Paper A). Tests conducted while the system remains within $K\varepsilon(\mathcal{O})$ are irrelevant to selection.

Trajectory Sensitivity. The experiment must probe single-run behavior or trajectory-level signatures, not ensemble averages alone.

Rate Sensitivity. The experiment must be capable of resolving timescales comparable to the predicted selection rate λ_{ij}^{-1} .

Only experiments satisfying all three can meaningfully constrain selection dynamics.

Test Map Summary.

The five tests below are ordered by what they target, not by experimental accessibility. Each test is independently meaningful.

BT1 — Order-of-Operations (B5.5).

Target: selection itself. Falsifies: selection as defined in B1.4. Method: continuously tune decoherence and check whether selection signatures appear before the system exits $K\varepsilon(\mathcal{O})$. Platform: superconducting qubits with tunable coupling to measurement cavity.

BT2 — Active Selection Signature (B5.2).

Target: existence of selection. Falsifies: selection is present in the tested regime. Method: compare single-trajectory statistics against all linear Lindblad models fitted to the same decoherence data. Observable: telegraph noise or diffusive wandering inconsistent with any CPTP unraveling. Platform: continuously monitored superconducting qubits or trapped ions with fluorescence readout.

BT3 — Null-Rate Regime (B5.3).

Target: gravity-limited hypothesis. Falsifies: F_G2 . Method: prepare decohered superpositions with $\Delta E_G = 0$ and monitor for selection. Observable: persistent multiplicity (no single-sector stabilization) versus rapid selection. Platform: nitrogen-vacancy centers in diamond, nuclear spin states with identical mass distributions.

BT4 — Rate-Bound Regime (B5.4).

Target: gravity-limited hypothesis. Falsifies: F_G1 . Method: create spatial superpositions of mesoscopic masses, measure selection timescale, compare against $\tau_min = \hbar/\Delta E_G$. Observable: selection faster or slower than bound. Platform: levitated nanoparticles (tungsten, $R \approx 100$ nm, $\tau_min \sim 1$ -10 s) in cryogenic vacuum.

BT5 — Born Boundary Condition (B3.6).

Target: Born-consistency of selection. Falsifies: the Born-consistent model class. Method: large ensembles of identically prepared, fully decohered

systems with single-shot readout. Observable: realized-sector frequencies deviating from $\{p_i\}$ beyond statistical tolerance. Platform: trapped-ion arrays or superconducting qubit arrays.

B5.2 — Signature of Active Selection

Selection is operationally distinct from decoherence.

An **active selection signature** is any trajectory-level behavior, occurring after operational irreversibility, that:

- (i) cannot be reproduced by any system-local linear CPTP evolution consistent with the independently characterized decoherence dynamics of the system, and
- (ii) enforces persistent confinement to a single record sector under all admissible system-local controls.

Examples of admissible signatures include:

- Irreversible loss of interference revival capacity despite full system-only control.
- Stochastic stabilization of record-sector behavior inconsistent with linear Lindblad dynamics fitted to the same decoherence data.
- Telegraph-like trajectory behavior resolving into a single sector with no subsequent switching within accessible timescales.

Absence of such signatures implies absence of selection in the tested regime.

B5.3 — Null-Rate Regime (Gravitational Degeneracy)

Consider record sectors that are operationally decohered but gravitationally indistinguishable: $\Delta E_G = 0$.

Under the gravity-limited hypothesis, the gravity-constrained contribution to the selection rate vanishes. The argument therefore predicts one of two outcomes:

Persistent multiplicity. No selection signatures appear within experimentally accessible timescales.

Non-gravitational selection. Selection occurs at a slower rate governed by an alternative limiter satisfying the requirements of B4.2.

Concrete example: a nitrogen-vacancy (NV) center in diamond.

Prepare the NV centre in a superposition of spin states $|m_s = +1\rangle$ and $|m_s = -1\rangle$. These states have identical mass distributions ($\Delta E_G = 0$) but are operationally distinguishable via microwave spectroscopy.

After environmental decoherence has suppressed spin coherence, the reduced state is

$$\rho = \frac{1}{2}|+1\rangle\langle+1| + \frac{1}{2}|-1\rangle\langle-1|.$$

Under the gravity-limited hypothesis, no gravitational contribution to selection exists.

If single-trajectory monitoring reveals rapid stabilization to one spin state inconsistent with any Lindblad model of the decoherence dynamics, the gravity-limited hypothesis is falsified.

Observation of rapid selection in this regime falsifies the gravity-limited hypothesis. Persistent multiplicity is consistent with it.

B5.4 — Rate-Bound Regime (Macroscopic Distinguishability)

For record sectors with significant gravitational distinguishability $\Delta E_G \gg \hbar/T$ — where T is the duration over which the experiment maintains sensitivity to trajectory-level behavior — the gravity-limited hypothesis predicts an upper bound:

$$\tau_{ij} \geq \hbar/\Delta E_G.$$

Experiments in this regime can test whether observed selection times respect the bound (hypothesis survives), approach the bound (gravity-limited selection likely active), or violate the bound (hypothesis falsified).

Candidate systems and estimates.

A tungsten nanoparticle ($R = 100 \text{ nm}$, $\rho \approx 19 \text{ g/cm}^3$) in spatial superposition with separation $\Delta x \sim R$ yields $\Delta E_G \sim Gm^2/R$, giving $\tau_{\min} = \hbar/\Delta E_G \sim 1\text{-}10$ seconds.

This is within reach of cryogenic levitation experiments maintaining coherence for seconds (cf. Paper A, Appendix D).

For silica ($\rho \approx 2 \text{ g/cm}^3$), $\tau_{\min} \sim 10^2\text{-}10^3$ seconds, at the boundary of current coherence times. High-density materials are strongly preferred for near-term tests.

Observation of selection with $\tau < \tau_{\min}$ falsifies the gravity-limited hypothesis (F_{G1}). Absence of selection within accessible timescales is consistent with the hypothesis but does not confirm it.

B5.5 — Order-of-Operations Test

Selection must not precede irreversibility.

Experiments that continuously tune environmental coupling can test whether selection signatures appear only after the system exits the recoverable set $K\varepsilon(\mathcal{O})$.

Specifically: if trajectory-level confinement to a single record sector is observed while the system remains within $K\varepsilon(\mathcal{O})$ as defined in Paper A (Definition D12), then selection as defined in B1.4 is falsified.

This test targets selection itself, not merely the gravity-limited hypothesis.

B5.6 — Outcome Classification

Experimental outcomes partition cleanly:

- **No selection observed.** Selection absent in the tested regime.
- **Selection observed, rate indeterminate.** Selection present; gravity-limited hypothesis neither confirmed nor falsified.
- **Selection observed within bound.** Selection present and consistent with the gravity-limited hypothesis.

- **Selection observed faster than bound.** Gravity-limited hypothesis falsified.
- **Selection observed in null-rate regime without alternative limiter.** Gravity-limited hypothesis falsified or incomplete.

No outcome retroactively rescues the hypothesis.

B5.7 — Scope Closure

This paper establishes what selection must be if it exists, what it must cost, how fast it may occur, and how it can be falsified.

It does not determine whether selection actually occurs in nature.

That question is empirical.

B6 — Conclusions and Program Status

This paper has treated selection as a physical exclusion process constrained by irreversibility, control limits, and rate bounds, without invoking interpretation, agency, or collapse mechanisms.

The results can be summarised as follows.

B6.1 — What Has Been Established

If selection exists, it must satisfy all of the following.

1. Post-Irreversibility Constraint. Selection cannot act before operational irreversibility is established. Any exclusion prior to exit from $K\epsilon(\mathcal{O})$ falsifies selection as defined here.

2. Trajectory-Level Character. Selection must act at the level of individual realizations while preserving linear ensemble evolution.

3. Record-Algebra Locality. Selection may act only on degrees of freedom that distinguish record sectors and may not reintroduce interference.

4. Absorbing Dynamics. Once a record sector is realized, sector membership is fixed under subsequent selection dynamics.

5. Contractivity of Multiplicity. Selection must monotonically reduce record-sector multiplicity along individual trajectories.

6. Cost and Rate Constraints. Selection requires physical resources and cannot occur arbitrarily fast.

7. Universal Rate Limiter (Hypothesis). Gravity provides a candidate universal upper bound on selection rates, expressible through gravitational distinguishability ΔE_G , and is falsifiable by explicit rate tests.

Each condition is necessary. None is assumed to be sufficient.

B6.2 — What Has Not Been Assumed

This paper has not:

- Assumed that selection must occur.
- Derived outcome statistics or the Born rule.
- Specified a concrete dynamical generator.
- Invoked observers, consciousness, or epistemic update.
- Claimed gravity *causes* selection.
- Extended irreversibility beyond what is established in Paper A.

Failure of any hypothesis in this paper leaves the foundations of Paper A intact.

B6.3 — Status of the Gravity-Limited Hypothesis

The gravity-limited hypothesis introduced in B4 is empirically motivated, dimensionally consistent, and experimentally falsifiable.

It stands or falls entirely on observation. Its failure would constrain the space of admissible selection mechanisms, not rescue them.

B6.4 — Programmatic Closure

Together with Paper A, this work completes the physics-level characterization of selection.

Paper A establishes irreversibility without definiteness.

Paper B establishes definiteness as costly, rate-limited exclusion, if it exists.

**No further progress on selection can be made by argument alone.
The remaining uncertainty is empirical.**

B6.5 — Forward Dependency

If selection is absent or constrained, the remaining question is not about definiteness. It is about structure.

How does behavior unfold within a single realized record sector under irreversible constraint?

That question concerns control under irreversibility, not the emergence of definiteness. It is addressed in Paper C, where agency is treated as constrained dynamics downstream of the physics established here.

End of Paper B. Selection as Irreversible Exclusion. Locked · Execution-Complete.

Paper C

Agency as Constrained Control

Depends on Papers A and B

You are an agent. You make choices. You maintain yourself against decay. You navigate a space of possibilities that narrows with every irreversible step. You have a budget that depletes. You face drift that never stops.

And somewhere ahead of you, invisible but real, is a boundary beyond which no choice you make can save you.

Everything you just read is geometry. Not philosophy. Not metaphor. Geometry — measurable, computable, falsifiable.

This paper strips the philosophy out of agency and replaces it with a number. The number measures the fraction of survivable states you can still reach from where you stand.

That number is more honest than any definition philosophy has ever produced, because it does not care about your intentions. It cares about your position in the state space and the size of your control set.

The rest is arithmetic.

Abstract.

This paper develops a control-theoretic account of agency under irreversible physics. Your agency, measured as a number.

Building on Papers A and B, agency is defined as a geometric property: the fraction of survivable states reachable from where you currently stand, using whatever control you have, within the record sector you actually occupy.

No new physical assumptions are introduced.

The results: if you stop maintaining, agency decays. At the no-return boundary, agency hits zero. High-variance or misaligned strategies waste

the budget faster than steady ones. None of this is surprising. All of it is now proven.

Control fatigue, noise, coupling, and exit are defined as consequences of constrained reachability rather than psychological or normative phenomena. The paper establishes necessary conditions for persistence of controlled behavior under irreversibility and provides falsifiers for the control framework.

What remains unresolved is empirical — and that is exactly as it should be. The geometry is proven. Which real systems instantiate it is nature’s answer.

C0 — Scope

C0.1 — Dependency Statement

This paper depends explicitly and exclusively on the physical results established in **Paper A** and **Paper B**.

It assumes as given:

- Irreversibility as loss of reachability under admissible control (Paper A).
- The existence of no-return surfaces induced by bounded capacity (Paper A).
- Selection, if it exists, as a costly, rate-limited exclusion process acting after irreversibility (Paper B).

Paper C requires only that selection produces confinement to a single record sector. It does not depend on the mechanism, rate, or statistics of selection.

No physical construct is redefined or rederived here.

Cross-reference note. The viability kernel $\text{Viab}(\mathbf{R})$ used throughout this paper (Paper A, Definition D7) corresponds, in the quantum setting, to the recoverable set $\text{K}\epsilon(\mathbf{O})$ (Paper A, Definition D12). Paper C operates entirely within a single realized record sector, so the relevant constraint set $\mathbf{R}$ is the

set of states accessible to the system after selection, not the full quantum state space.

C0.2 — Purpose

Paper C addresses a question that is not physical in origin, but structural in consequence:

Given irreversible physics and costly definiteness, how can controlled behavior persist within a single realized record sector?

Agency is treated not as intention, belief, or choice, but as a control property — a number you can compute of a system evolving under irreversible constraints.

C0.3 — Hard Non-Claims

This paper does not:

- Introduce new physical laws.
- Modify or reinterpret quantum mechanics.
- Explain why selection occurs.
- Invoke psychology, motivation, ethics, or meaning.
- Provide prescriptions or normative guidance.

Failure of Paper C does not invalidate Papers A or B.

C1 — Agency as a Geometric Control Quantity

C1.1 — Definition of Agency

Within a single realized record sector, define **agency** as the fraction of the viability kernel reachable from the current state under admissible control.

Let $x(t)$ denote the system state confined to a realized record sector. Let $\text{Viab}(R)$ be the viability kernel defined in Paper A relative to admissible

controls, and let $\text{Reach}(x)$ be the set of states reachable from x under those controls. Define

$$\mathcal{M}(x) \equiv \mu(\text{Reach}(x) \cap \text{Viab}(R)) / \mu(\text{Viab}(R)),$$

where μ is the natural volume measure induced by the state-space metric and $\text{Viab}(R)$ has finite positive measure.

The normalization ensures $\mathcal{M} \in [0, 1]$. $\mathcal{M} = 1$ when the entire viability kernel is reachable. $\mathcal{M} = 0$ at the no-return surface where no viable future remains.

That is agency — measured.

Monotonicity and regularity assumptions. The measure μ is monotone with respect to set inclusion: if $S_1 \subseteq S_2$ then $\mu(S_1) \leq \mu(S_2)$. The analysis assumes $\text{Reach}(x)$ varies continuously with x in the Hausdorff metric on compact subsets of the state space, ensuring $\mathcal{M}$ is continuous. These are standard regularity conditions in viability theory (Aubin, 1991) and are not additional physical assumptions.

C1.2 — Control Authority

Let admissible controls $u(t) \in U$ be bounded by physical and energetic constraints. Control authority is determined by three quantities:

Bandwidth. The maximal rate at which control can counteract irreversible drift.

Reachability. The remaining volume of $\text{Viab}(R)$ accessible from $x(t)$.

Slack. The time-to-boundary from $x(t)$ under zero control. (Defined formally in C8.1.)

Limit condition. By definition of $\Sigma_{\text{NR}} = \partial \text{Viab}(R)$ and continuity of $\mu(\text{Reach}(\cdot) \cap \text{Viab}(R))$ as a function of state (guaranteed by the Hausdorff regularity assumption above):

$$\lim_{x \rightarrow \Sigma_{\text{NR}}} \mathcal{M}(x) = 0.$$

At the boundary, only a single future trajectory remains.

C2 — Drift as a Consequence of Irreversibility

C2.1 — Irreversible Drift

For open systems with nonzero irreversible drift, ordered states decay toward loss of structure in the absence of sustained control.

This follows directly from bounded control capacity and the operator-horizon result of Paper A (Theorem T2). It is not an independent axiom.

You have felt this. The garden untended. The body unrested. The relationship neglected. Drift is what reality does to structure when you stop maintaining it.

C2.2 — Baseline Dynamics

Absent control ($u = 0$), the system evolves as

$$dx/dt = f(x),$$

where $f(x)$ is the irreversible drift field pointing toward an attractor of structural loss (equilibrium, failure, or saturation).

The scalar decay model of Paper A ($dx/dt = -ax + u$) is a special case of this general form.

Proposition C2.1 (Agency decay under drift).

This is the mathematical expression of what you already know: everything falls apart without maintenance.

Let $x(t)$ evolve under $dx/dt = f(x) + u$ with $u(t) \in U$, and suppose the drift field f points inward toward an attractor $x \notin Viab(R)$. If $|f(x)| \geq a\|x - x^*\|$ for some $a > 0$ (linear lower bound on drift), and if $\mu(\text{Reach}(x) \cap Viab(R))$ is Lipschitz in x with constant L , then along any trajectory with $|u(t)| \leq u_{\max}$:

$$dM/dt \leq L(u_{\max} - a\|x - x^*\|) / \mu(Viab(R)).$$

When the system is far from the attractor ($\|x - x^*\| > u_{\max}/a$), the right-hand side is strictly negative: agency decreases regardless of control.

This reproduces the operator horizon result of Paper A (Theorem T2) in the agency framework and quantifies the rate of agency loss beyond the horizon.

Proof sketch.

$d\mathcal{M}/dt = (d/dt)[\mu(\text{Reach}(x) \cap \text{Viab}(R))] / \mu(\text{Viab}(R))$. By Lipschitz continuity, $|\Delta\mu| \leq L|\Delta x|$. The state velocity is $|dx/dt| = |f(x) + u| \leq |f(x)| + |u|$.

The drift pushes toward x (*reducing Reach*), while control pushes away (*expanding it*). Net rate: $d\mathcal{M}/dt \leq L(|u| - |f(x)|) / \mu(\text{Viab}(R)) \leq L(u_{\max} - a\|x - x^*\|) / \mu(\text{Viab}(R))$.

C3 — Necessary Conditions for Agency Preservation

C3.1 — Continuous Control Cost

For open systems with $f(x) \neq 0$ away from fixed points, maintaining distance from Σ_{NR} requires continuous expenditure of control effort.

Except at exact fixed points of f , no finite intervention permanently arrests drift. Such fixed points, if they exist, may themselves lie outside $\text{Viab}(R)$ or require sustained control to reach. Their existence does not generally provide a cost-free maintenance strategy.

There is no rest. There is only steadier maintenance.

C3.2 — Variance-Conditioned Control Effectiveness

Proposition C3.2 (conditional).

For admissible control systems in which instantaneous control cost $c(u)$ is convex in $|u|$, low-variance control trajectories preserve $\mathcal{M}(x)$ more effectively than high-variance or impulsive strategies with the same mean control effort.

Proof sketch. For convex c , Jensen's inequality gives $\mathbb{E}[c(u)] \geq c(\mathbb{E}[u])$. Variable control with fixed mean effort therefore incurs greater cumulative cost than constant control at the mean level, depleting the control budget $B(t)$ (defined in C5.1) more rapidly and thereby reducing reachable viability volume.

Steady wins. Bursts cost more. The mathematics says exactly what every athlete, every operator, every long-distance runner already knows.

Corollary C3.1a (Maintenance condition).

For the scalar system $dx/dt = -ax + u$ with $a > 0$ and $u \in [0, u_{\max}]$, the agency $\mathcal{M}(x)$ is maintained ($d\mathcal{M}/dt = 0$) if and only if $u = ax$ — control exactly balances drift.

This requires $x \leq u_{\max}/a = x_h$ (the operator horizon). For $x > x_h$, no admissible control can maintain $\mathcal{M}$, and $d\mathcal{M}/dt < 0$ strictly.

The maintenance condition is the agency-framework restatement of Paper A's Theorem T2: the horizon is the boundary between maintainable and inevitably decaying agency.

C4 — No-Return Geometry Within a Realized Sector

C4.1 — Horizon Geometry

The operator horizon from Paper A (Theorem T2, Definition D9) applies strictly within a realized record sector.

Crossing this boundary removes states from $\text{Viab}(\mathbf{R})$.

C4.2 — Ruin as Absorbing State

Ruin is defined as

$\mathbf{x} \notin \text{Viab}(\mathbf{R})$.

Once this occurs, recovery is impossible under admissible control. Ruin is a geometric property of state space, not a subjective condition.

The word *ruin* is precise here. It does not mean failure. It does not mean despair. It means: the set of admissible trajectories that return you to viability is empty. You cannot get back from where you stand.

Worked Example: No-Return Geometry in a 2D Linear System

Consider a two-dimensional system with state $x = (x_1, x_2) \in \mathbb{R}^2$, drift $f(x) = (-a_1x_1, -a_2x_2)$ with $a_1, a_2 > 0$, and control $u = (u_1, u_2) \in [0, u_1^{\max}] \times [0, u_2^{\max}]$. The constraint set is $R = \{x : x_1 \geq 0, x_2 \geq 0\}$.

The viability kernel is the rectangle

$\text{Viab}(R) = [0, x_{1h}] \times [0, x_{2h}]$, where $x_{ih} = u_i^{\max} / a_i$ is the per-axis operator horizon.

The no-return surface Σ_{NR} is the boundary of this rectangle. Any state with $x_1 > x_{1h}$ or $x_2 > x_{2h}$ is in the capture basin and will be driven to the boundary regardless of control.

Agency computation.

For a state $x = (x_1, x_2)$ inside $\text{Viab}(R)$, the reachable set within $\text{Viab}(R)$ is

$\text{Reach}(x) \cap \text{Viab}(R) = [0, \min(x_1 + u_1^{\max}/a_1, x_{1h})] \times [0, \min(x_2 + u_2^{\max}/a_2, x_{2h})]$ (for steady-state reachability).

The normalized agency is $\mathcal{M}(x) = \mu(\text{Reach}(x) \cap \text{Viab}(R)) / \mu(\text{Viab}(R))$.

At the origin, $\mathcal{M} = (u_1^{\max}/a_1)(u_2^{\max}/a_2) / (x_{1h} \cdot x_{2h}) = 1$. The full viability kernel is reachable.

At the corner (x_{1h}, x_{2h}) , Reach shrinks to the single point and $\mathcal{M} \rightarrow 0$.

The example illustrates three features. The no-return surface is axis-separable in the linear case. Agency varies continuously from 1 to 0 across the viability kernel. Position within $\text{Viab}(R)$ determines how much future flexibility remains, independent of the system's history.

You can compute this for any system that admits the geometry. That is the entire claim.

C5 — Control Budgets and Fatigue

C5.1 — Control Budget

Define the **control budget**

$$\mathbf{B}(t) = \mathbf{B}_0 - \int_0^t c(\mathbf{u}(s)) \, ds,$$

where $c(\mathbf{u})$ is the instantaneous control cost and $\mathbf{B}_0 > 0$ is the initial budget. Admissible control requires $\mathbf{B}(t) \geq 0$.

The budget is what you have. The cost is what you spend. When the budget reaches zero, control ceases. There is no controlled behavior on an empty tank.

C5.2 — Control Fatigue

Control fatigue occurs as $\mathbf{B}(t) \rightarrow 0$. High-frequency or high-magnitude control accelerates depletion of $\mathbf{B}(t)$, reducing $\mathcal{M}(\mathbf{x})$.

By Proposition C3.2, impulsive strategies with convex cost deplete the budget strictly faster than steady control at the same mean effort.

Theorem C5.1 (Survival time bound).

Let $\mathbf{x}(t)$ evolve under $d\mathbf{x}/dt = \mathbf{f}(\mathbf{x}) + \mathbf{u}$ with $\mathbf{u}(t) \in U$ and control cost $c(\mathbf{u}) \geq c_{\min} > 0$ for all $\mathbf{u} \neq 0$. Let $\mathbf{B}(t) = \mathbf{B}_0 - \int_0^t c(\mathbf{u}(s)) \, ds$ be the control budget.

Define the survival time T as the first time at which either $\mathbf{B}(T) = 0$ or $\mathbf{x}(T) \notin \text{Viab}(\mathbf{R})$. Then

$$T \leq \mathbf{B}_0 / c_{\min}.$$

Proof. Since $c(\mathbf{u}) \geq c_{\min}$ for any nonzero control, the budget depletes at rate $d\mathbf{B}/dt = -c(\mathbf{u}) \leq -c_{\min}$ whenever the system is actively controlled. If the system requires continuous control to remain in $\text{Viab}(\mathbf{R})$ (i.e., $\mathbf{f}(\mathbf{x})$ points outward at $\mathbf{x}$ for all $\mathbf{x}$ on the trajectory), then control must be nonzero for the entire survival period, giving $\mathbf{B}(T) = \mathbf{B}_0 - \int_0^T c(\mathbf{u}) \, ds \leq \mathbf{B}_0 - c_{\min} \cdot T$. Setting $\mathbf{B}(T) = 0$ yields $T \leq \mathbf{B}_0/c_{\min}$.

The bound is tight for constant minimal-cost control.

It establishes that finite budgets imply finite survival. No system with bounded resources can maintain agency indefinitely against persistent drift.

The bound does not depend on the drift rate a , only on the control cost floor. Faster drift depletes the budget faster (higher u needed), but the absolute bound is set by the budget-to-cost ratio.

Worked example (scalar system).

For $dx/dt = -ax + u$ with $a = 1$, $u_{\max} = 2$, $c(u) = u$ (linear cost), $B_0 = 10$, and initial state $x_0 = 1.5$ (inside $\text{Viab}(R) = [0, 2]$):

- Maintaining $x = 1.5$ requires $u = ax = 1.5$, costing $c = 1.5$ per unit time.
- Survival time: $T = B_0/c = 10/1.5 \approx 6.67$ time units.
- After budget depletion, $u = 0$ and x decays exponentially toward 0. The system crosses into ruin when it can no longer reach any target within $\text{Viab}(R)$.

C6 — Noise and Silence

C6.1 — Noise

Noise is exogenous or stochastic input to the system dynamics that is not under admissible control and that consumes control bandwidth without increasing $\text{Reach}(x) \cap \text{Viab}(R)$.

Formally, noise is any perturbation $\xi(t)$ added to the drift field, $f(x) \rightarrow f(x) + \xi(t)$, where ξ is not an element of the admissible control set U .

Proposition C6.1 (Noise-induced agency decay).

Let the system evolve under $dx/dt = f(x) + u + \xi(t)$ where ξ is a zero-mean stochastic forcing with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[|\xi|^2] = \sigma^2$. If the system must expend additional control Δu to compensate for ξ , then the effective budget depletion rate increases:

$dB/dt = -c(u + \Delta u) \leq -c(u) - \alpha\sigma^2$, for some $\alpha > 0$ depending on the convexity of c .

Consequently, noise reduces survival time:

$$T_{\text{noisy}} \leq B_0 / (c_{\text{min}} + \alpha \sigma^2) < T_{\text{quiet}}.$$

Noise taxes the control budget without expanding reachable viability volume. Every distraction, every interruption, every forced response to something that wasn't going to threaten you anyway — pure tax. Pure depletion with no expansion of what you can reach.

C6.2 — Silence

Withholding response ($u(t) = 0$) is an admissible control action.

When the drift field $f(x)$ is slow or favorable (directed away from Σ_{NR}), silence preserves control budget at no agency cost.

This is not inaction in the colloquial sense. It is the optimal control policy when the marginal agency cost of intervention exceeds the agency cost of drift.

Formally, silence is preferred when

$$c(u) / |\partial \mathcal{M} / \partial u| > |d\mathcal{M}/dt|_{\{u=0\}},$$

i.e., when the budget cost per unit of agency preservation exceeds the drift-induced agency loss rate.

In noise-dominated regimes, silence may also prevent noise-amplifying feedback loops in which control effort introduces additional disturbance.

The mathematics confirms what discipline already knew. Sometimes the best move is no move.

C7 — Coupling and Rescue

C7.1 — Coupled Systems and Agency Transfer

When systems are coupled, their drift fields combine and control capacities load jointly.

Agency transfer occurs when, under coupled dynamics,

$$d\mathcal{M}_A/dt > 0 \text{ while } d\mathcal{M}_B/dt < 0,$$

indicating expansion of reachable viability for system A at the expense of system B.

The total agency of the coupled system is not conserved.

This is not an assumption. It follows from the geometry of $\text{Viab}(R)$ under coupling.

C7.2 — Rescue Instability (Sufficient Condition)

Rescue is coupling a stabilized system A to a divergent system B to offset B's drift using A's control capacity.

Let $|\cdot|$ denote the norm induced by the coupled dynamics on the joint state space. A **sufficient condition** for joint loss of viability is

$$|f_A| + |f_B| > |u_A|_{\max} + |u_B|_{\max}.$$

Under this condition, the total drift magnitude exceeds the total available control, and the coupled system approaches Σ_{NR} faster than either system in isolation.

This is sufficient, not necessary. Directional alignment of drift and control fields may permit stability even when this scalar inequality holds.

Non-conservation: two examples.

(1) *Cooperative coupling.* Two scalar systems with drift $a = 1$, $u_{\max} = 1$ each, coupled so that each contributes control to the other. If the coupling allows total control capacity to be shared: effective $u_{\max}$ per system = 2, x_h doubles for both. $\mathcal{M}_A + \mathcal{M}_B$ increases. Coupling creates agency.

(2) *Parasitic coupling.* System A ($a = 1$, $u_{\max} = 2$) is coupled to system B ($a = 3$, $u_{\max} = 0$). B diverts A's control capacity: effective $u_{\max}$ for A drops to 1, while B still cannot sustain itself ($3 > 1$). Both systems lose agency: $\mathcal{M}_A + \mathcal{M}_B$ decreases. Coupling destroys agency.

These examples demonstrate that agency transfer is not zero-sum. The coupling topology and the relative drift-to-control ratios determine whether joint agency expands, contracts, or redistributes. No conservation law governs total agency.

C8 — Slack and Robustness

C8.1 — Slack

Slack is the minimum time to reach Σ_{NR} under zero control:

$$s(\mathbf{x}) \equiv \inf \{ t \geq 0 : \phi_t(\mathbf{x}) \in \Sigma_{\text{NR}} \},$$

where ϕ_t is the uncontrolled flow generated by $f(\mathbf{x})$.

Slack measures time-to-boundary, not Euclidean distance, and is the operationally relevant quantity for assessing control margin. Greater slack increases the time window available for corrective control and absorbs perturbations.

Proposition C8.1 (Scalar Slack-Agency Correspondence).

For the scalar system $dx/dt = -ax + u$ with $u \in [0, u_{\text{max}}]$, the slack $s(x) = x/a$ (time to reach $x = 0$ under zero control) and the horizon $x_h = u_{\text{max}}/a$ satisfy:

$\mathcal{M}(\mathbf{x})$ is monotonically increasing in $s(\mathbf{x})$ for $\mathbf{x} \in [0, x_h]$.

Greater slack implies greater agency.

You have felt this — the difference between having three months of savings and having three days. The same drift, the same income, the same expenses. The slack is the entire difference between agency and emergency.

At $s = 0$ (boundary), $\mathcal{M} = 0$. At $s = x_h/a$ (maximal slack at origin), $\mathcal{M} = 1$. Slack is the operationally measurable proxy for agency in systems where direct computation of $\text{Reach}(\mathbf{x}) \cap \text{Viab}(\mathbf{R})$ is intractable.

In higher dimensions or systems with non-convex constraints, slack is necessary but not sufficient for agency. A state may have large time-to-boundary under zero control yet be surrounded by regions from which no viable trajectory exists — geometric cul-de-sacs.

C8.2 — Redundancy

A system has **redundancy** $r \geq 1$ with respect to a target state $x \in \text{Viab}(R)$ if there exist at least r distinct admissible control trajectories reaching x from the current state while remaining within $\text{Viab}(R)$.

Redundancy reduces sensitivity of viable trajectories to perturbations in $f(x)$ and $u(t)$. Higher redundancy increases robustness at the cost of efficiency, since maintaining multiple viable pathways consumes control capacity that could otherwise extend reachability.

C9 — Exit as a Control Outcome

C9.1 — Withdrawal

When $\mathcal{M}(x(t))$ decreases monotonically under all admissible controls in a coupled system, decoupling preserves more reachable viability volume than continued coupling.

This holds when the coupled drift exceeds the joint control capacity (C7.2), so that decoupling removes the excess drift load.

Exit is therefore a control outcome implied by reachability geometry, not a prescription.

C9.2 — Agency-Dissipative Environments

An environment is **agency-dissipative** if, for all admissible controls,

$$d\mathcal{M}/dt < 0.$$

Persistence in such an environment strictly reduces reachable viability volume. This is a geometric characterization, not a recommendation.

Proposition C9.1 (Decoupling condition).

Let systems A and B be coupled with joint dynamics. Decoupling is agency-preserving for A if and only if the coupled drift acting on A exceeds A's isolated drift:

$$|f_{\text{coupled},A}(x)| > |f_A(x)|.$$

Decoupling is preferred when the coupling increases the effective drift on A beyond what A experiences in isolation.

You know this. The relationship that costs more energy to maintain than it provides is a relationship that increases your drift. The mathematics says: leave. Not because leaving is morally right. Because the geometry of your viability kernel contracts while you stay.

This is a necessary and sufficient condition for decoupling to instantaneously increase $d\mathcal{M}_A/dt$. It does not account for future recoupling opportunities or transient effects.

C10 — Falsifiability and Closure

C10.1 — Falsifiers

Paper C is falsified if any of the following are observed.

F_C1. $\mathcal{M}(x)$ increases without corresponding control expenditure (violates C5.1).

F_C2. Irreversible loss of reachability is reversed without external intervention violating the admissibility constraints of Paper A.

F_C3. Stable control persists beyond Σ_{NR} under admissible control (violates C4.2).

F_C4 (Free lunch). A system maintains $\mathcal{M}(x) > 0$ indefinitely with B_0 finite and no external resource input, in the presence of persistent nonzero drift. Violates Theorem C5.1.

F_C5 (Resurrection). A system recovers $\mathcal{M}(x) > 0$ after reaching $\mathcal{M} = 0$ (ruin) without external intervention that violates the admissibility constraints of Paper A. Violates C4.2.

C10.2 — Closure

Paper C introduces no new physics. It applies the irreversible and selective constraints of Papers A and B to controlled dynamics within a realized record sector.

Identification of concrete systems instantiating these constraints — biological, engineered, or otherwise — is treated separately. No extension of this argument is possible without new physical assumptions.

Experimental Instantiation

The definitions and propositions of Paper C are abstract control-theoretic structures. They become empirically meaningful when instantiated in concrete systems.

Two candidate systems are outlined below — one biological, one engineered — to demonstrate that the argument makes operationally testable predictions.

System 1: Bacterial chemotaxis

A bacterium navigating a nutrient gradient instantiates the scalar control model.

- **State.** Nutrient concentration at cell location.
- **Drift.** Diffusion-driven nutrient depletion ($a > 0$).
- **Control.** Flagellar motor switching ($u \in \{\text{run}, \text{tumble}\}$).
- **Budget.** Metabolic energy store (ATP).
- **Viability kernel.** Nutrient concentrations supporting growth.
- **No-return surface.** Starvation threshold below which metabolic shutdown is irreversible.

Testable prediction. Survival time scales with initial metabolic reserve divided by maintenance metabolic rate (Theorem C5.1).

Noise. Brownian rotational diffusion acts as stochastic forcing ξ , taxing the control budget (Proposition C6.1). **Observable:** mean survival time decreases with increasing environmental noise, controlling for nutrient availability.

Falsifier (C10.1, F_C1). If a non-motile mutant ($u_{\max} = 0$) maintains its position in the gradient without external intervention, agency as defined here is falsified — reach grew without control expenditure.

Falsifier (C10.1, F_C4). If a motile cell with finite ATP reserve persists indefinitely in a persistent nutrient gradient flow, the survival time bound (Theorem C5.1) is falsified.

The experiments are within the capability of standard microfluidics laboratories. Every construct in Paper C — drift, control, horizon, budget, fatigue, noise, slack, ruin — maps to a measurable variable in this system.

System 2: Autonomous robotic navigation

A battery-powered robot avoiding obstacles instantiates the 2D control model.

- **State.** (Position, battery level) $\in \mathbb{R}^2 \times \mathbb{R}_+$.
- **Drift.** Gravitational or terrain slope.
- **Control.** Motor torque ($u \in U$, bounded by motor capacity).
- **Budget.** Battery charge (B_0).
- **Viability kernel.** States from which the robot can reach a charging station before battery depletion.
- **No-return surface.** States where remaining battery is insufficient to reach any charger under optimal control.

Testable prediction. The robot's reachable viable set shrinks monotonically as battery depletes (Proposition C2.1).

Slack. Time-to-boundary under zero motor input = coasting distance / terrain slope.

Observable. Optimal control policies should exploit silence (zero motor) on favorable slopes, consistent with C6.2.

These instantiations are not metaphors. Each maps the abstract quantities ($\mathcal{M}$, B , s , Σ_{NR}) to physically measurable variables with quantitative predictions. Failure of the predictions in either system falsifies the corresponding propositions of Paper C.

C11 — Structural Closure

The trilogy establishes a layered, one-way dependency chain.

Paper A — irreversibility as loss of reachability under bounded control. Defines Actualization State, proves monotonicity under decohering dynamics, and establishes no-return surfaces. Independent of Papers B and C.

Paper B — selection as costly, rate-limited, irreversible exclusion of alternative record sectors, if it exists. Derives structural requirements and a falsifiable gravitational rate bound. Depends on Paper A. Independent of Paper C.

Paper C — agency as normalized reachable viability volume under constrained control within a single realized record sector. Establishes propositions on agency decay under drift (C2.1), survival time bounds (C5.1), noise-induced depletion (C6.1), non-conservation under coupling (C7), and slack-agency correspondence (C8.1). Provides worked examples and experimental instantiations in biological and engineered systems. Depends on Paper A. Uses the outcome of Paper B but not its mechanism.

Failure of Paper C does not invalidate Paper B. Failure of Paper B does not invalidate Paper A. Each layer is independently falsifiable.

What remains is empirical. Which systems realize these structures, and how closely.

End of Paper C. Agency as Constrained Control. Locked · Execution-Complete.

Forward reference. Paper C's agency apparatus measures what the operator can do on the records-side of the now — the reach within the viability kernel, the control authority against drift, the no-return surface as absorbing boundary. AP43 (The Gravity of Possibilities) develops the structural dual on the pre-state side: the four-term system of coupling capacity, pattern, pattern coherence, and tracking coherence. AP43 reads what the body leans into ahead of actualisation; Paper C measures what the body has already accumulated as agency on what is written. The two are duals across the eye. AP43 opens Notebook VII of the work and inherits

AP01's apparatus as the records-side anchor for the operator-interface sector.

Paper D

Coupled Viability

Structural Conditions for Multi-Agent Persistence Under Irreversible Dynamics

Depends on Papers A, B, and C

You have lived inside this paper your entire life.

Every family. Every workplace. Every alliance, every rivalry, every marriage, every team. Every institution that has held together and every one that has fallen apart.

What looked like sociology, or politics, or psychology — was geometry. The geometry of viability kernels overlapping in a shared environment, with drift that never stops and control budgets that deplete.

Paper C measured agency as a number. Paper D extends that number to systems of agents. The configurations that persist are not the ones that are wisest, or fairest, or most intentional. They are the ones whose geometry permits persistence.

Hierarchy. Cooperation. Deterrence. Cascade failure. Every structural feature you have ever encountered in human systems has a geometric definition in this paper. Not as a metaphor — as the actual claim.

No new physics is introduced. The trilogy already supplied everything needed. Paper D is the four-paper spine completing itself.

D0 — Dependency, Scope, and Non-Overlap

D0.1 — Dependency Statement

This paper depends explicitly and exclusively on the results of Papers A, B, and C.

It assumes as given:

- Actualization State as an operational measure of record-structured irreversibility (Paper A).
- Selection as costly, rate-limited exclusion to definiteness, if it exists (Paper B).
- Agency as normalized reachable viability volume under constrained control within a single realized record sector (Paper C).

No construct from Papers A, B, or C is redefined or rederived here.

Failure of Paper D does not invalidate any prior paper.

D0.2 — Purpose

Paper D addresses a single question:

Given multiple agents, each described by Paper C’s formalism, operating within shared constraint environments under irreversible physics, what are the structural conditions for persistent joint dynamics, and what forms of emergent order are admissible?

This is a question about the geometry of coupled viability kernels under drift.

It is not a question about society, cooperation, or morality.

D0.3 — Positioning Relative to Existing Literature

Multi-agent viability theory exists. Aubin, Bayen, Saint-Pierre, and colleagues have developed the mathematics of viability kernels for coupled systems, differential games, and multi-agent control.

The contribution of this paper is not in proving new viability theorems. It is in applying viability theory to the specific structure of irreversibility (Paper A), selection (Paper B), and agency (Paper C). The results are constraints derived from physical irreversibility and record structure, not abstract control theory.

Paper D is not evolutionary game theory. It does not invoke fitness, replication, or selection pressure.

Paper D is not multi-agent reinforcement learning. It does not invoke reward signals, policy gradients, or learning.

Paper D is not mechanism design. It does not invoke incentive compatibility, revelation principles, or social welfare functions.

Paper D is viability geometry applied to physically irreversible, record-structured, agency-bearing coupled systems.

D0.4 — Hard Non-Claims

This paper does not:

- Introduce new physical laws.
- Modify or reinterpret quantum mechanics.
- Invoke psychology, motivation, ethics, value, meaning, or consciousness.
- Assume rationality, optimization, or fitness maximization.
- Model communication, signaling, or strategic negotiation.
- Derive evolutionary fitness.
- Propose normative guidance or prescriptions.
- Claim emergent structures are designed, intended, or purposeful.
- Claim that social structures emerge from viability constraints alone.

Failure of Paper D does not invalidate Papers A, B, or C.

D0.5 — Loaded Terms: Geometric Definitions

Several terms in this paper carry normative or sociological connotations in ordinary language. Each receives a strict geometric definition at first appearance. No connotation beyond the definition is implied.

Cooperation. A geometric condition where mutual record externalities expand joint viability. No intention, reciprocity, or payoff implied.

Hierarchy. Asymmetric coupling where higher-capacity agents' record externalities dominate the constraint landscape of lower-capacity agents. A consequence of scale asymmetry.

Deterrence. A coupling configuration where the cost of unilateral decoupling exceeds the cost of continued coupling for both agents. A viability geometry.

Impedance. The ratio of control authority to drift rate: $Z = u_{\max} / a$. Two agents are impedance-matched when their operator horizons are comparable.

Resonance. Frequency and phase compatibility between coupled control strategies. Constructive resonance expands joint viability; destructive resonance contracts it.

Read these definitions. The words do not mean what they normally mean. They mean exactly what is written here.

D1 — Shared Constraint Environments

D1.1 — Shared Viability Domain

When multiple agents operate within a common physical environment, their individual viability kernels may overlap.

The joint state space is the product of individual state spaces. The joint dynamics are defined by the individual drift fields, the individual control sets, and the coupling terms that transmit the effect of one agent's actions onto another's drift.

The **joint viability kernel** is the set of joint states from which there exists a joint admissible control strategy that maintains all agents within their individual viability constraints for all future time.

The **shared viability domain** is the projection of the joint viability kernel onto the shared constraint dimensions. It is the region of the common environment in which joint persistence is geometrically possible.

Structural assumption. Agents share constraint dimensions. They operate in a common physical environment whose state is affected by the actions of all agents. This is the defining condition for being in a shared constraint environment.

If agents' state spaces are fully orthogonal — no shared dimensions — they are uncoupled and Paper D does not apply.

This assumption is not a theorem. It is a scope condition. The paper analyses systems that satisfy it.

D1.2 — Constraint Coupling

Paper C (C7) treats coupling as direct energy and control transfer between two systems. Paper D introduces a second coupling mode: **constraint coupling**.

When Agent A's actions modify the shared environment in a way that alters Agent B's drift field, control set, or viability kernel, the agents are constraint-coupled. No direct energy exchange is required. The coupling operates through the shared constraint landscape.

Example. Robot A occupies the charging station. Robot B's admissible trajectories contract — it cannot charge. No energy flowed from A to B. But B's reachable set changed, because A's action modified the shared environment.

This is the everyday form of coupling. The seat someone takes. The lane someone occupies. The opportunity someone holds. None of it transfers energy. All of it modifies your viability kernel.

Scope. Pairwise coupling as the base analysis. Network effects modeled as cascades through pairwise links. Testable with three-agent systems (the minimal non-trivial network).

D1.3 — Record Externalities (Geometric Exclusion Principle)

Definition (Record-Writing Action).

An irreversible action by Agent A whose recorded environmental change lies in the shared constraint coordinates (e), and which modifies B's admissible dynamics $f_B(\cdot; e)$ or admissible control set $R_B(e)$.

The shared constraint coordinates are the dimensions of the environment state space that appear in B's dynamics or constraints.

Geometric Exclusion Principle.

For coupled agents with $K_A \cap K_B \neq \emptyset$, if Agent A performs a record-writing action that changes the shared constraint coordinates on which B's viability depends, then K_B changes, and $\mu(K_B)$ changes generically (Corollary D1.3).

Non-degeneracy assumption.

The map $e \mapsto K_B(e)$ is non-degenerate: the viability kernel boundary ∂K_B depends smoothly on e , and the constraint surface intersects the kernel boundary transversally. This excludes pathological cases where the environmental change lies entirely within the kernel interior, producing no boundary change.

Proof.

(1) A's record-writing action irreversibly modifies the shared constraint coordinates $e \rightarrow e'$ (by definition of record-writing action and Paper B's irreversibility).

(2) B's viability kernel is a function of the shared constraint coordinates: $K_B = K_B(e)$. Since B's admissible dynamics or control set depend on e (by definition), and the kernel boundary depends smoothly on e (by the non-degeneracy assumption), changing e changes the set of states from which B can persist.

(3) Under the non-degeneracy assumption, $K_B(e') \neq K_B(e)$. The transversality condition ensures the boundary moves under perturbation of e .

The change may be positive (expansion) or negative (contraction), depending on the direction of $e \rightarrow e'$ relative to B's constraint surface.

(4) **Sign classification.** If e' tightens B's constraints (reduces B's control set or increases B's drift), K_B contracts — negative externality. If e' loosens B's constraints, K_B expands — positive externality.

(5) $\mu(K_B)$ changes generically: by the transversality theorem, the set of e' for which $K_B(e') \neq K_B(e)$ but $\mu(K_B(e')) = \mu(K_B(e))$ — volume-preserving

deformations — has measure zero in the space of admissible environmental changes. This is the content of Corollary D1.3.

Corollary D1.3 (Genericity of Non-Neutrality).

In smooth families of coupling maps, the set of record-writing actions that produce exactly zero change in $\mu(K_B)$ has measure zero. Neutral externality requires parameter-level fine-tuning. This holds under the same non-degeneracy assumption stated above.

In plain language. Almost every action you take in a shared environment changes your neighbour's viability. Neutral actions are the exception, not the rule, and they are exceptional in the strict mathematical sense. The default condition of coupled life is non-neutrality.

Falsifier D1 (No Free Survival).

If Agent A exerts a negative record externality on Agent B (measured as decrease in $\mu(K_B)$), and Agent B increases its agency $\mathcal{M}_B$ (Paper C measure) without:

- (a) severing coupling,
 - (b) increasing its control budget $u_{\{B, \max\}}$, or
 - (c) receiving compensating positive externalities from a third agent,
- the argument is falsified.

Scope boundary. The measure-zero neutrality result (Corollary D1.3) depends on smoothness of the coupling map and regularity (C^2) of the viability kernel boundary. If the coupling map is non-smooth or the kernel boundary contains cusps, corners, or discontinuities, the transversality argument may fail and non-neutral actions could have positive measure.

This is an explicit scope limitation: Paper D's genericity claims apply to smooth families of coupling maps with regular kernel boundaries.

A further exception arises in systems with continuous symmetries — rotational invariance, for instance — where record-writing actions correspond to symmetry operations that preserve the viable volume by

construction. Outside of such symmetry-protected subspaces, non-neutrality is generic.

D2 — Composition of Agency

D2.1 — Joint Agency and Non-Additivity

Paper C established that agency is non-conservative under coupling (C7.1). Paper D extends this to N agents.

Definition (Joint Agency).

Joint agency $\mathcal{M}_{\text{joint}}$ is defined as the viability volume of the joint state space under joint admissible controls, normalized by the total joint viability kernel. The measure $\mathcal{M}$ is inherited from Paper C's definition, applied to the product state space $X_1 \times X_2 \times \dots \times X_N$.

Proposition D2.1.

Joint agency is non-additive: $\mathcal{M}_{\text{joint}} \neq \sum \mathcal{M}_i$ in general.

The non-additivity term depends on (a) alignment of individual drift fields, and (b) compatibility of individual control sets.

Joint agency is **superadditive** ($\mathcal{M}_{\text{joint}} > \sum \mathcal{M}_i$) when drift fields are anti-aligned (agents face complementary threats) and control sets are compatible.

Joint agency is **subadditive** ($\mathcal{M}_{\text{joint}} < \sum \mathcal{M}_i$) when drift fields are co-aligned (agents face the same threat simultaneously) or control sets conflict.

Proof sketch (by construction).

Superadditive witness. Two scalar systems, each with drift $a = 1$ and $u_{\text{max}} = 1$. Uncoupled, each has $\mathcal{M}_i = \text{viability volume of its individual kernel}$. Coupled cooperatively (Paper C, C7.1, cooperative example), shared control capacity yields effective $u_{\text{max}} = 2$ per system. The joint viability kernel expands: $\mathcal{M}_{\text{joint}} > \mathcal{M}_1 + \mathcal{M}_2$. This is the cooperative coupling case from Paper C.

Subadditive witness. System A ($a = 1$, $u_{\max} = 2$) coupled parasitically to system B ($a = 3$, $u_{\max} = 0$). B diverts A's control capacity. Effective $u_{\max}$ for A drops to 1. The joint viability kernel contracts: $\mathcal{M}_{\text{joint}} < \mathcal{M}_1 + \mathcal{M}_2$. This is the parasitic coupling case from Paper C.

Since both strict inequality directions are realizable, $\mathcal{M}_{\text{joint}} \neq \Sigma \mathcal{M}_i$ in general. The sign depends on drift alignment and control compatibility.

Whether coupling creates agency or destroys it is not a question about intentions. It is a question about geometry — drift alignment and control compatibility. The geometry decides.

D2.2 — Impedance Matching

Definition.

Impedance $Z_i \equiv u_{\{i,\max\}} / a_i$, where $u_{\{i,\max\}}$ is maximum control authority and a_i is drift rate. This mirrors the operator horizon (Paper A, Theorem T2).

Drift timescale.

$\tau_i = 1/a_i$. The characteristic time for Agent i 's drift to carry it a significant distance toward its no-return surface.

When $Z_i \neq Z_j$, three distinct failure modes arise.

Failure mode A (primary): Deadline mismatch.

The low- Z agent has short slack — small τ , little time-to-ruin. Help must arrive before the no-return surface is crossed.

If the high- Z agent's response delay exceeds the low- Z agent's remaining slack, the help arrives after no-return. Control effort applied after no-return produces zero viability gain.

This is the primary failure mode because no-return geometry makes time windows hard and asymmetric. Once missed, the effect is permanently zero.

The help that comes too late is not partial help. It is no help. Every parent of a teenager, every doctor in an ER, every aid worker in a famine has met this

geometry. The window closes. Effort applied after the window does not partially save what was lost.

Failure mode B: Absorption bottleneck.

Even if help arrives in time, the low-Z agent may not be able to convert it into viability. If the limiting constraint is $u_{\max}$, then energy that does not increase $u_{\max}$ does not change Z. Resources delivered to a system whose bottleneck is not resources produce zero reachability gain.

Failure mode C (secondary): Budget drain on the high-Z agent.

To rescue a short-slack agent, the high-Z agent must accelerate its response. The high-Z agent's budget drains faster.

This is secondary because the drain is usually caused by failure mode A: the high-Z agent spends, but the effect arrives too late or cannot be converted.

Proposition D2.2 (qualitative).

Coupling efficiency between agents i and j degrades as impedance ratio $|Z_i/Z_j|$ deviates from unity. Waste increases with mismatch. The primary waste mechanism is control effort applied outside the low-Z agent's viable intervention window.

Conjecture D2.2 (quantitative).

For a minimal interface model with transfer delay τ , conversion factor κ , and slack s , coupling efficiency η is bounded by the fraction of control effort deliverable within the slack window. A candidate functional form is

$$\eta \leq 1 / (1 + |Z_i/Z_j - 1|).$$

This requires derivation in the specific interface model and is not claimed as a universal result.

D2.3 — Resonance and Phase

System class.

Linear periodically forced agents with scalar state x_i , symmetric pairwise coupling κ , identical drift a , and sinusoidal control $u_i(t) = U \sin(\omega_i t + \phi_i)$.

Stability condition: $a > \kappa$ (drift exceeds coupling strength; if $\kappa \geq a$ the sum mode is unstable and both agents diverge regardless of phase).

Theorem D2.3 (Toy Model).

In the above system class, the measure of the joint viability set — the set of joint initial conditions from which both agents persist indefinitely — is maximized when $\omega_1 = \omega_2$ and $\varphi_1 - \varphi_2 = 0$ (in-phase resonance).

The joint viability set contracts monotonically as $|\varphi_1 - \varphi_2|$ increases from 0 to π .

Proof.

Dynamics: $dx_i/dt = -a \cdot x_i + u_i(t) + \kappa \cdot x_j$ ($i \neq j$), with viability constraint $x_i(t) \geq 0$.

For identical frequencies $\omega_1 = \omega_2 = \omega$, define $\Delta\varphi = \varphi_1 - \varphi_2$. Define sum mode $S = x_1 + x_2$ and difference mode $D = x_1 - x_2$. These decouple.

By trigonometric identities:

$$\sin(\omega t + \varphi_1) + \sin(\omega t + \varphi_2) = 2 \cdot \cos(\Delta\varphi/2) \cdot \sin(\omega t + (\varphi_1 + \varphi_2)/2),$$

$$\sin(\omega t + \varphi_1) - \sin(\omega t + \varphi_2) = 2 \cdot \cos(\omega t + (\varphi_1 + \varphi_2)/2) \cdot \sin(\Delta\varphi/2).$$

The effective control amplitude on the sum mode is $2U \cdot \cos(\Delta\varphi/2)$. The effective control amplitude on the difference mode is $2U \cdot |\sin(\Delta\varphi/2)|$.

The viability constraint $x_i \geq 0$ translates to $S \geq |D|$ — both components are non-negative if and only if their sum exceeds the absolute value of their difference.

Joint viability therefore requires:

- (a) S remains large, which requires maximum control amplitude on the sum mode;
- (b) $|D|$ remains small, which requires minimum forcing on the difference mode.

Condition (a) is optimized when $\cos(\Delta\phi/2) = 1$, i.e., $\Delta\phi = 0$. Condition (b) is optimized when $\sin(\Delta\phi/2) = 0$, i.e., $\Delta\phi = 0$. Both conditions are simultaneously optimized at $\Delta\phi = 0$ (in-phase resonance).

Monotonic contraction. As $|\Delta\phi|$ increases from 0 to π , $\cos(\Delta\phi/2)$ decreases monotonically from 1 to 0 (sum-mode control weakens) and $|\sin(\Delta\phi/2)|$ increases monotonically from 0 to 1 (difference-mode forcing strengthens). Both effects reduce the set of initial conditions from which both agents persist.

At $\Delta\phi = \pi$: $\cos(\pi/2) = 0$ (zero sum-mode control) and $|\sin(\pi/2)| = 1$ (maximum difference-mode forcing). This is the worst case.

Anti-phase coupling does worse than no coupling. The trigonometry confirms what every coordinated team and every dysfunctional pair already knows. When you push as your partner pulls, the joint kernel collapses.

Conjecture D2.3 (General).

For broader classes of periodically controlled coupled agents, joint viability is generically maximized under frequency commensurability and phase alignment.

The empirical signature: perturbing a persistent coupled system's phase relationship should contract the joint viability margin (measured as minimum slack over one control cycle).

Falsifier for Conjecture D2.3.

If a persistent coupled system is shown to have maximum joint viability at anti-resonant phase ($\phi_1 - \phi_2 = \pi$) under standard coupling, the conjecture is falsified.

D3 — Stable Configurations Under Drift

D3.1 — Compositional Equilibrium

A **compositional equilibrium** (CE) is a joint state-control configuration in which all agents maintain positive agency ($\mathcal{M}_i > 0$ for all i) indefinitely, given their joint drift field and joint control constraints.

CE does not invoke rationality. CE does not invoke payoff maximization.

CE is a geometric fixed-point condition: the joint system remains within the interior of the joint viability kernel.

Mathematical obstacle. The two-agent sinusoidal proof above succeeds because the viability constraint is linear in the sum/difference decomposition and the control is purely periodic. The general case resists proof for three identified reasons.

(i) Nonlinear coupling terms — multiplicative or saturating interactions — break the sum/difference decoupling that enables the trigonometric argument.

(ii) Non-convexity of the joint viability kernel under general control policies means the viability volume does not decompose into independent modal contributions.

(iii) For non-periodic control strategies, the phase relationship is not a well-defined scalar parameter, and the optimisation landscape may have local maxima at non-zero phase offset.

Any proof of the general conjecture must either restrict the coupling class — to affine or monotone coupling, for example — or establish a variational principle on the joint viability volume that is monotone in phase alignment.

Until such a proof exists, the conjecture is supported by the two-agent result and by the specified falsifier.

CE $\neq$ NE: The Operator-Required Charger

This is the load-bearing example of the entire paper. It demonstrates, with arithmetic only, that compositional equilibrium and Nash equilibrium are not the same configuration — that a unilateral local improvement can be a strict path to extinction.

System. Two robots, each with battery $b_i(t) \in [0, 100]$. Ruin at $b_i \leq 10$.

Drift. Battery drains at 12 units/hour (always on).

Coupling condition. To charge, the other robot must crank (operate the charger). Charging rate: +30 units/hour. Cranking cost: additional -4 units/hour (total -16 /hour for the cranking robot). If no one cranks, no one charges.

Compositional Equilibrium: Alternation.

Hour 0-1: R1 charges, R2 cranks. R1: $100 \rightarrow 100$ (capped). R2: $100 \rightarrow 84$.

Hour 1-2: R2 charges, R1 cranks. R2: $84 \rightarrow 100$ (capped). R1: $100 \rightarrow 84$.

Cycle repeats. Both oscillate between 84 and 100. Both remain far above the ruin threshold (10). Persistence is indefinite. This is CE.

Nash-type defection: Refuse to crank.

At any cranking turn, defection costs only -12 /hour instead of -16 /hour. Net gain: $+4$ units/hour. Strict local improvement.

If R1 defects every cranking turn but accepts cranking from R2:

Hour 0-1: R1 charges, R2 cranks. R1: 100, R2: 84.

Hour 1-2: R1 refuses. Nobody charges. R1: 88, R2: 72.

Hour 2-3: R1 charges, R2 cranks. R1: 100, R2: 56.

Hour 3-4: R1 refuses. R1: 88, R2: 44.

Hour 4-5: R1 charges, R2 cranks. R1: 100, R2: 28.

Hour 5-6: R1 refuses. R1: 88, R2: 16.

Hour 6–6.375: R2 crosses ruin threshold while cranking. R2 dead.

Hour 6.375–13.5: R1 cannot charge (no cranker). Drifts to ruin at 12/hr. R1 dead.

Result. CE (alternation) produces indefinite survival. The Nash move (refuse to crank) is a strict local improvement (+4/hour). The Nash move kills the partner at $t \approx 6.375$ hours. Without a partner, the defector cannot charge and dies at $t \approx 13.5$ hours.

Conclusion. CE is not Nash-stable under immediate-payoff incentives. The Nash-type unilateral improvement exits the joint viability kernel.

Viability $\neq$ utility. A rational agent can calculate its way into extinction by ignoring the operator horizon.

This is the paper's most important sentence. It is also the most important sentence about coupled human systems that has ever been derived from physics.

Every short-term optimisation that ignores the joint viability kernel ends here. Every relationship, every team, every institution that treats local improvement as the test of rationality is operating under a model that is provably wrong on contact with irreversible physics. The Nash move can be the dying move. The geometry is what it is.

D3.2a — Necessary Conditions for Persistence (under alignment)

Assumption (Monotonic Alignment).

All agents face drift toward the same boundary of the viability kernel. No agent's drift partially compensates another's drift without control expenditure.

Assumption (Regularity).

The viability kernel boundary is smooth (C^2). No agent's state is exactly on a cusp or non-differentiable point of the kernel boundary.

Proposition D3.2a (Necessary Conditions).

Under the monotonic-alignment and regularity assumptions, a multi-agent configuration that persists indefinitely must satisfy:

(N1) Aggregate control capacity exceeds aggregate drift (joint operator horizon condition).

(N2) Impedance compatibility: agents' drift timescales are close enough that help can arrive within the low-Z agent's slack window.

(N3) No agent's record externalities push another past its no-return surface faster than that agent can compensate (measured as: drift increase from externality $\leq$ compensating control authority).

(N4) Joint control budget is sufficient to maintain all agents above ruin (total energy expenditure $\leq$ total available budget over any finite horizon).

Failure of any condition implies at least one agent reaches its no-return surface in finite time, under the stated assumptions.

Proof of necessity (sketch).

N1 violated (aggregate drift exceeds aggregate control). Under monotonic alignment, all drift vectors point toward the same boundary. The joint system's state moves toward Σ_{NR} at net rate $\Sigma a_i - \Sigma u_{\{i,max\}} > 0$. Since the viability kernel has finite diameter, the boundary is reached in finite time bounded by $\text{diam}(K) / (\Sigma a_i - \Sigma u_{\{i,max\}})$. At least one agent exits.

N2 violated (impedance incompatibility). Under monotonic alignment, the low-Z agent's drift carries it toward Σ_{NR} with slack $s_{low} = d(x_{low}, \Sigma_{NR}) / a_{low}$. If the high-Z agent's minimum response delay $\tau_{response} > s_{low}$, the intervention arrives after no-return. By Paper A's absorbing-state result, the low-Z agent cannot recover. Its survival time T is bounded by s_{low} .

N3 violated (externality exceeds compensation). Agent i 's record-writing action increases j 's effective drift by Δa_j . If $\Delta a_j > u_{\{j,max\}} - a_j$ (the remaining control margin), then j 's net drift becomes positive toward Σ_{NR} . By the same finite-diameter argument as N1, j reaches the boundary in finite time.

N4 violated (budget exhaustion). Each agent's control expenditure rate is at least a_i (the maintenance cost from Paper C, Corollary C3.1a). If $\sum a_i >$ total budget rate, the aggregate budget depletes to zero in finite time. Once budget is exhausted, all agents are subject to uncontrolled drift and reach Σ_{NR} in finite time.

Scope of necessity.

Under non-aligned drift — where agents' drift fields partially cancel — configurations may persist while violating $N2$, because natural cancellation reduces the effective impedance mismatch. The necessity claim is strictly conditional on the alignment assumption.

D3.2b — Sufficient Conditions for Persistence (without alignment)

Proposition D3.2b (Sufficient Conditions).

The following are sufficient conditions for a multi-agent configuration to persist under irreversible drift, with no alignment assumption required:

(S1) Each agent independently satisfies its single-agent viability condition (Paper C): $u_{\{i,max\}} > a_i$ and budget $>$ maintenance cost.

(S2) All pairwise record externalities are non-negative (no agent's actions contract any other agent's kernel).

(S3) Coupling is impedance-compatible (D2.2).

(S4) Joint control budget exceeds joint maintenance cost.

Under these conditions, all agents persist. No agent's kernel contracts due to coupling, and each has sufficient resources to maintain itself.

Proof sketch.

By S1, each agent satisfies Paper C's single-agent viability condition independently: $u_{\{i,max\}} > a_i$ and budget exceeds maintenance cost.

By S2, no agent's record externalities contract any other agent's kernel, so coupling does not degrade any individual viability condition.

By S3, coupling is impedance-compatible, so control transfers between agents arrive within viable intervention windows.

By S4, the joint budget sustains the joint maintenance cost indefinitely.

Each agent therefore remains within its individual viability kernel for all time, and the joint state remains within the joint viability kernel.

Note. D3.2a and D3.2b are categorically different claims.

D3.2a tells you what persistent configurations *must* look like (under alignment). D3.2b tells you what configurations *will* definitely persist. Neither subsumes the other.

A configuration can satisfy D3.2b without violating D3.2a. But violating D3.2a does not imply violating D3.2b, because D3.2a requires alignment assumptions that D3.2b does not.

D3.3 — Instability and Cascade Failure

When Agent i exits its viability kernel ($\mathcal{M}_i = 0$), and the coupling was such that i 's control actions were partially compensating j 's drift, then j 's effective drift increases by the lost coupling contribution.

Proposition D3.3.

Failure of Agent i propagates to Agent j if the removal of i 's control contribution increases j 's effective drift beyond j 's remaining control margin ($u_{\{j, \max\}}$). If the new effective drift exceeds $u_{\{j, \max\}}$, j is pushed beyond its operator horizon and cascades toward its own no-return surface.

Containment. Cascade stops at Agent j if j has sufficient slack (s_j from C8.1) to absorb the shock before the increased drift pushes it past Σ_{NR} .

Falsifiable test. Three-agent system: A coupled to B coupled to C. Terminate A. Measure survival times T_B and T_C .

The cascade condition predicts: if removal of A's contribution increases B's drift past $u_{\{B,max\}}$, T_B is finite. If B's failure increases C's drift past $u_{\{C,max\}}$, T_C is finite. Propagation stops when the drift increase at an agent is less than that agent's remaining control margin.

You have watched this cascade. The supplier that fails. The institution that depended on it. The systems further down the chain that depended on the institution. Slack is what stops the wave. Without slack, the wave keeps going.

D4 — Emergent Order Without Design

D4.1 — Null Model and Order Metric

Before claiming emergent order, the paper establishes what the absence of order looks like.

Null model.

A population of agents with random (uncorrelated) control policies under the same drift field, coupling topology, initial condition distribution, and environment noise. Only survivors are analyzed.

The null model must match every confound except control policy coordination.

Order metric (Slack Correlation).

For surviving agents, measure individual slack $s_i(t)$ = time-to-ruin if controls are frozen at time t . Compute pairwise cross-correlation

$$\rho_{ij} = \text{corr}(s_i(t), s_j(t))$$

over time.

Random survivors: $\rho \approx 0$ (independent fluctuations). Coordinating agents: ρ significantly positive (slack levels move together).

Statistical test.

Compute $\bar{\rho}_{\text{obs}}$ = mean pairwise correlation among survivors in the observed system. Build null distribution $\{\bar{\rho}_{\text{null}}\}$ from N simulation runs with random control policies and survivorship filtering. Compute empirical p-value:

$$p = (1 + \#\{\text{null runs with } \bar{\rho} \geq \bar{\rho}_{\text{obs}}\}) / (1 + N).$$

Declare order if $p < 0.05$.

Structural precondition. Overlap ratio $O(t) = \mu(\cap K_i) / \mu(\cup K_i)$ measures geometric capacity for coordination. Slack correlation measures actual coordination. Paper D uses both.

Falsifier D4 (Order Indistinguishable from Noise).

If observed persistent configurations cannot be statistically distinguished from the null ($p \geq 0.05$ for empirical slack correlation), D4 is falsified.

Observable: empirical p-value.

D4.2 — Structural Filtering of Configurations

Under irreversible drift, configurations that violate the necessary conditions of D3.2a — under the stated alignment and regularity assumptions — are eliminated.

Survivors are biased toward configurations satisfying these conditions. Not because they were selected for. Because everything else exited the viability kernel.

Proposition D4.2.

Under irreversible drift and the monotonic-alignment and regularity assumptions of D3.2a, the long-run support of persistent multi-agent configurations is contained in the set of configurations satisfying the necessary conditions N1-N4. No optimization, fitness function, or teleology is required.

The structure is what is left. That is the entire mechanism. There is no designer. There is no goal. There is only the geometry of what persists when irreversibility runs.

Conjecture D4.2.

Under additional assumptions (ergodicity, stationary drift, specified policy update process), the distribution of persistent configurations converges to the set of compositional equilibria.

This conjecture requires explicit specification of the stochastic process and is not claimed as a theorem.

D4.3 — Hierarchy as Constraint Geometry

When agents have asymmetric capacity (different Z values), stable configurations generically exhibit hierarchical structure. Higher-capacity agents' record externalities dominate the constraint landscape of lower-capacity agents.

Regularity assumption. The coupling map is smooth and the constraint surface is non-degenerate (as in D1.3). Agent impedances are distinct: $Z_i \neq Z_j$.

Proposition D4.3.

In a coupled system with asymmetric agent impedances under the above regularity assumption, persistent configurations exhibit hierarchical coupling: the high- Z agent's record externalities alter $\mu(K_j)$ for the low- Z agent more than the reverse.

Proof sketch.

The high- Z agent has a larger viability kernel. Higher $u_{\max}/a$ ratio means more reachable viable states, by Paper A's operator horizon.

Under the regularity assumption (smooth coupling, non-degenerate constraint surface), the kernel expansion projects non-trivially onto the shared constraint coordinates. The additional reachable states include states that differ in the shared dimensions, not only in private dimensions.

A larger footprint in the shared constraint coordinates means that the high-Z agent's record-writing actions generically — on a set of full measure in the space of coupling parameters — modify more of the shared constraint space than the low-Z agent's actions.

By the Geometric Exclusion Principle (D1.3), this produces larger changes in the partner's kernel.

The hierarchy is geometric, not intentional.

Hierarchy is geometry. Not intention. Not malice. Not will. Geometry.

The agent with the larger viability kernel projects more onto the shared constraint coordinates. Its actions reshape the joint environment more than the lower-Z agent's actions can. This is the structural origin of hierarchy as a feature of coupled systems.

It does not require any agent to want to dominate. It does not require any agent to mean to dominate. It happens because the impedances are asymmetric and the geometry follows.

Falsifier D4.3 (Hierarchy Inversion).

If, in a system with $Z_i \gg Z_j$ (impedance ratio $> 10:1$) satisfying the regularity assumption, the low-Z agent's record externalities dominate the constraint landscape of the high-Z agent — $\Delta\mu(K_i)$ from j 's actions $> \Delta\mu(K_j)$ from i 's actions, measured over equivalent action magnitudes — D4.3 is falsified.

Observable: $\Delta\mu(K_i)$ and $\Delta\mu(K_j)$ per unit record-writing action.

D4.4 — Cooperation and Deterrence as Structural Outcomes

Proposition D4.4a (Cooperation).

Cooperative equilibria exist when mutual record externalities expand each agent's viability kernel more than coupling cost contracts it. The operator-required charger (D3.1) is an instance.

Observable: $\mathcal{M}_{\text{joint}} > \sum \mathcal{M}_i$.

Falsifier D4.4a (Cooperation Nonexistence).

If, in every tested coupled system where mutual record externalities are positive (each agent's actions expand the other's kernel), $\mathcal{M}_{\text{joint}} \leq \sum \mathcal{M}_i$ (joint agency never exceeds the sum of individual agencies), D4.4a is falsified.

Observable: $\mathcal{M}_{\text{joint}}$ and $\sum \mathcal{M}_i$ computed from the joint and individual viability kernels.

Proposition D4.4b (Deterrence).

Deterrence equilibria exist when unilateral decoupling cost exceeds continued coupling cost for both agents.

Observable: for each agent, $\mathcal{M}_i(\text{coupled}) > \mathcal{M}_i(\text{decoupled})$. Neither agent can improve its viability by exiting the coupling.

This is a geometric fixed-point, not a threat.

Deterrence in this paper does not require any agent to threaten anything. It does not require strategy or will. It requires that each agent's viability kernel is larger inside the coupling than outside it. Once this geometric condition holds, neither agent benefits from leaving — and the configuration persists.

What ordinary language calls deterrence — between rivals, between nations, between competitors — has this geometry whenever it persists. The threats are surface. The coupling-vs-decoupling viability calculation is the substrate.

Falsifier D4.4b (Deterrence Exit).

If an agent in a coupled system with $\mathcal{M}_i(\text{coupled}) > \mathcal{M}_i(\text{decoupled})$ for all i can unilaterally decouple and increase its agency ($\mathcal{M}_i(\text{after decoupling}) > \mathcal{M}_i(\text{coupled})$), the characterization of the configuration as a deterrence equilibrium is falsified.

Observable: $\mathcal{M}_i$ before and after decoupling.

Both are geometric. Neither is normative.

D5 — Experimental Instantiations and Falsifiers

D5.1 — Worked Examples

System 1: Microbial ecology (chemostat).

- *Shared viability domain*. Nutrient-population configuration space.
- *Record externalities*. Waste products altering pH/nutrient availability (irreversible environmental modification).
- *Impedance matching*. Metabolic rate compatibility between species.
- *Slack*. Time-to-washout at current dilution rate and population density.
- *Cascade failure*. Trophic coupling propagation.

Each construct maps to a measurable variable with a quantitative prediction.

System 2: Operator-required charger (two robots).

- *Shared viability domain*. Joint (position, battery) space with shared charging infrastructure.
- *Record externalities*. Station occupation.
- *Coupling condition*. Charging requires partner's cranking.
- *Impedance matching*. Battery capacity and discharge rate compatibility.
- *Slack*. Time-to-ruin at current battery level and discharge rate.
- *CE $\neq$ NE demonstration*. D3.1.

Each construct maps to a measurable variable with a quantitative prediction.

D5.2 — Falsifiers

Global Falsifier F0 (Kill Switch).

If a multi-agent system persists indefinitely (survival time $T = \infty$ for all agents) while violating *all* necessary conditions N1–N4 of D3.2a, under a configuration satisfying the monotonic-alignment and regularity assumptions, Paper D is falsified.

Observable: survival time T for each agent; verification of N1–N4 violation; verification that alignment and regularity assumptions hold.

Falsifier D1 (No Free Survival). Defined in D1.3. Observable: $\mu(K_B)$ and $\mathcal{M}_B$ before and after A's record-writing action.

Falsifier D2.1 (Additivity Under Coupling). If joint agency equals the sum of individual agencies ($\mathcal{M}_{\text{joint}} = \Sigma \mathcal{M}_i$) in a coupled system with non-zero coupling terms (non-orthogonal shared constraint coordinates), D2.1 is falsified. Observable: $\mathcal{M}_{\text{joint}}$ and $\Sigma \mathcal{M}_i$ in coupled vs uncoupled configurations.

Falsifier D2.2 (Impedance-Independent Efficiency). If coupling efficiency (measured as viability transfer per unit control effort) does not degrade as impedance ratio $|Z_i/Z_j|$ deviates from unity, D2.2 is falsified. Observable: viability transfer rate at impedance ratios 1:1, 2:1, 5:1, and 10:1 under matched conditions.

Falsifier D2.3 (Anti-Resonant Optimality). Defined in D2.3. Observable: joint viability margin at phase offsets $0, \pi/4, \pi/2, 3\pi/4, \pi$.

Falsifier D3.3 (Cascade Non-Propagation). Defined in D3.3. Observable: survival times T_B and T_C after termination of A in a 3-agent chain.

Falsifier D4 (Order Indistinguishable from Noise). Defined in D4.1. Observable: empirical p-value for slack correlation. If $p \geq 0.05$ for all candidate systems, D4 is falsified.

Falsifier D4.2 (Persistent Violators). If a multi-agent configuration persists indefinitely while violating one or more of N1–N4, under a system satisfying the monotonic-alignment and regularity assumptions, D4.2 is falsified.

This differs from F0, which requires violation of *all four* conditions. D4.2 claims the long-run support is contained in the satisfying set; a single persistent violator of any condition falsifies it. Observable: persistence time T and verification of individual N1–N4 conditions for each surviving configuration.

Falsifier D4.3 (Hierarchy Inversion). Defined in D4.3. Observable: $\Delta\mu(K_i)$ and $\Delta\mu(K_j)$ per unit record-writing action in impedance-asymmetric systems.

Falsifier D4.4a (Cooperation Nonexistence). Defined in D4.4. Observable: $\mathcal{M}_{\text{joint}}$ and $\Sigma \mathcal{M}_i$ in systems with mutually positive externalities.

Falsifier D4.4b (Deterrence Exit). Defined in D4.4. Observable: $\mathcal{M}_i$ before and after unilateral decoupling.

Every proposition has at least one testable falsifier with a specified observable. Falsifiers are independent of A, B, C. Failure of any proposition leaves all prior papers intact.

D5.3 — Scope Closure

Paper D establishes:

- What multi-agent composition must look like under the trilogy's constraints.
- What persistent configurations require.
- What destroys them.
- How to test these claims.

It does not determine whether specific configurations are realized in nature. That question remains empirical.

D6 — Structural Closure

Paper A. Irreversibility as loss of reachability. Independent of B, C, D.

Paper B. Selection as costly exclusion, if it exists. Depends on A.
Independent of C, D.

Paper C. Agency as constrained control. Depends on A; uses outcome of B.
Independent of D.

Paper D. Coupled viability under multi-agent constraint. Depends on A, B, C. Extends coupling (C7), introduces shared constraint environments, derives structural filtering, hierarchy, cooperation, and deterrence as geometric consequences.

The one-way dependency is preserved. Failure of D does not invalidate C, B, or A.

Each layer adds structure. None adds physics.

Together, the four papers constitute the spine. The narrative of P0 is the intuition that motivated them; the spine itself stands without it.

What you have just read is the geometry underneath every coupled human system you have ever lived inside. The terms changed — kernel for *room*, drift for *decay*, slack for *time*, ruin for *the moment after which nothing helps* — but the structure is the same. It was always geometry.

Forward reference. Paper D treats binary and chained multi-agent couplings (D1 through D4). The structural account of multipartite dynamics — operations between three or more operators reading each other's slopes simultaneously in shared viability domains — is open future work in The 420 Code. The higher-order dynamics inherit AP43's four-term system (coupling capacity, pattern, pattern coherence, tracking coherence) and Paper D's joint-viability apparatus. The structural account of teams operating at full resolution, of pair-bonds at high pattern coherence, of communities and institutions at operational scale, all sit inside the architecture this paper establishes and AP43 extends. The specific multipartite formalism is open.

Appendix E — Exploratory: Reclamation and Renewal (Non-Load-Bearing)

Paper A's optional module (A6) addresses capacity saturation and restoration for single systems. This appendix extends that to coupled systems: joint saturation, partial reclamation, the multi-agent loop.

It inherits the speculative status of A6. Explicitly non-load-bearing. No proposition in the main body depends on it. Included for structural completeness and intellectual honesty.

End of Paper D. Coupled Viability. Locked · Execution-Complete.

6 — Kill Switches

Twenty-four kill switches are engaged in this paper. Each is operationally testable. Each has a specified recovery position — if the kill switch fires, the work retreats to a named earlier position rather than collapsing entirely. Numbering follows the master registry: KS-V.1 through KS-V.24. Legacy labels (F0, F1-F3, G1-G3, B2, FC1-FC5, FD0-FD4.4b) are annotated at first mention and remain attached to the formal blocks in the body of the Papers.

KS-V.1 is the global kill switch. If it fires, the entire structure fails — AS is not well-defined and every subsequent result rests on a quantity that has no meaning. KS-V.2 through KS-V.7 target the selection postulates and the gravitational rate limiter. KS-V.8 targets the selection-irreversibility ordering. KS-V.9 through KS-V.13 target the agency apparatus of Paper C. KS-V.14 through KS-V.24 target the coupled-viability apparatus of Paper D.

KS-V.1 — Operational invariance of AS (F0)

Claim. AS values computed from any two co-admissible coarse-grainings of the same system agree within experimental tolerance. The Actualization State is a property of the system and its physical coupling, not an artefact of the analyst's choice of basis. This is the global kill switch (legacy label F0, formalised at D5 in Paper A).

Test. Prepare a calibration state. Compute AS in two distinct physically realizable coarse-grainings $\mathcal{O}_1$ and $\mathcal{O}_2$ that both satisfy D1 for the same physical coupling. The values must agree within the experimental noise floor δ_{exp} . Failure: persistent disagreement beyond δ_{exp} between two co-admissible coarse-grainings of the same physical system.

Status. LIVE — EMPIRICAL. GLOBAL.

Recovery. If KS-V.1 fires, AS is not a well-defined operational quantity and the entire structure fails. There is no graceful retreat — every theorem in Papers A through D is stated in terms of AS or quantities that inherit from AS's well-definedness. The work would have to be rebuilt on a different operational measure of record-structured irreversibility, or the project's central claim — that irreversibility can be measured operationally — would be retracted.

KS-V.2 — Pointer-basis targeting (F1)

Claim. The objective-actualisation channel respects the record algebra $\mathcal{O}$. Selection lands on the pointer states selected by the system-environment coupling, not on bases orthogonal to the pointer basis. Legacy label F1.

Test. In systems where the pointer basis is not position — superconducting qubits, photon polarisation, spin ensembles — measure where definiteness consistently appears. Failure: definiteness consistently appears in position despite $\mathcal{O} \neq \text{position}$ (or, more generally, in a basis not selected by the coupling).

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.2 fires, the objective-actualisation-channel postulate (P in A4.2) fails. The work retreats to decoherence-without-objective-selection:

the ensemble dynamics of Paper A's irreversibility results survive, but the trajectory-level selection account in Paper B has to be reformulated. Paper C's agency apparatus depends only on the existence of a single realized record sector and survives the retreat.

KS-V.3 — Born violation (F2)

Claim. Ensemble statistics of realised branches deviate from the Born distribution $\{p_i\}$ predicted by the pre-selection density matrix. The objective-actualisation channel preserves Born statistics at the ensemble level. Legacy label F2.

Test. Run repeated selections on prepared states. Compare the empirical distribution of realised branches against the Born-predicted distribution. Failure: systematic deviation from the Born distribution beyond statistical noise across many trials.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.3 fires, the boundary condition BC1 (B3.6) and the ensemble-consistency requirement (B3.5) fail. The work retreats to a position where selection produces individual realizations whose statistics do not match the Born prediction. AS as a quantity survives; Paper A's irreversibility theorems survive; only the postulate that the objective channel respects the Born rule fails.

KS-V.4 — Context dependence (F3)

Claim. Selection depends on observer intervention rather than objective dynamics. Whether and when selection occurs is a function of whether an external system has been brought into interaction with the target system, beyond what decoherence-with-environment-coupling already supplies. Legacy label F3.

Test. Run identical decoherence dynamics with and without explicit observation. Measure whether the timing or basis of selection differs in ways not predicted by the standard decoherence-environment-coupling

account. Failure: detection of selection signatures whose timing or basis depends on observer intervention beyond what the environmental coupling predicts.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.4 fires, the objective-dynamics requirement on the selection channel fails. The work retreats to a position closer to the Copenhagen interpretation, where selection is observer-relative. Paper A's record-algebra-based irreversibility results survive at the ensemble level; Paper B's individual-realization account would require observer-relative formulation.

KS-V.5 — Selection rate exceeds gravitational bound (G1)

Claim. Selection rates obey the bound $\lambda_{ij} \leq \Delta E_G[i,j]/\hbar$ for gravitationally distinguishable record sectors. The gravity-limited hypothesis (postulate G in A4.3, formalised in B4) sets the rate ceiling. Legacy labels G1 / F_G1.

Test. Create spatial superpositions of mesoscopic masses with computable ΔE_G . Measure selection timescales. Compare against the predicted bound $\tau_{\min} = \hbar/\Delta E_G$. Platform: levitated nanoparticles (tungsten, $R \approx 100$ nm, $\tau_{\min}$ on the order of seconds) in cryogenic vacuum. Failure: observation of selection with $\tau < \tau_{\min}$ for systems with well-defined ΔE_G .

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.5 fires, the gravity-limited rate hypothesis fails. The work retreats to the weaker position that selection is rate-limited by some universal limiter not yet identified. Papers A and C survive intact. Paper B's structural requirements (B3) survive; only the gravitational-specific identification of the limiter in B4 fails.

KS-V.6 — Selection in gravitationally degenerate regime (G2)

Claim. Selection does not occur between records with $\Delta E_G = 0$. In gravitationally degenerate configurations, multiplicity persists indefinitely (or at least at timescales much longer than the gravity-limited prediction would allow). Legacy labels G2 / F_G2.

Test. Prepare decohered superpositions with $\Delta E_G = 0$ — for example, nitrogen-vacancy centres in diamond, or nuclear spin states with identical mass distributions. Monitor for single-sector stabilisation versus persistent multiplicity. Failure: rapid selection in the absence of any alternative limiter satisfying B4.2.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.6 fires, the gravitational identification of the rate limiter fails — selection is happening when $\Delta E_G = 0$, so something else is driving it. The work retreats to the search for the actual limiter. Papers A, C, D survive; B retreats to structural requirements on selection without identifying gravity as the specific limiter.

KS-V.7 — Non-gravitational rate scaling (G3)

Claim. Selection rates do not scale universally with non-gravitational parameters across macroscopic records. If selection rates depend systematically on, for example, electromagnetic coupling strength or thermodynamic state across regimes where the gravity-limited bound is not saturated, the universal-limiter requirement fails. Legacy labels G3 / F_G3.

Test. Compare selection rates across systems with similar ΔE_G but different non-gravitational characteristics (charge distribution, temperature, internal coupling). Failure: systematic scaling of selection rates with non-gravitational parameters at rates where the gravitational bound is not yet saturated.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.7 fires, gravity is not the universal limiter. The work retreats to a multi-mechanism account or to a different universal limiter not yet identified. Same retreat as KS-V.5 and KS-V.6 — Papers A, C, D survive; B's gravitational-specific claim fails.

KS-V.8 — Pre-irreversibility selection (B2)

Claim. Selection cannot precede operational irreversibility. The selection deviation $\delta_{\text{sel}}(\rho)$ is zero on the pre-irreversibility manifold of states. Definiteness emerges only after the record algebra has become operationally definite. Legacy label B2 (B2.5).

Test. Look for selection signatures (single-trajectory variance, individual-realisation distinguishability) in states that have not yet undergone operational decoherence to the record algebra. Failure: detection of selection signatures prior to operational irreversibility — $\delta_{\text{sel}}(\rho) \neq 0$ on states where the dephasing map $\Delta_{\mathcal{O}}$ has not yet acted.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.8 fires, the post-irreversibility activation requirement (B3.1) fails. The work retreats to a position where selection and irreversibility may be coincident rather than sequentially ordered. The cost-of-selection result (B2.4) survives at the ensemble level; the structural ordering between Papers A and B reverses or merges.

KS-V.9 — Agency increase without control (FC1)

Claim. Agency cannot increase without control expenditure. If reach grows without active control input — for example, a non-motile mutant maintaining position in a gradient — agency as defined in C1 has been violated. Legacy label FC1.

Test. Observe a system where $\text{reach}(x)$ grows or remains stationary in a drifting environment without active control. The canonical case is a non-motile mutant ($u_{\text{max}} = 0$) maintaining position in a nutrient gradient. Failure: reach grew or held without control expenditure.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.9 fires, the definition of agency as reach-within-viability-under-admissible-control fails. The work retreats to a position where agency includes passive structural-position effects (e.g. anchoring) alongside active control. Papers A and B are unaffected. Paper D's coupling apparatus would require reformulation for the agency-transfer subsections.

KS-V.10 — Irreversible loss reversed (FC2)

Claim. Once a system has passed a no-return surface, it cannot return — even with unlimited control input. Irreversibility is operational: no admissible control trajectory exists from the post- Σ_{NR} region back to the viability kernel. Legacy label FC2.

Test. Identify a system trajectory that crosses a no-return surface (as defined in D9) and then demonstrates return to the viability kernel through control. Failure: observation of passive or controlled return from beyond a verified no-return surface.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.10 fires, the no-return surface concept fails operationally — what looked irreversible was reversible under sufficient control. The work retreats to a position where no-return surfaces are defined only relative to specific control budgets. Paper C’s operator-horizon theorem (T2) survives at bounded-control regimes; the absolute-irreversibility claim weakens.

KS-V.11 — Stable control past no-return surface (FC3)

Claim. Control authority approaches zero as the state approaches a no-return surface (C1.2). Stable controlled behaviour cannot be sustained on Σ_{NR} or beyond. Legacy label FC3.

Test. Demonstrate a system maintaining stable controlled behaviour in a region the formalism identifies as past a no-return surface. The system holds its position or completes controlled trajectories from the post- Σ_{NR} region. Failure: stable controlled behaviour from beyond the predicted no-return surface.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.11 fires, the control-authority result ($\lim \mathcal{M}(x) = 0$ as $x \rightarrow \Sigma_{\text{NR}}$) fails. The work retreats to a position where some no-return surfaces are operational only against specific drift regimes, not universal.

KS-V.12 — Free lunch (FC4)

Claim. Control expenditure carries non-zero cost (C5). A system with finite control budget B_0 cannot exceed the survival time bound $T \leq B_0/c_{\min}$ derived in theorem C5.1. Legacy label FC4.

Test. Observe a motile cell with finite ATP reserve persisting indefinitely in a persistent nutrient gradient flow. Or any system that exceeds the predicted survival-time bound while verifiably operating with finite control budget. Failure: persistence beyond $T_{\max}$ with verified finite B_0 .

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.12 fires, the control budget formulation fails. The work retreats to a position where some control regimes are cost-free in the relevant operational sense (perhaps under specific energy-harvesting boundary conditions). Theorem C5.1 retreats to a conditional statement scoped to non-harvesting regimes.

KS-V.13 — Resurrection (FC5)

Claim. Once an agent is in the ruin set ($x \notin \text{Viab}(R)$, absorbing per C4.2), no admissible control trajectory returns the agent to the viability kernel. Legacy label FC5.

Test. Observe an agent in the ruin set returning to the viability kernel through control. The ruin set is operationally absorbing; failure means it isn't.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.13 fires, the absorbing-ruin-set definition fails. The work retreats to a position where ruin is reversible under specific control regimes. Most of Paper C's results survive at the bounded-control level; the irreversibility-of-ruin claim weakens.

KS-V.14 — Multi-agent persistence violation (FD0)

Claim. Multi-agent persistence requires satisfaction of structural conditions N1-N4 (D3.2a). A multi-agent configuration cannot persist indefinitely while

violating any of these conditions, under the regularity and monotonic-alignment assumptions. Legacy label FD0 (global D-paper falsifier from D5.2).

Test. Identify a multi-agent configuration that persists for a duration much longer than the drift timescale while clearly violating at least one of the N1–N4 conditions. The configuration must satisfy the monotonic-alignment and regularity assumptions. Failure: long-run persistence with verified N-condition violation.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.14 fires, the necessary-conditions theorem (proposition D3.2a) fails. The work retreats to a position where some N-condition violations are tolerable under the alignment assumption. Sufficient conditions (D3.2b, without alignment) may still hold for narrower scopes.

KS-V.15 — No free survival (FD1)

Claim. Record-writing actions by one agent in a shared viability domain change the viability measure available to other agents. The Geometric Exclusion Principle (D1.3) holds: there is no record-writing action that leaves all other agents' viability strictly unchanged. Legacy label FD1.

Test. Identify a record-writing action by agent A that leaves both $\mu(K_B)$ and $\mathcal{M}_B$ unchanged for all other agents B operating in the same shared viability domain. The action must be non-trivial (it must have actually written records). Failure: existence of a non-trivial free-survival action.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.15 fires, the Geometric Exclusion Principle fails. The work retreats to a position where some record-writing actions are externality-free. The structural-filtering (D4.2) and hierarchy (D4.3) results would need to be reformulated; cooperation and deterrence (D4.4) survive if recast in terms of strict-Pareto rather than generic externalities.

KS-V.16 — Additivity under coupling (FD2.1)

Claim. Joint agency is non-additive under coupling. In a coupled system with non-zero coupling terms (non-orthogonal shared constraint coordinates), $\mathcal{M}_{\text{joint}} \neq \Sigma \mathcal{M}_i$. Legacy label FD2.1.

Test. Measure $\mathcal{M}_{\text{joint}}$ and $\Sigma \mathcal{M}_i$ in coupled versus uncoupled configurations. If they coincide across coupled systems with non-orthogonal shared coordinates, additivity holds and the claim fails. Failure: $\mathcal{M}_{\text{joint}} = \Sigma \mathcal{M}_i$ across non-trivial coupling regimes.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.16 fires, the non-additivity result fails and the analysis of impedance, anti-resonance, and cascade effects loses its structural ground. The work retreats to a position where coupling effects are treated additively at first order with corrections fitted empirically. Paper D's emergent-order results (D4) would require reformulation.

KS-V.17 — Impedance-independent efficiency (FD2.2)

Claim. Coupling efficiency — viability transfer per unit control effort — degrades as the impedance ratio $|Z_i/Z_j|$ deviates from unity. Impedance mismatch is a real cost. Legacy label FD2.2.

Test. Measure viability transfer rate at impedance ratios 1:1, 2:1, 5:1, and 10:1 under matched conditions. Failure: efficiency does not degrade with impedance mismatch — transfer rate is independent of ratio.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.17 fires, the impedance-matching subsection fails. Paper D retreats to a position where coupling efficiency is governed by mechanisms other than impedance. The deadline-mismatch and absorption-bottleneck failure modes named in D2.2 would require alternative structural accounts.

KS-V.18 — Anti-resonant optimality (FD2.3)

Claim. Joint viability margin is maximised away from phase-aligned (resonant) coupling and approaches a minimum at phase offsets that align constraint excursions. Anti-resonance — the offset that distributes constraint excursions — outperforms resonance. Legacy label FD2.3.

Test. Measure joint viability margin at phase offsets $0, \pi/4, \pi/2, 3\pi/4, \pi$ in coupled oscillatory systems with shared constraint. Theorem D2.3 (toy model) predicts the optimum is off-resonance. Failure: maximum joint viability at phase offset 0 (perfectly resonant).

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.18 fires, the anti-resonance optimality claim fails. The work retreats to a position where resonant coupling is optimal — which would invert the design intuition for stable multi-agent systems. Most of Paper D’s framework survives; the specific phase result fails.

KS-V.19 — Cascade non-propagation (FD3.3)

Claim. In a chain $A \rightarrow B \rightarrow C$ with appropriate coupling and shared constraint, termination of A leads to degradation of B and C on predictable timescales. Cascade failures are real and geometrically constrained. Legacy label FD3.3.

Test. Measure survival times T_B and T_C after termination of A in a 3-agent chain. Compare against predictions from D3.3. Failure: B and C do not degrade after A’s termination — the chain does not cascade.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.19 fires, the cascade-failure account in D3.3 fails. The work retreats to a position where multi-agent chains are more resilient than the structural account predicts — each link is somehow buffered. The structural-filtering claim (D4.2) may need to be scoped to non-chained configurations.

KS-V.20 — Order indistinguishable from noise (FD4)

Claim. Persistent multi-agent configurations satisfy structural conditions (N1–N4 or similar) at rates significantly above what null-model noise produces. Emergent order is structurally filtered, not random. Legacy label FD4.

Test. Compute empirical p-value for slack correlation across observed multi-agent configurations against a null model. Failure: $p \geq 0.05$ for all candidate systems — observed configurations are statistically indistinguishable from null-model noise.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.20 fires, the structural-filtering claim fails — what looked like emergent order is indistinguishable from random distribution of configurations. The work retreats to a position where Paper D’s results describe the space of possible configurations but do not predict observed-frequency biases.

KS-V.21 — Persistent violators (FD4.2)

Claim. A multi-agent configuration cannot persist indefinitely while violating one or more of N1–N4 conditions, under the monotonic-alignment and regularity assumptions. Legacy label FD4.2.

Test. Identify a multi-agent configuration that persists at timescales much longer than the drift timescale while a single N-condition is verifiably violated. Failure: long-run persistence with verified single-condition violation under the alignment assumption.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.21 fires, the necessary-conditions theorem (D3.2a) is weaker than stated — some N-condition violations are individually tolerable. The work retreats to a stricter statement: persistence requires joint satisfaction of the conditions in some specific logical structure (e.g. at least three of the four) rather than all four.

KS-V.22 — Hierarchy inversion (FD4.3)

Claim. In impedance-asymmetric systems, the lower-impedance agent gains more reach per unit record-writing action than the higher-impedance agent. Hierarchies emerge as a consequence of impedance asymmetry, not from imported authority. Legacy label FD4.3.

Test. Measure $\Delta\mu(K_i)$ and $\Delta\mu(K_j)$ per unit record-writing action in impedance-asymmetric systems. Failure: high-impedance agents gain more reach than low-impedance agents per unit action — hierarchy inverts from the geometric prediction.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.22 fires, the hierarchy-as-impedance-geometry claim fails. The work retreats to a position where hierarchies may have geometric origins but the impedance-asymmetry mechanism is not the principal driver. Paper D's structural-filtering and cooperation results survive; the specific hierarchy mechanism would need reformulation.

KS-V.23 — Cooperation nonexistence (FD4.4a)

Claim. In systems with mutually positive externalities, joint agency exceeds the sum of individual agencies — cooperation is structurally favoured. Legacy label FD4.4a.

Test. Measure $\mathcal{M}_{\text{joint}}$ and $\Sigma \mathcal{M}_i$ in systems with verified mutually positive externalities. Failure: $\mathcal{M}_{\text{joint}} \leq \Sigma \mathcal{M}_i$ — cooperation produces no joint-agency benefit in systems where the structural account says it should.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.23 fires, the cooperation-as-geometric-outcome claim fails. The work retreats to a position where cooperation in observed systems is sustained by mechanisms other than joint-agency benefit (e.g. reciprocal altruism, evolved norms). Paper D's deterrence and hierarchy results survive; the cooperation-specific result fails.

KS-V.24 — Deterrence exit (FD4.4b)

Claim. Unilateral decoupling from a stable deterrence configuration reduces the decoupling agent’s individual agency. Deterrence is a geometric fixed-point, not a threat: leaving it costs the leaver. Legacy label FD4.4b.

Test. Measure $\mathcal{M}_i$ before and after unilateral decoupling from a verified stable deterrence configuration. Failure: the decoupling agent’s $\mathcal{M}_i$ increases or remains unchanged after exit.

Status. LIVE — EMPIRICAL.

Recovery. If KS-V.24 fires, the deterrence-as-geometric-fixed-point claim fails. The work retreats to a position where deterrence configurations are sustained by mechanisms other than joint-agency-loss-on-exit. Paper D’s other results survive; the deterrence-specific result fails.

Why these twenty-four

The kill switches engage every load-bearing claim in Papers A through D. KS-V.1 is global — if AS is not operationally well-defined, every theorem downstream rests on a quantity without meaning. KS-V.2 through KS-V.4 engage the postulate that selection respects the record algebra and the Born distribution. KS-V.5 through KS-V.7 engage the gravitational rate-limiter postulate. KS-V.8 engages the structural ordering between irreversibility and selection. KS-V.9 through KS-V.13 engage the agency apparatus. KS-V.14 through KS-V.24 engage the coupled-viability apparatus.

Each kill switch has a specified recovery position. If a kill switch fires, the work does not collapse — it retreats to a named earlier position with the specific theorems that fall explicitly identified. KS-V.1 is the exception: there is no graceful retreat from a framework-level failure of AS’s operational well-definedness.

Smaller claims throughout the paper — specific numerical estimates in the experimental-feasibility appendices, the worked qubit example in Paper A’s F appendix, the specific platform candidates in Paper B’s B5, the chemostat and operator-required-charger worked examples in Papers C and D — could each be tested independently. The work does not list them as formal kill

switches because they are illustrative rather than load-bearing. If a specific worked example fails on its specific parameters, the paper still stands. If KS-V.1 through KS-V.24 hold, the load-bearing structure holds.

7 — What This Establishes and What Remains Open

7.1 — What this paper closes

The four formal Papers of the work's operational spine, derived from {S, B, R, C}.

Paper A installs the Actualization State — the operational measure of record-structured irreversibility, definable on any system that admits a physically realizable coarse-graining. T1 proves AS is monotone under decohering dynamics with a specified rate bound. T2 proves the operator horizon as the structural form of no-return at the manifold level.

Paper B installs selection as a costly, rate-limited, irreversible exclusion process. Selection cannot precede operational irreversibility (B3.1). The selection cost $\delta_{\text{sel}}(\rho)$ is non-zero, conditioned by ensemble consistency (B3.5) and the Born boundary condition (B3.6). The gravity-limited rate hypothesis (B4) sets a falsifiable rate ceiling with three kill switches.

Paper C installs agency as the operator-relative ratio of reach-within-viability to viability, with control authority approaching zero at no-return surfaces. The control-budget result (C5) and the survival-time theorem (C5.1) make the cost of agency operational. Slack, redundancy, noise, silence, coupling, rescue, and exit each receive structural treatment with their own kill switches.

Paper D extends to coupled multi-agent systems. The Geometric Exclusion Principle (D1.3) shows that record-writing actions in a shared viability domain are never externality-free. Joint agency is non-additive under coupling. Impedance matching, anti-resonance, and cascade-failure dynamics are formalised. Structural filtering, hierarchy, cooperation, and deterrence fall out as geometric consequences, each with a falsifier and a kill switch.

The work's operational vocabulary — actualisation, irreversibility, record algebra, selection cost, agency, control authority, viability kernel, no-return surface, slack, coupling, impedance, joint viability, structural filtering — is installed. Every later derivational paper inherits its definitions from one of the four Papers here.

7.2 — What this paper does not close

No structural debts are outstanding from this paper. AP01's load-bearing claims are derived within the paper or are framed as postulates with explicit kill switches. The gravity-limited hypothesis (postulate G in A4.3, B4) is a falsifiable postulate carrying its own kill switches (KS-V.5 through KS-V.7), not a structural debt. The objective-actualisation channel postulate (postulate P in A4.2) is a falsifiable postulate carrying KS-V.2 through KS-V.4 as its kill switches, not a structural debt. If either postulate fires, the work retreats cleanly to a named earlier position.

The relativistic extension of Paper A's quantum-mechanical instantiation is sketched only. The full relativistic form of the gravity-limited rate bound, and the relativistic form of the no-return surface in curved spacetime, are open work for AP08 (gravity-from-records) and the future relativistic-field-equation paper that closes AP43's D1.

The integration of AP01's records-side apparatus with AP43's pre-state-side apparatus is open work. AP01 measures how irreversibly the body's records have been written; AP43 reads the slope of what could be written next. The two are duals across the now. The formal integration — the field equation that produces both AS-on-records and the gravity-of-possibilities as faces of one structure — is the immediate next major paper in the operator-interface sector.

7.3 — Items held open without claiming debt status

Three items are bracketed inside this paper. They are real and live, but they are not structural debts — they are operational specifications and parameter identifications rather than unresolved derivations.

The first-principles derivation of the coupling parameters in Papers C and D — the control budget B_0 , the cost function $c(u)$, the impedance Z_i , the resonance phase. These are parameterised in the structural account so the theorems can be stated and the falsifiers specified. Their physical instantiations in specific embodied systems (cells, organisms, robots, economies, institutions) are open work for domain experts collaborating with the work.

The full operational specification of the gravitational self-energy $\Delta E_G[i, j]$ across regimes. The work uses the Newtonian form (Appendix C) and notes the relativistic extension as open. For mesoscopic platforms (levitated nanoparticles, NV centres) the Newtonian form suffices for current experimental tests. For black-hole-scale or cosmological tests, the relativistic form is open work.

The structural status of multipartite operator dynamics — operations between three or more operators reading each other's slopes simultaneously in shared viability domains. Paper D treats binary and chained couplings; the higher-order structural dynamics are open work, inheriting AP43's four-term system once the integration with AP01 is formalised. The structural account of teams, communities, institutions at operational resolution is future work in The 420 Code and does not block this paper's closure.

7.4 — Where the philosophical-register implementations live

This paper develops the work's operational apparatus at the formal register. The Ø Models catalogue treats the same structural facts at the philosophical register, in several places with greater philosophical apparatus than this paper carries. The work reads at both registers; this subsection names where the philosophical-register companions live.

Dissolutions Chapter 12 (The Measurement Problem) is the philosophical-register companion to Papers A and B. Where Paper A defines AS as the operational measure of record-structured irreversibility and Paper B installs selection as costly irreversible exclusion, Dissolutions Chapter 12 dissolves the observer/observed binary that the measurement problem has held as primitive — observer and observed turn out to be two positions in one structure rather than two domains. The chapter installs the work's name for what forces commitment: the forcing function. A coupling of sufficient structural capacity forces non-optionality on the trajectories it couples with. Paper B's structural requirements on selection are the formal-register apparatus that the forcing-function vocabulary names philosophically.

Dissolutions Chapter 6 (Other Minds) and Dissolutions Chapter 7 (Free Will and Determinism) are the philosophical-register companions to Paper C. Chapter 6 derives operator-sameness through one-interior — the inside-reading at every coupling is the same structural position expressed locally wherever coupling occurs, not many separate insides. Chapter 7 dissolves the libertarian-versus-determinist binary through the structural fact that the separate chooser the binary trades on does not exist; what stands is *freedom under constraint* — the operator coupling within a field of constraint, neither free of constraint nor determined by it. Paper C's agency apparatus — reach-within-viability under admissible control, with control authority approaching zero at no-return surfaces — is the formal-register apparatus that Dissolutions Chapter 7's freedom-under-constraint names philosophically. Where Paper C carries the geometric formalism, Chapter 7 carries the philosophical dissolution of the binary that the formalism dissolves geometrically.

Resolutions Chapter 8.5 (Choice) installs two paired locked formulations. The lived/embodied stress-test: "Choice is the freedom to turn left when every possible computation tells you to turn right — that ability to see the constraints within which you are acting and then decide to pursue a sub-optimal way forward." The structural register: "Choice is irrational coupling capacity manifested as sub-optimal coupling based on all computation results to date." Both formulations are G's locked phrasings in Resolutions Chapter 8.5. The chapter installs choice as operator-mediated commitment among live trajectories, with override-capacity as the structural feature that proves the operator is not identical to the computed gradient. Paper C's

agency apparatus carries the geometric measure of the live-trajectories field; Resolutions Chapter 8.5 names what the operator does within that field at the philosophical register.

Resolutions Chapter 10 (Other Intelligences) is the philosophical-register companion to Paper D's coupling apparatus and to the open question of the four-term system's silicon-substrate implementation (AP43 D6). The chapter installs the substrate-independence settlement at the operator-hood level: any architecture meeting the four structural conditions — *Does it couple? Does it write records about its own state? Does it model its coupling-geometry? Does it have valenced self-state?* — is an operator regardless of substrate. The four-question working specification at line-resolution in Resolutions Chapter 10 is the philosophical-register companion to Paper D's joint-viability apparatus and to the structural-filtering result. Where Paper D carries the geometric formalism for multi-agent couplings, Resolutions Chapter 10 settles the question of which architectures qualify as operators within those couplings — at the general operator-hood level, substrate-independently.

This paper was drafted before AP40, AP43, and the Ø Models catalogue's relevant chapters were complete. Each of those companions has subsequently been locked at the philosophical register. AP01 is the formal-register foundation; the philosophical-register treatments stand on it and extend it. The direction-of-acknowledgment is two-way: AP01 supplies the operational apparatus the philosophical-register treatments inherit; the philosophical-register treatments in some places (notably the structural-conditions specification in Resolutions Chapter 10 and the freedom-under-constraint formulation in Dissolutions Chapter 7) go further than this paper carries. The work reads at both registers.

7.5 — Structural relationships to subsequent work

AP01 opens the work's formal spine. Every later derivational paper that uses the words actualisation, irreversibility, selection, agency, viability, coupling, joint viability, no-return surface, or operator horizon inherits its definitions from one of the four Papers here.

AP07 (The Record Measure) inherits Paper A's record-algebra coarse-graining framework and specialises it to the electromagnetic-coupling-dominant chemical-biological scale, establishing the electron as the unique writer of configurable records at that scale.

AP08 (The Identity — Einstein Field Equations) inherits Paper A's irreversibility theorems and derives the field equations on the manifold side. AP08 is the gravity-from-records sector. AP43 (The Gravity of Possibilities) is the structural dual on the pre-state side, inheriting AS as the records-side anchor and developing the four-term system as the operator's interface with the amplitude landscape.

AP25 (The Measure) inherits Paper A's operational coarse-graining apparatus and the dephased-entropy structure, applying Gleason's theorem to derive the Born rule on the amplitude landscape. The Born boundary condition (B3.6) of this paper is the ensemble-statistics version of what AP25 derives from non-contextuality.

The future relativistic-field-equation paper for the gravity-of-possibilities (closing AP43's D1) inherits both AP01's records-side apparatus and AP43's pre-state-side apparatus, producing the unified equation that both AS-on-records and the gravity-of-possibilities are partial closures of.

7.6 — Placement in The 420 Code

This paper is Chapter 2 of Notebook I — The Premise. Notebook I carries three chapters: AP40 (The Irrational), AP01 (The Actualization State), AP20 (The Proof). AP40 sits first, AP01 second, AP20 third.

The placement satisfies the four structural requirements. First, dependencies: AP01's load-bearing dependencies are the axiom as primitive (no upstream paper required) and the axioms {S, B, R, C} (formal statement carried at AP20 section 3.3). AP40 sits upstream as the axiom's reflexive examination but is not load-bearing for AP01's derivations. Second, foundation: AP01 installs the four operational quantities (AS, selection cost, agency, joint viability) that every later derivational paper in the work inherits. Third, arc: the work reads premise → spacetime → quantum mechanics → forces → particles → cosmology → operator-interface → applications. AP01's operational apparatus sits in the Premise where it

supplies the vocabulary every later Notebook uses. Fourth, register pairing: AP01 is the formal-register foundation for the philosophical-register treatments in Dissolutions Chapters 6, 7, and 12, and in Resolutions Chapters 8.5 and 10. The pairing is named explicitly in section 7.4.

7.7 — How the work reads now

With this paper in place at Chapter 2 of Notebook I, the work reads as follows.

Notebook I (The Premise) is now read as carrying the foundational triad: AP40 names what the axiom is (its irrational structural character); AP01 installs the operational apparatus the axiom produces under $\{S, B, R, C\}$; AP20 proves the apparatus holds in physical reality. The three chapters answer three questions in order: what is the axiom, what does it produce operationally, and does it hold.

The work's later derivational papers stand on the apparatus this paper installs. The operational vocabulary — actualisation, irreversibility, selection, agency, viability, coupling, no-return surface, operator horizon — is canonical from this paper forward. Every theorem in every Notebook II through VIII paper that uses these words inherits its definitions from one of the four Papers here.

The formal-register/philosophical-register pairing is made visible at the spine. Readers who first encounter the measurement question through Dissolutions Chapter 12's dissolution of observer/observed can be pointed to Papers A and B's operational apparatus. Readers who first encounter agency through Resolutions Chapter 8.5's locked choice formulations can be pointed to Paper C's geometric formalism. Readers who first encounter joint life through Dissolutions Chapter 6's one-interior derivation, or substrate-independence through Resolutions Chapter 10's four operator-hood conditions, can be pointed to Paper D's coupling apparatus. The work reads at both registers; AP01 makes the formal-register ground legible.

7.8 — The closing

The four operational quantities are installed. Actualization State. Selection cost. Agency. Joint viability under coupling.

Each is defined operationally — it can be measured. Each obeys specified theorems with their conditions stated and their proofs declared locally. Each is tied to at least one kill switch with a specified observable. Twenty-four kill switches in total. KS-V.1 is the global kill switch — if AS is not operationally well-defined, the framework fails.

The dependency is one-way. Paper A defines AS without depending on B, C, or D. Paper B depends on Paper A's irreversibility apparatus but not on C or D. Paper C depends on A and B but not on D. Paper D depends on A, B, and C. Failure at any later paper does not invalidate the earlier ones. The kill switches are independent. Each can be checked against its observable.

Paper 0 is the narrative that motivated the work — the structural story of how nothing becomes structure, how structure once written cannot be unwritten, and how the loop running at the actualising now is what every later paper measures. Paper 0 is not load-bearing. The papers stand or fall on their own logic, independent of the narrative.

The work's operational apparatus is now installed at the spine. From this paper forward, every derivation inherits its operational vocabulary from one of the four Papers here. Every theorem rests on a kill switch. Every kill switch has an observable. The work stands open to falsification at twenty-four specified points.

The work is the work. The apparatus is in place.

Claim Summary

section 0 (Dependency and Scope). Contract with the reader.

Dependencies named (axiom as primitive; AP40 as the prior chapter in Notebook I; AP43 as structural dual; the Ø Models companions). Axiom mapping across S, B, R, C. Epistemic status per section. Twenty-four kill switches summarised. No outstanding structural debts. Three items held

open without claiming debt status. Structural relationships to AP07, AP08, AP25, AP43, and the Ø Models companions named.

Paper 0 (Foundations). NARRATIVE. The structural story that motivated the work — nothing becomes structure; structure once written cannot be unwritten; the loop runs at the actualising now. Explicitly non-load-bearing. The four formal Papers stand or fall on their own logic, independent of the narrative.

Paper A (Actualization State: An Operational Measure of Record-Structured Irreversibility). DEFINITION + DERIVATION. Five core definitions (D1–D5). Theorem T1 (monotonicity of AS under decohering dynamics) with rate corollary T1a and converse T1b. Theorem T2 (operator horizon — no-return as an inequality). No-return surfaces and operational irreversibility (D6–D10, P3.3). Quantum-mechanical instantiation (D11–D16, postulate P, postulate G, P4.1). Seven falsifiers: F0 (global, KS-V.1), F1–F3 (selection postulate, KS-V.2–V.4), G1–G3 (gravitational, KS-V.5–V.7). Optional turn module (A6) and appendices.

Paper B (Selection as Irreversible Exclusion). STRUCTURAL DERIVATION. Definiteness reframed as operational individual-realization status. Nonlinearity requirement and selection cost made operational (δ_{sel} formalised in B2.3, B2.4). Six structural requirements on selection dynamics (B3.1–B3.6). Universal rate constraints with gravity as candidate limiter (B4). Five experimental regimes (BT1–BT5). Kill switch B2 (KS-V.8) on pre-irreversibility selection. Programmatic closure: no further progress on selection can be made by argument alone; the remaining uncertainty is empirical.

Paper C (Agency as Constrained Control). DEFINITION + DERIVATION. Agency as operator-relative ratio of reachable-within-viability to viability (C1). Control authority approaching zero at no-return surfaces. Drift under irreversibility (C2). Necessary conditions for agency preservation (C3). No-return geometry within a realised sector (C4). Control budgets and fatigue (C5 with theorem C5.1). Noise and silence (C6). Coupling and rescue (C7). Slack and robustness (C8). Exit as control outcome (C9). Falsifiers FC1–FC5 (KS-V.9–V.13). Structural closure with experimental instantiation (bacterial chemotaxis, autonomous robotic navigation).

Paper D (Coupled Viability). DEFINITION + DERIVATION. Shared constraint environments with Geometric Exclusion Principle (D1.3). Composition of agency: non-additivity, impedance matching, anti-resonance (D2). Stable configurations under drift with necessary (D3.2a) and sufficient (D3.2b) conditions; cascade failure (D3.3). Emergent order without design: structural filtering (D4.2), hierarchy (D4.3), cooperation and deterrence as geometric outcomes (D4.4). Eleven falsifiers FD0–FD4.4b (KS-V.14–V.24). Structural closure with worked examples and Appendix E flagged as exploratory and non-load-bearing.

section 6 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-V.1 (global kill switch on AS operational invariance, LIVE — EMPIRICAL). KS-V.2–V.4 (selection postulate, LIVE — EMPIRICAL). KS-V.5–V.7 (gravitational rate limiter, LIVE — EMPIRICAL). KS-V.8 (pre-irreversibility selection, LIVE — EMPIRICAL). KS-V.9–V.13 (agency apparatus, LIVE — EMPIRICAL). KS-V.14–V.24 (coupled-viability apparatus, LIVE — EMPIRICAL). Each kill switch in Claim · Test · Status · Recovery form. Legacy labels F0/F1–F3/G1–G3/B2/FC1–FC5/FD0–FD4.4b annotated at first mention.

section 7 (What This Establishes and What Remains Open). SYNTHESIS. Four operational quantities installed. No structural debts outstanding. Three items held open without claiming debt status (coupling parameter first-principles derivations, relativistic extension of the gravitational rate bound, multipartite operator dynamics). Philosophical-register companions named: Dissolutions Chapter 12 (Measurement Problem), Dissolutions Chapter 6 (Other Minds), Dissolutions Chapter 7 (Free Will and Determinism — freedom-under-constraint), Resolutions Chapter 8.5 (Choice — both locked formulations), Resolutions Chapter 10 (Other Intelligences — four operator-hood conditions). Placement in The 420 Code: Chapter 2 of Notebook I — The Premise.

Conditionality Footer

Dependencies

The axiom $1:1 + 1 \times \varepsilon$ @ AS as primitive (foundational to The 420 Code; no upstream paper required for AP01's own work).

The axioms {S, B, R, C} (formal statement carried at AP20 section 3.3; completeness and minimality obligations fenced at KS-P.1/KS-P.2). Load-bearing for every derivation in Papers A through D.

AP40 (The Irrational — the prior chapter in Notebook I, examining the structural character of the axiom; not load-bearing for AP01's derivations but named for completeness).

Standard tools: dephasing maps and decoherence dynamics (open quantum systems); doubly stochastic operations (Schur–Horn); viability theory (Aubin's reach-and-kernel formalism); control theory (admissible control and control budgets); Gleason-type non-contextuality (invoked in Paper B's boundary condition; formally derived in AP25).

Dependents

AP07 (The Record Measure) — inherits Paper A's record-algebra coarse-graining.

AP08 (The Identity — Einstein Field Equations) — inherits Paper A's irreversibility theorems.

AP25 (The Measure) — inherits Paper A's coarse-graining apparatus and dephased-entropy structure.

AP43 (The Gravity of Possibilities) — inherits AS as the records-side anchor and develops the structural dual on the pre-state side.

Every derivational paper in Notebooks II through VIII inherits operational vocabulary from one of the four Papers here.

The Ø Models catalogue's philosophical-register treatments of measurement (Dissolutions Chapter 12), other minds (Dissolutions Chapter 6), free will

and determinism (Dissolutions Chapter 7), choice (Resolutions Chapter 8.5), and other intelligences (Resolutions Chapter 10) stand on this paper's apparatus.

Kill switches closed by this paper

None explicitly closed. This paper opens twenty-four new kill switches (KS-V.1 through KS-V.24) and does not retire any prior kill switches in The 420 Code.

Kill switches that remain live in this paper

KS-V.1 through KS-V.24. See section 6 for the full register with Claim, Test, Status, and Recovery for each.

Structural debts owed

None outstanding from this paper.

Items held open without claiming debt status

First-principles derivation of the coupling parameters in Papers C and D (control budget B_0 , cost function $c(u)$, impedance Z_i , resonance phase). Parameterised in the structural account; physical instantiations open work for domain experts.

Full relativistic extension of Paper A's quantum-mechanical instantiation, including the relativistic form of the gravitational rate bound and the no-return surface in curved spacetime. Sketched only; open work for AP08 and the future relativistic-field-equation paper.

Structural status of multipartite operator dynamics — operations between three or more operators reading each other's slopes simultaneously in shared viability domains. Paper D treats binary and chained couplings; higher-order dynamics inherit AP43's four-term system and are future work in The 420 Code.

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